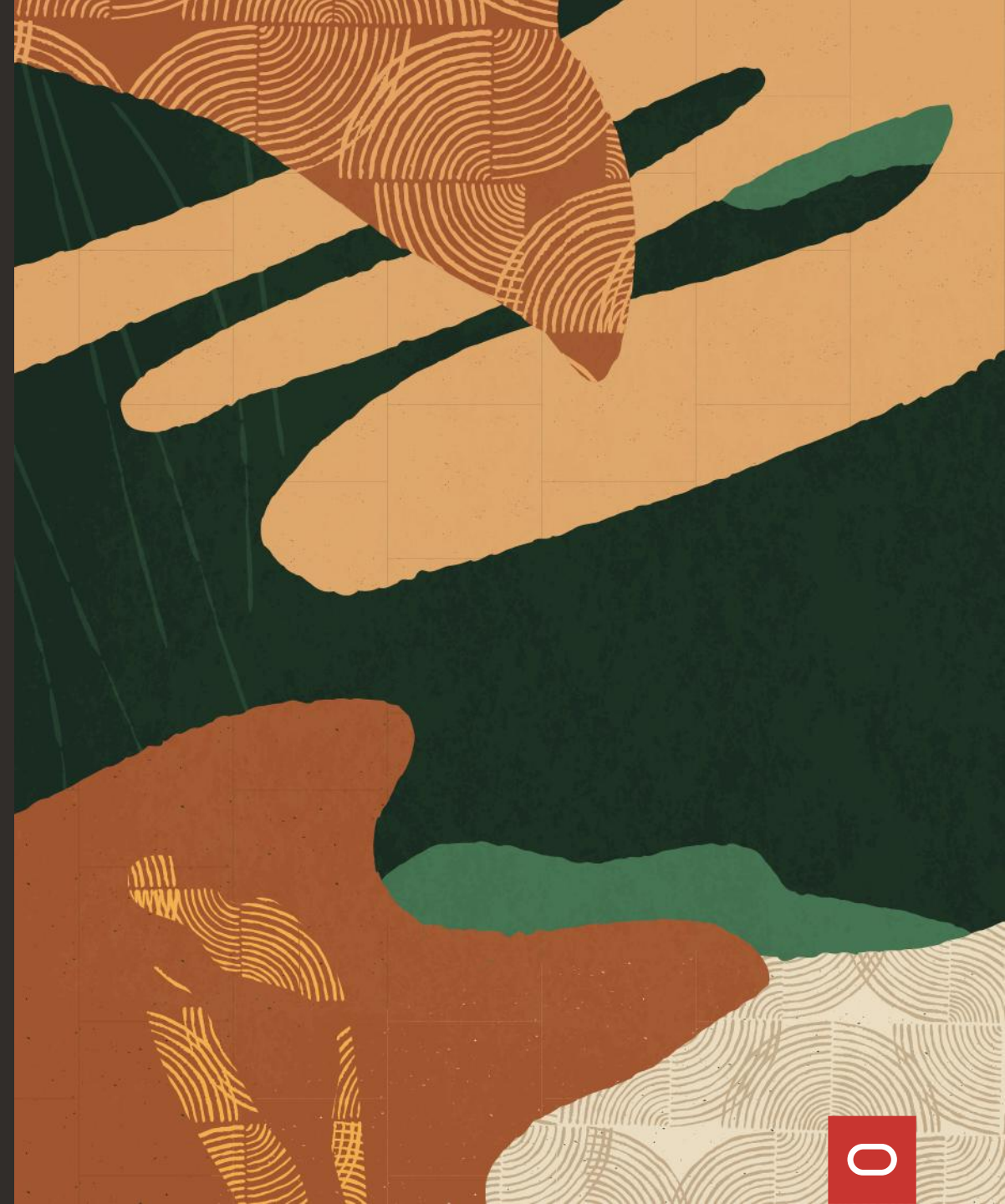


# Oracle Database@Azure

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Technical Deck

**OCI Multicloud**



# Safe Harbor Statement

The following is intended to outline our general product direction. It is intended for information purposes only and may not be incorporated into any contract. It is not a commitment to deliver any material, code, or functionality, and should not be relied upon in making purchasing decisions. The development, release, timing, and pricing of any features or functionality described for Oracle's products may change and remains at the sole discretion of Oracle Corporation.

Statements in this presentation relating to Oracle's future plans, expectations, beliefs, intentions and prospects are "forward-looking statements" and are subject to material risks and uncertainties. A detailed discussion of these factors and other risks that affect our business is contained in Oracle's Securities and Exchange Commission (SEC) filings, including our most recent reports on Form 10-K and Form 10-Q under the heading "Risk Factors." These filings are available on the SEC's website or on Oracle's website at <http://www.oracle.com/investor>.



# Agenda

01

Service Overview

02

Exadata Database Service (Exa-D)

03

Autonomous Database (ADB-S)

04

Architecture, Network Topologies

05

Purchase / Enablement, Provisioning

06

HA-DR, Backup

07

Migration & Observability

08

Identity, Security and Compliance

09

Roadmap

09

Automation

10

Support



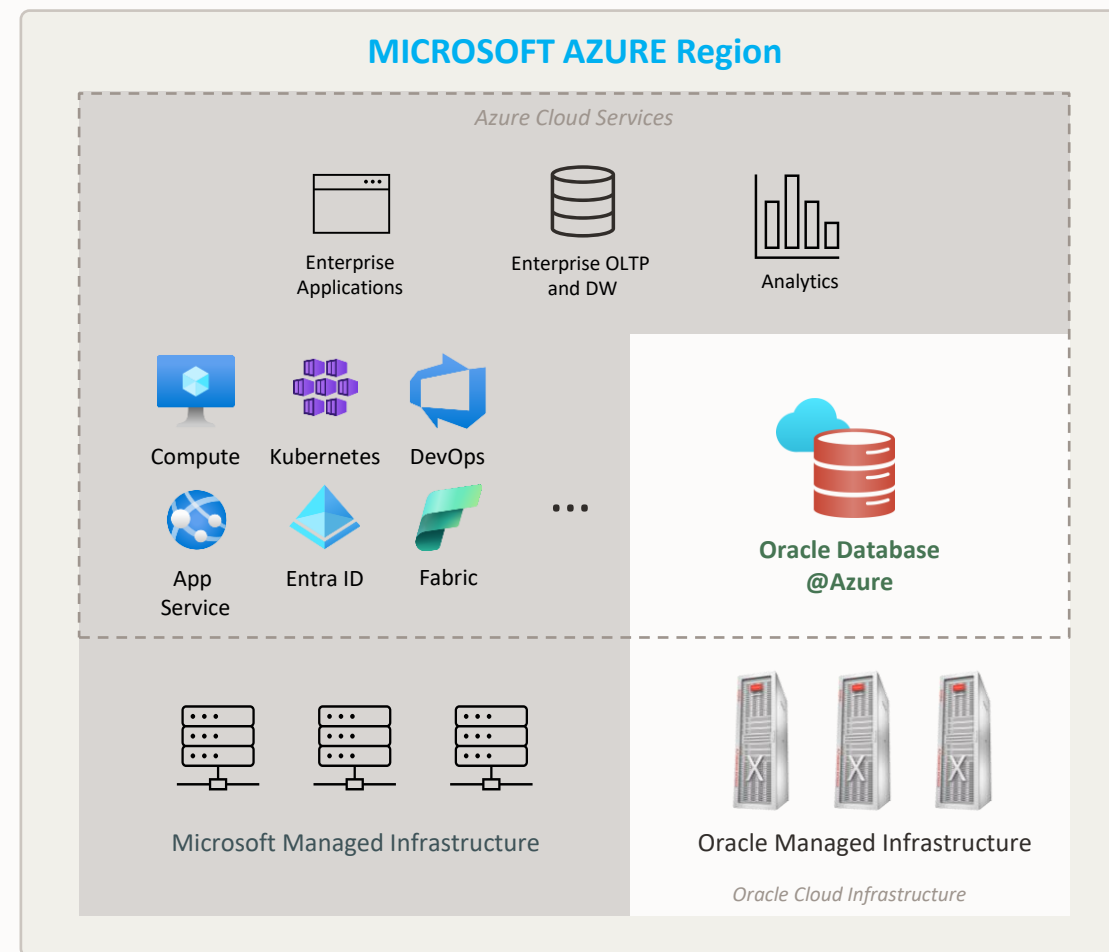
# Service Overview



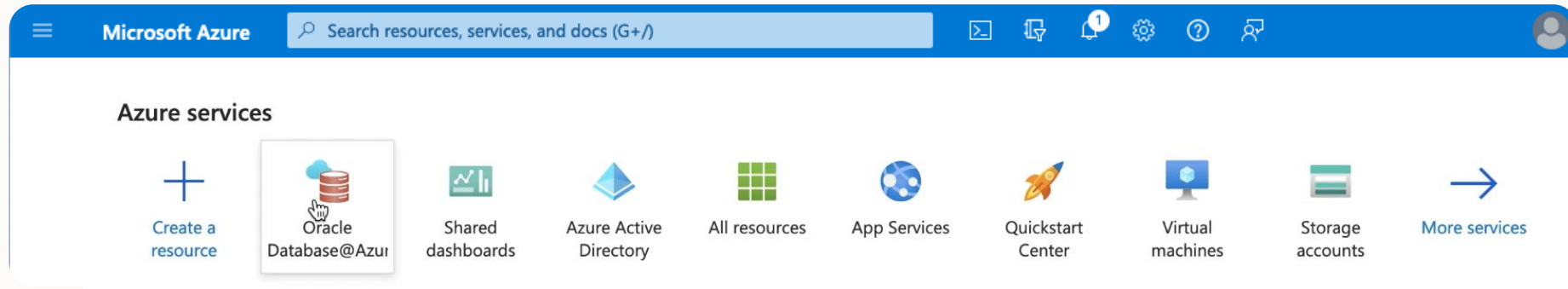
# Oracle Database@Azure – Multicloud done right

Oracle database services on OCI in Microsoft Azure datacenters enable you to:

- Migrate and modernize Oracle databases in the cloud
- Innovate with OCI and Azure services
- Simplify purchasing and operations



# Oracle Database@Azure offerings



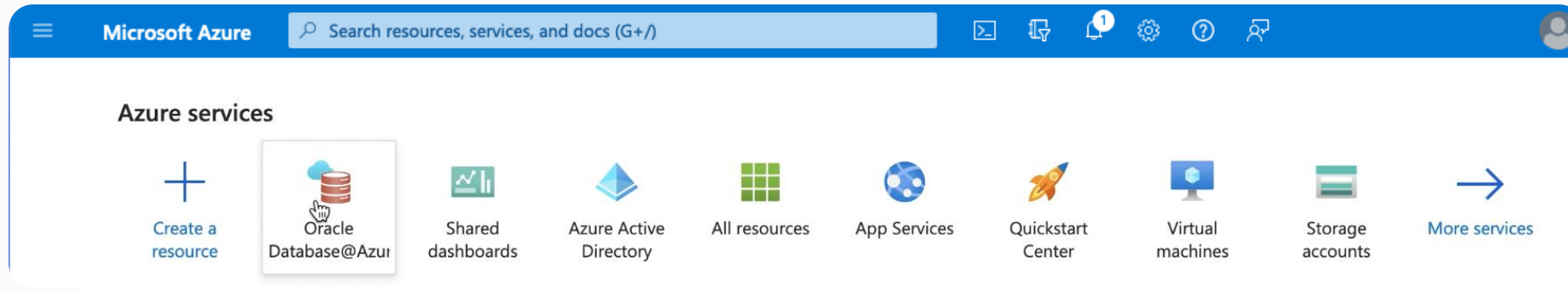
## Exadata Database Service

- Dedicated, Oracle-managed infrastructure for most challenging workloads and performance/isolation
- Scale-out design, database-optimized compute, networking, storage
- Full control of database management and control plane
- Algorithm-driven OLTP, analytics, and consolidation

## Autonomous Database

- Built on Oracle Exadata infrastructure
- Fully managed and automated provisioning, patching, upgrades, hardware refreshes with enterprise-class business continuity
- Elastic resources and pay-for-use billing
- Built-in low-code APEX, ELT, AI/ML, graph and spatial analytics

# Oracle Database@Azure offerings



## Zero Dataloss Autonomous Recovery Service

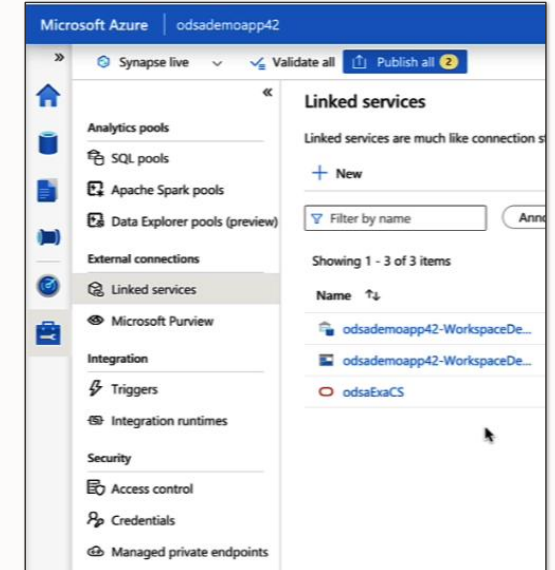
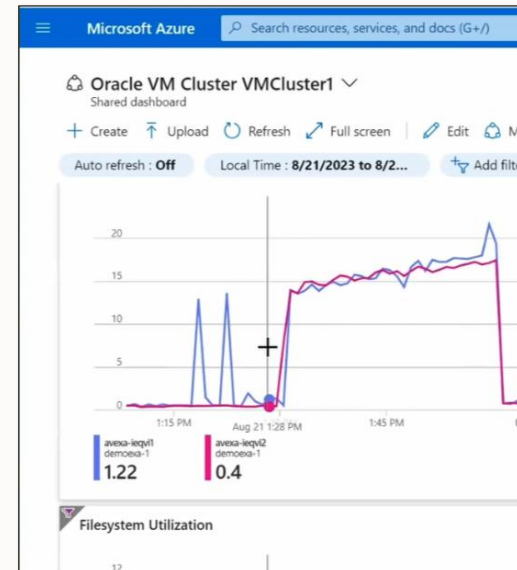
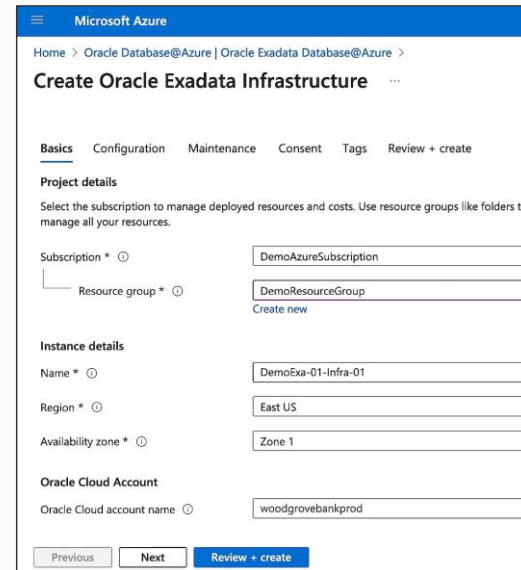
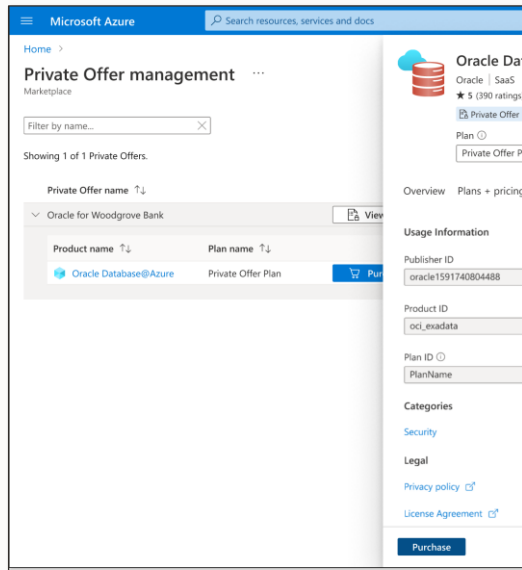
- Oracle Database Zero Data Loss Autonomous Recovery Service is a fully managed data protection service for Oracle databases running on Oracle Database@Azure Exadata Database Service
- Backup automation and enhanced data protection capabilities to offload all backup processing and storage requirements to Oracle Database Autonomous Recovery Service,
- Eliminating backup infrastructure costs and manual administration overhead. control plane

# The best of Oracle and Microsoft, all in one place

Purchase in Azure Marketplace and parity with OCI

Deploy, manage, and monitor through Azure Portal and APIs

Combine with your choice of Azure services

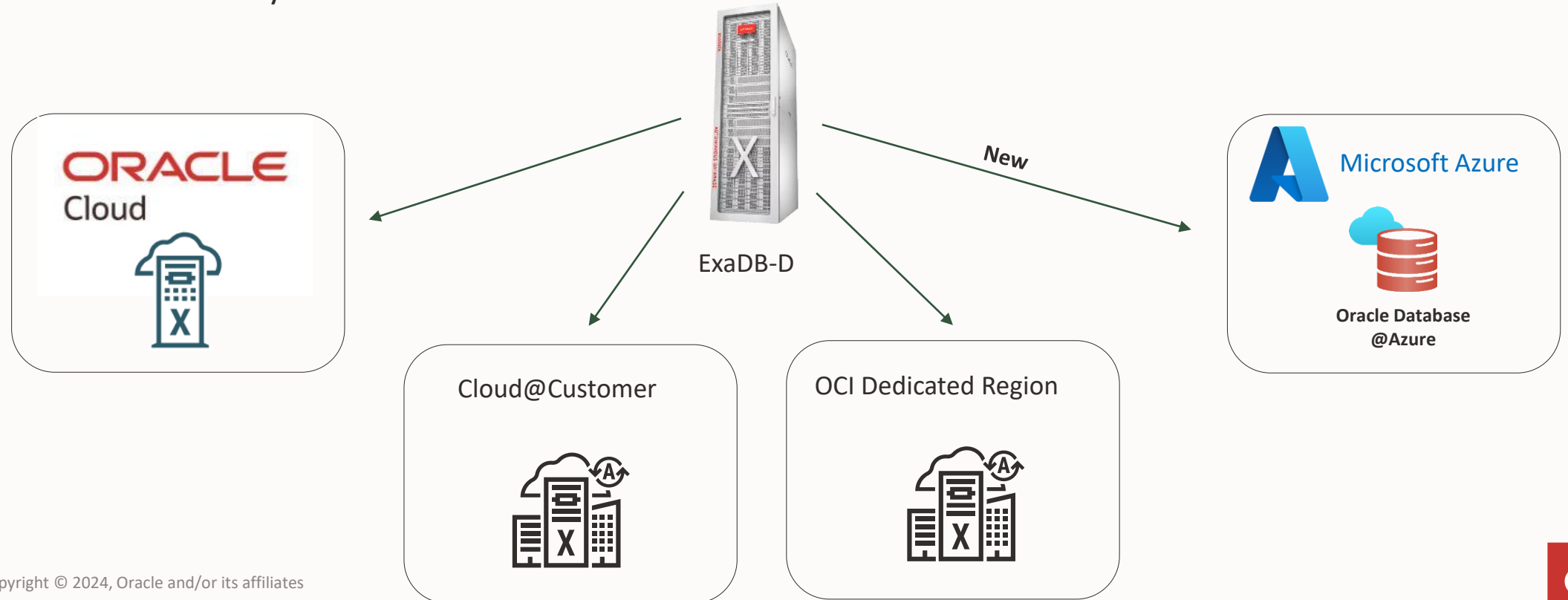


# Exadata Database Service on Dedicated Infrastructure (ExaDB-D)



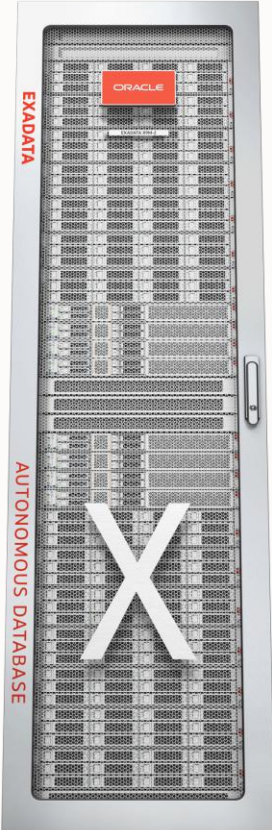
# Deployment models

- ExaDB-D that is available in Oracle Database@Azure is the same that is offered in other deployment models (OCI, Cloud@Customer and Dedicated Region)
- This new solution is optimized for those customer who need the performance, availability and resiliency of an ExaDB-D but want to continue with application tier in Azure and need a very short network latency to database



# Exadata Vision

Extreme Performance and Availability, Lowest Cost, Available Everywhere



## Ideal Database Hardware

Scale-out, database optimized compute, networking, and storage

## Database Aware System Software

Unique algorithms vastly improve OLTP, Analytics, and Consolidation

## Automated Management

Fully automated and optimized end-to-end

## Available

Oracle Cloud

Oracle  
Cloud@Customer

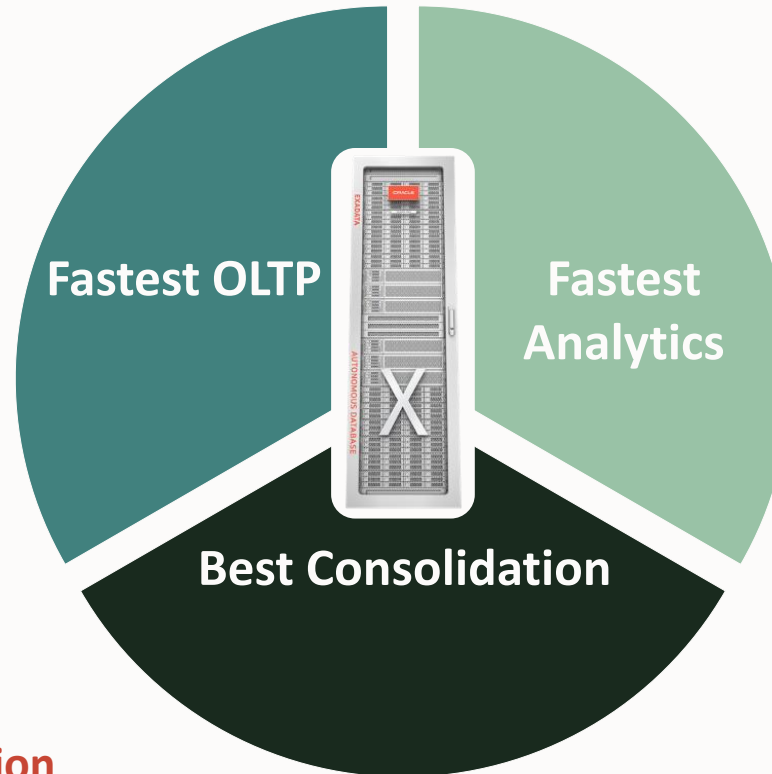
Oracle  
Database@Azure

On Premises

# Exadata Smart System Software: OLTP | Analytics | Consolidation

## Fastest OLTP

- **Fastest OLTP I/O** with scale-out storage, RDMA, shared memory in storage servers, and NVMe flash
- **Fastest scale-out** with unique RDMA algorithms for inter-node cluster coordination
- **Fastest recovery** from unplanned downtime and component failure



## Fastest Analytics

- **Unique Smart Scan** automatically offloads data intensive SQL operations to storage
- **Unique Smart Flash Cache** and **Storage Index** automatically accelerate database I/O
- **Unique Columnarization** automatically converts data to fast In-Memory Columnar format in flash

## Best Consolidation

- **Unique Prioritization** of latency sensitive or important workloads
- **Unique Workload Isolation** of multiple tenants or workloads

# Exadata Database Service on Dedicated Infrastructure

All the Power and Functionality of Oracle Database and Exadata in the Public Cloud

## Enterprise-Grade Oracle Database as a Service

- Same architecture as on-premises Exadata
  - Runs Oracle Real Applications Clusters
- Dedicated Exadata platform for performance/isolation
- Includes all Oracle Database options or Bring Your Own License
- 100% compatible with existing applications that use Oracle Database

## Simplicity, Elasticity, Flexibility

- Infrastructure managed by Oracle experts
- Root access to customer VM
- Customer controlled cloud automation
- Online, elastic resource scaling
- Pay-per-use database cores



### Exceptional Outcomes

- Performance
- Availability
- Security
- Consolidation



### Elastic Scaling

- Add CPU and Storage Online
- No DB/Workload too Large



### Highest Value

- Best Price/Performance
- Pay-per-use Pricing
- One DB Service for all workloads



# Exadata X9M Infrastructure in Public Cloud

Fully-elastic compute and storage provisioning and expansion

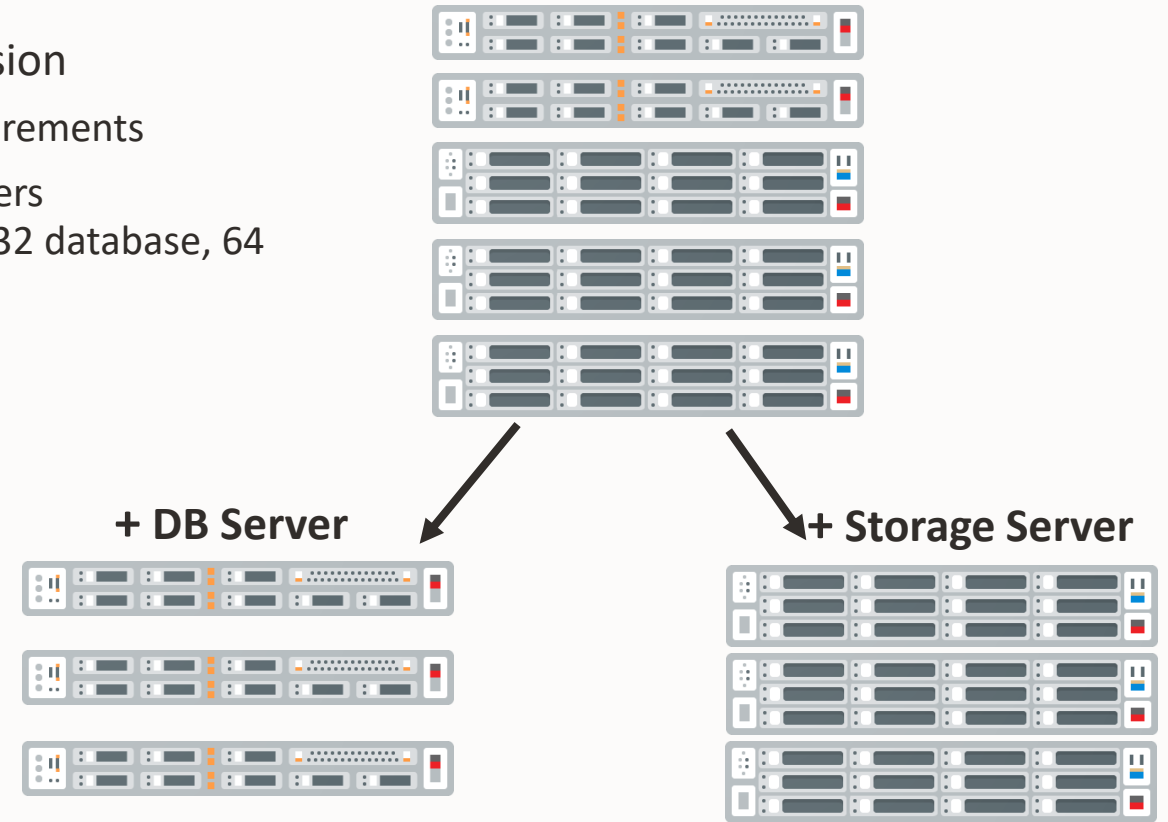
- Properly size hardware configurations to meet workload requirements
- Minimum configuration of 2 Database Servers, 3 Storage Servers
  - Add as little as one database or one storage server up to 32 database, 64 storage servers
- Dedicated to you – no noisy or malicious neighbors

## Scale-Out X9M Database Servers

- 126 Maximum OCPUs
- 1,390 GB Memory
- 50 Gbps Network (Shared for Client/Backup)

## Scale-Out Intelligent X9M Storage Servers

- 48 processor cores
- 1.5 TB PMem
- 25.6 TB NVMe Flash
- 63.6 TB HDD Usable Storage



**Add Database or Storage Servers as needed**



# Exadata Database Service on Dedicated Infrastructure Architecture

## Exadata Components in the Cloud

- Scale-out database and storage servers connected by secure fabric
- Database clustering with Real Applications Clusters
- Fully redundant hardware and software for high availability

## Cloud Controls and Networking

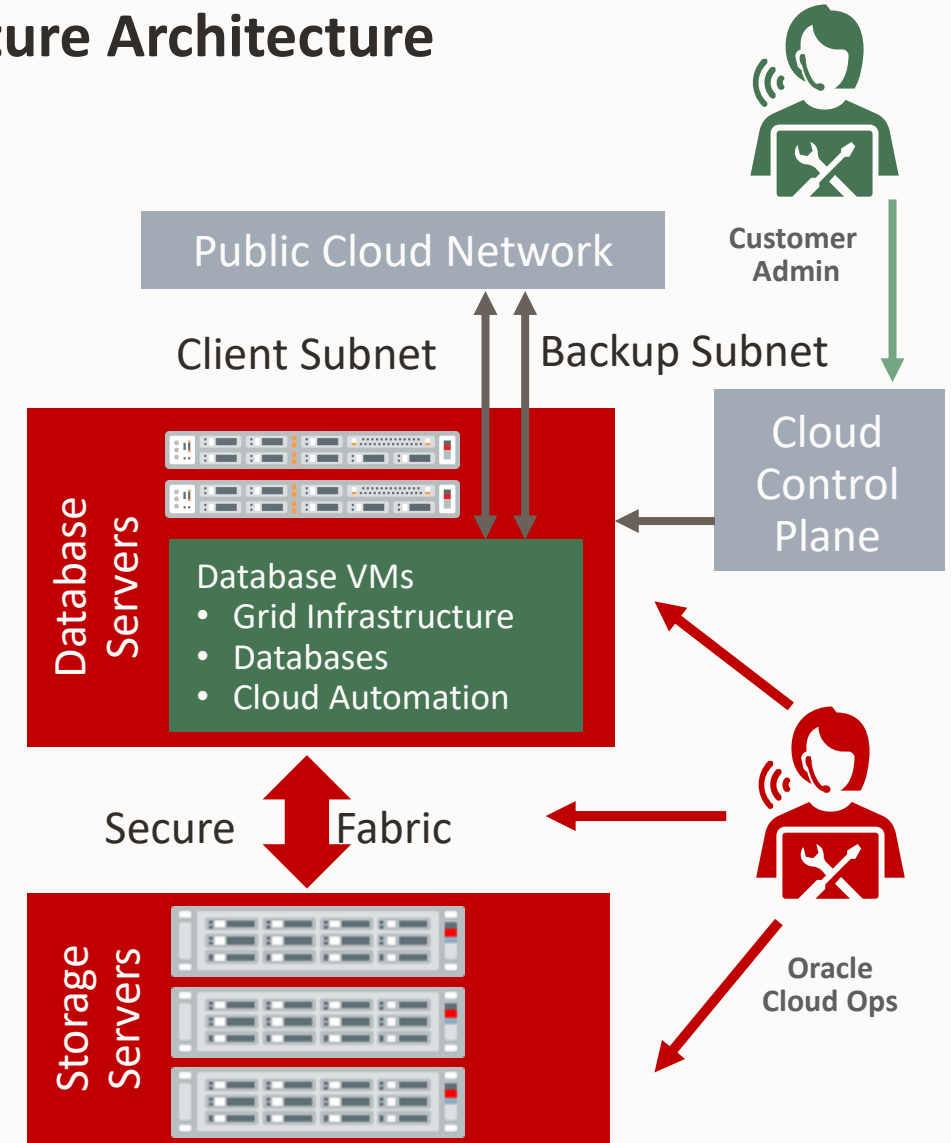
- Identity management and policies to control access
- Software defined networks with isolation

## Secure Isolation

- Databases run in secure virtual machines isolated from underlying infrastructure

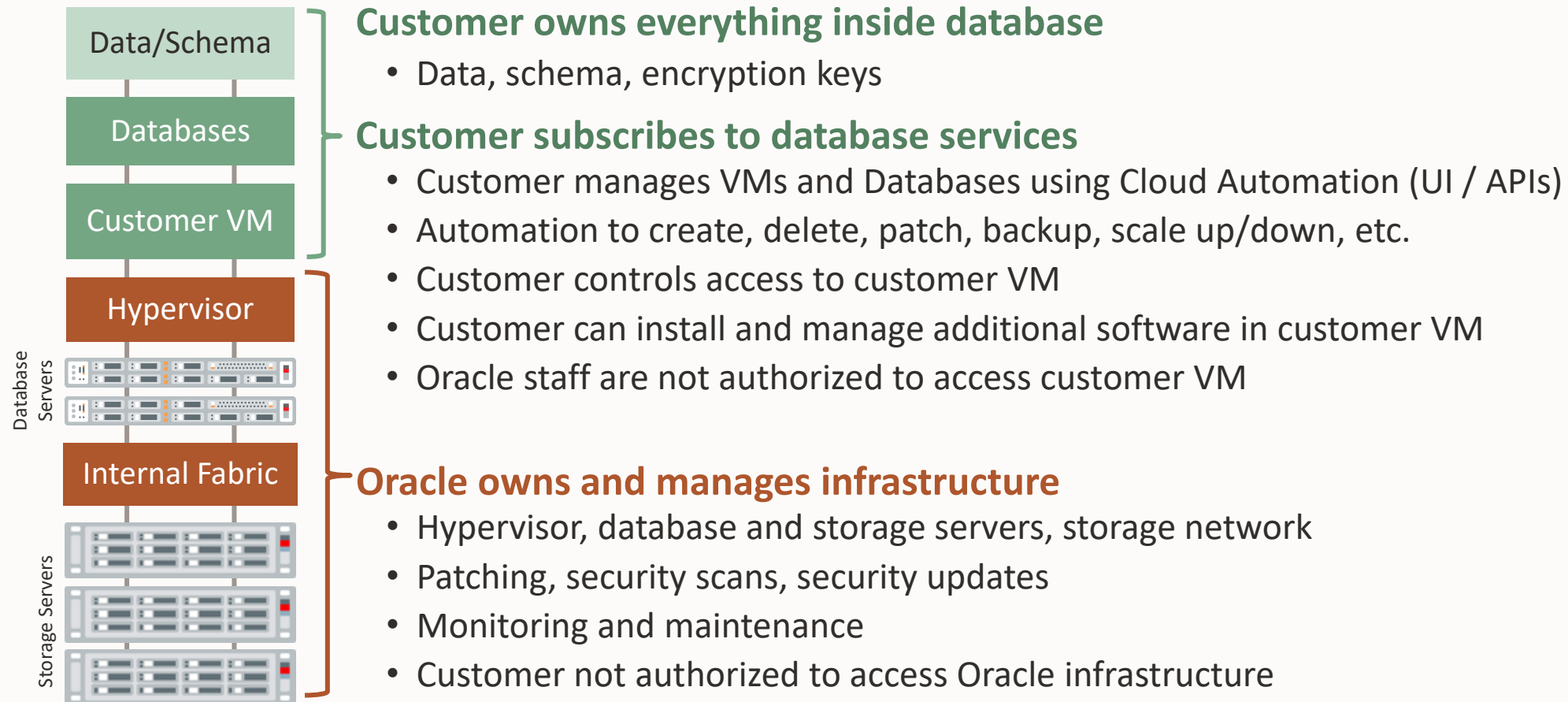
## Control Plane

- Controls and manages Exadata infrastructure





# Simple Cloud Management Model in Public Cloud



# Benefits of Oracle Database@Azure with ExaDB-D

1

Run your applications with the best platform for Oracle databases in Azure

2

Achieve fastest performance and elastic scalability with Exadata

3

Attain maximum availability uniquely featuring Oracle RAC in Azure

4

Consolidate on Exadata to Reduce Costs

5

Migrate to cloud with no disruption and zero downtime



# Oracle Database@Azure Autonomous Database - Serverless

# Oracle Autonomous Database is complete and simple

Fully-managed, cloud native, Database as a Service

## Oracle Data Management Vision:

Make it **simple** to **develop** and **run** modern operational and analytic apps for **all** use cases at **any** scale



**Autonomous  
Database**

**Autonomous Database** delivers the complete Converged Database

- Best price, best performance
- Most cost-effective for all use cases at any scale
- Simplest for operations, developers, and analysts

# Oracle Autonomous Database delivers the best data experience

With the lowest total cost of ownership



**Mission-Critical  
Database**



**High Performance  
Engineered System**



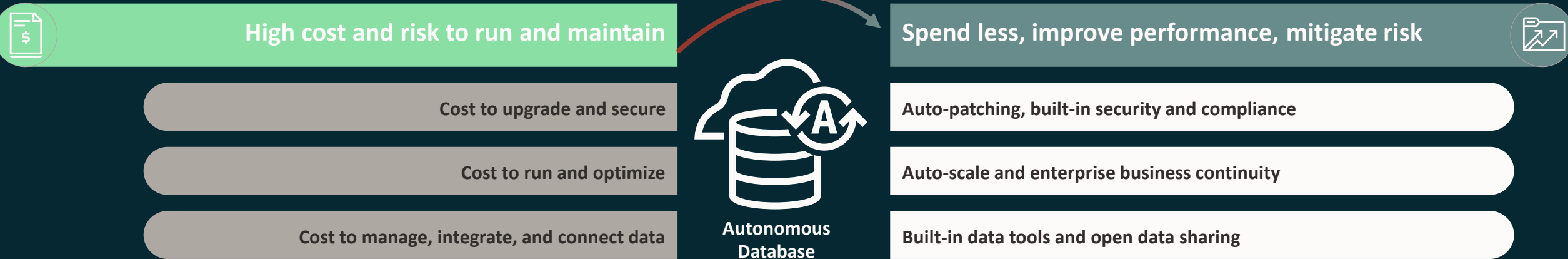
**Automation,  
Best Practices  
and Data Tools**



**Autonomous  
Database**

# Oracle Autonomous Database is complete and simple

Fully-managed, cloud native, Database as a Service



# Autonomous Database automatically secures, tunes, and scales



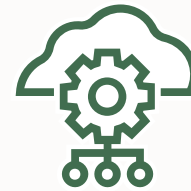
**Automated  
patching and  
security**



**Automated  
scaling**



**Automated  
backups**



**Automated  
provisioning**



**Automated  
configuration**



**Automated  
tuning**



Eliminates manual  
tasks



Eliminates human  
error



Eliminates  
downtime



Eliminates scaling  
complexity



Eliminates  
performance  
tuning complexity





# Simplification of day-to-day operational tasks

## Automated security



- Automatically and frequently applies the latest and greatest security patches
- Automatically encrypts all data at rest, in motion, and when backed up to protect from bad actors or malware
- Automatically protects against unauthorized user access to sensitive data as well as unauthorized database changes
- Offers built-in Database Vault for meeting industry, regulatory, and corporate security standards

## Automated scaling



- Automatically scale for changing workloads
- Zero downtime with no impact to performance
- Lower costs—pay-for-use model
- Shut off compute for idle systems, restart instantly
- Schedule automatic shutdown-startup

## Automated backups



- No manual backups required
- Fully-managed, automated backups
- Full backup every 60 days along with weekly cumulative and daily incremental
- Restore or recover a database to any point-in-time within the last 60 days

# Simplification of day-to-day operational tasks

## Auto-provisioning



- Deploys mission-critical databases which are fault-tolerant and highly available
- Automatically scales compute resources when needed and enables true pay-per-use
- Option to select patch level when provisioning ADB
- View maintenance and patch history

## Auto-configuration



- Automatically configures the database to optimize for data warehouse workloads
- Automatic partitioning and automatic indexing simplify and automate design decisions - removes guess-work from data management
- Predefined maintenance windows for automatically applying patches

## Automated tuning

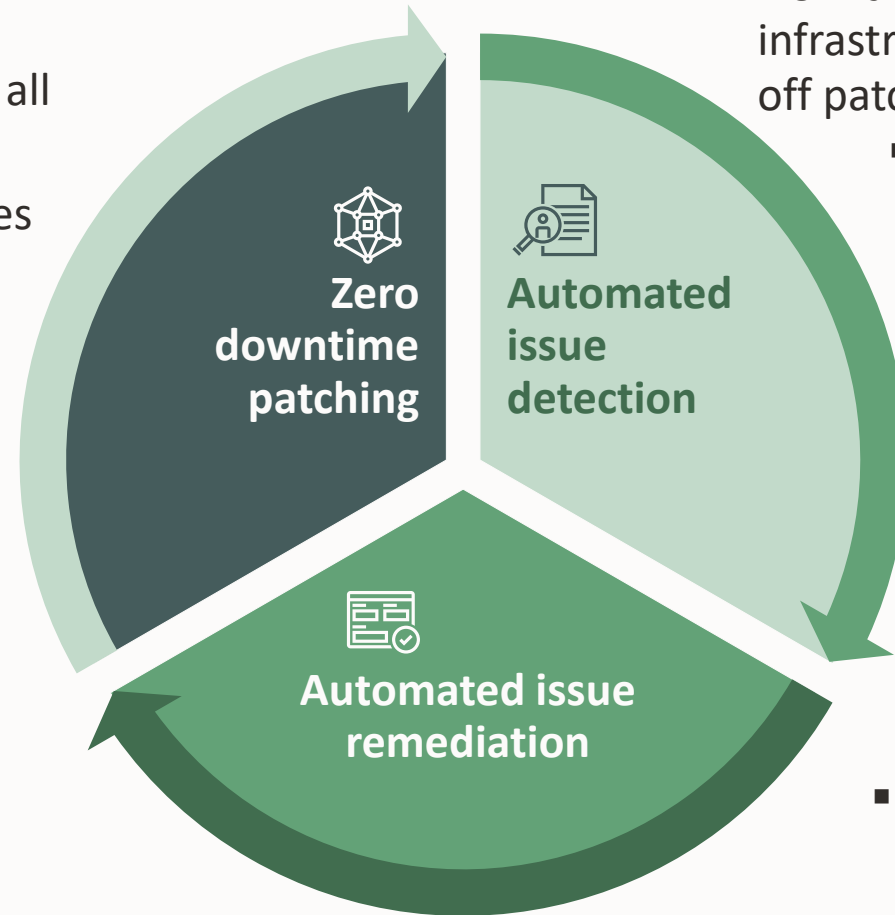


- ADB captures, identifies, verifies and monitors SQL statement execution
- Automatically optimizes SQL statements for better performance
- Use Performance Hub to view time range for performance data (ie Active Session History (ASH) Analytics details across different dimensions

# Automated security and bug fixes - Auto-patching

Reduce costs and risks of manual patching

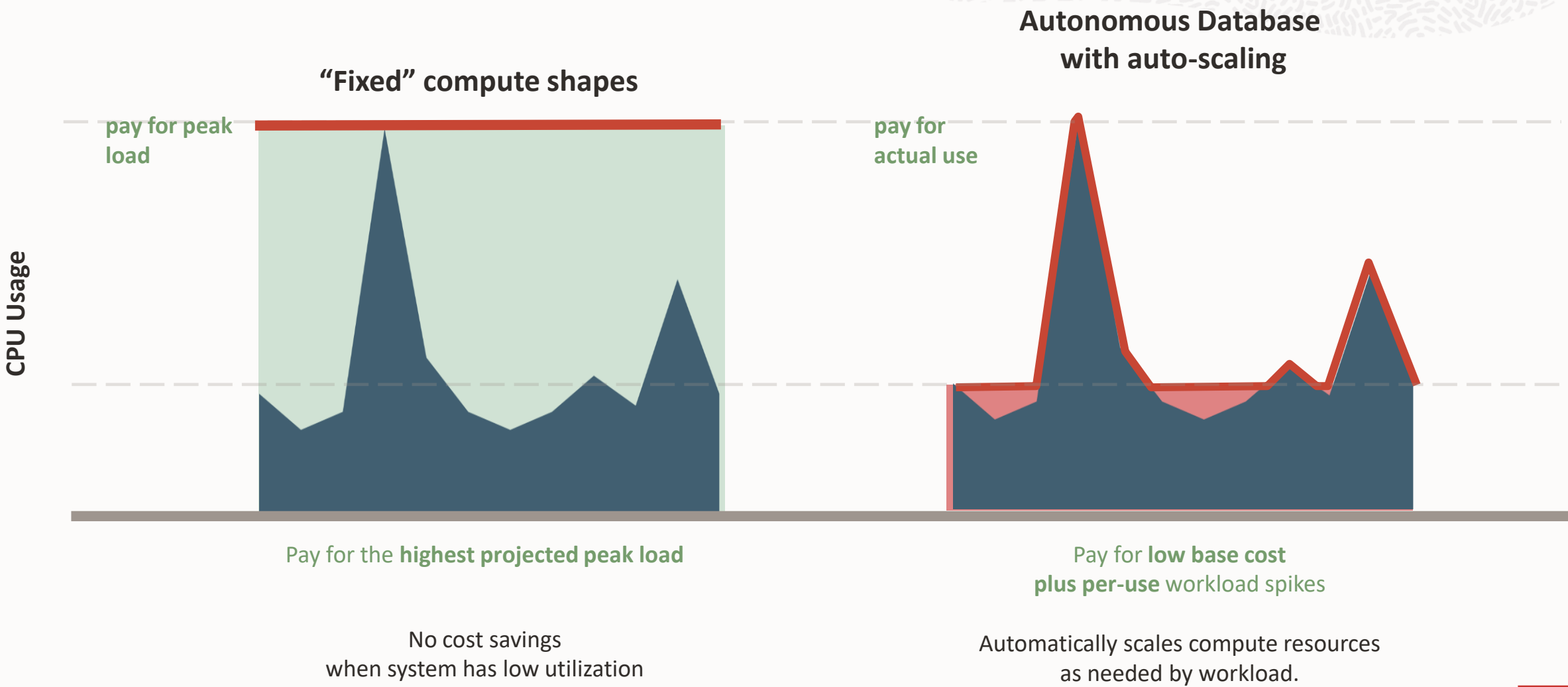
- **Zero downtime and no interruption of service** – all systems continue to run
- Continuous testing of fixes through CI/CD pipeline



- **No manual patch management** for infrastructure, release update or one-off patches
  - Continuously monitors the entire infrastructure for security issues and bugs
    - Automatically generates Service Requests for each issue
- **88% of issues are automatically resolved** with zero customer actions
- Immediate investigation and resolution by Cloud Ops
- Root-cause analysis for every issue

# Lower costs with elastic resources and pay-for-use billing

Comparing fixed compute shapes to auto-scaling



# Lower costs with flexible licensing

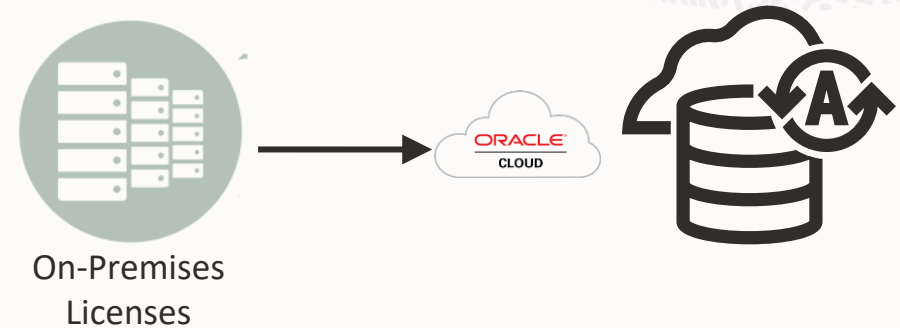
Up to **75% cost savings**



Everything is built in

## 1. License Included

- Ideal for organizations seeking to build new applications or expand functionality
- Includes Oracle Database, analytics, operations and management packs at one low price
- **Pay-per-use pricing, all paid using Universal Credits**



## 2. Bring Your Own License

- **Ideal for organizations moving existing workloads**
- Utilize **existing on-premises licenses** and benefit from 75% cost savings
- **Oracle Support Rewards** further reduces OCI costs
  - Cloud consumption provides credits toward technology support
- Includes Transparent Data Encryption, Data Safe, Partitioning, Advanced Compression, and select management packs at **no additional cost**



# Higher reliability and availability through automation

## Highly-available, fully automated infrastructure

- Automatic recovery for compute node failures
- Triple-mirrored disks for disk failures

## Online patching for complete protection

- Frequent software and security patching cycle
- Patches applied with zero database downtime

## Automatic backups

- Fully automated daily backups to Cloud Object Storage
- Point in time recovery to anytime in last 60 days



*"I'd rather build new features and functionality than worry about our business operating. Oracle Autonomous Database provides autoscaling, self-tuning, self-repairing, and self-encryption so **we have zero downtime to securely process mission-critical financial transactions** while we focus on delivering new capabilities to our customers at the same time."*

Ray Grove, Vice President, Product Management,  
Thomson Reuters

# Automation removes time spent on operational tasks

DBA's now focus on innovation and increasing business value

## Tasks and responsibilities to **generate business value**

- Architecture, planning, data modeling
- Data security and lifecycle management
- Application related tuning
- End-to-end service level management

## Tasks and responsibilities for ~~maintenance and administration~~

- ~~• Configuration and tuning of systems, network, storage~~
- ~~• Database provisioning, patching~~
- ~~• Database backups, H/A, disaster recovery~~
- ~~• Internal helpdesk and call center~~

**Welcome to the future: Autonomous Database**

## Value Scale

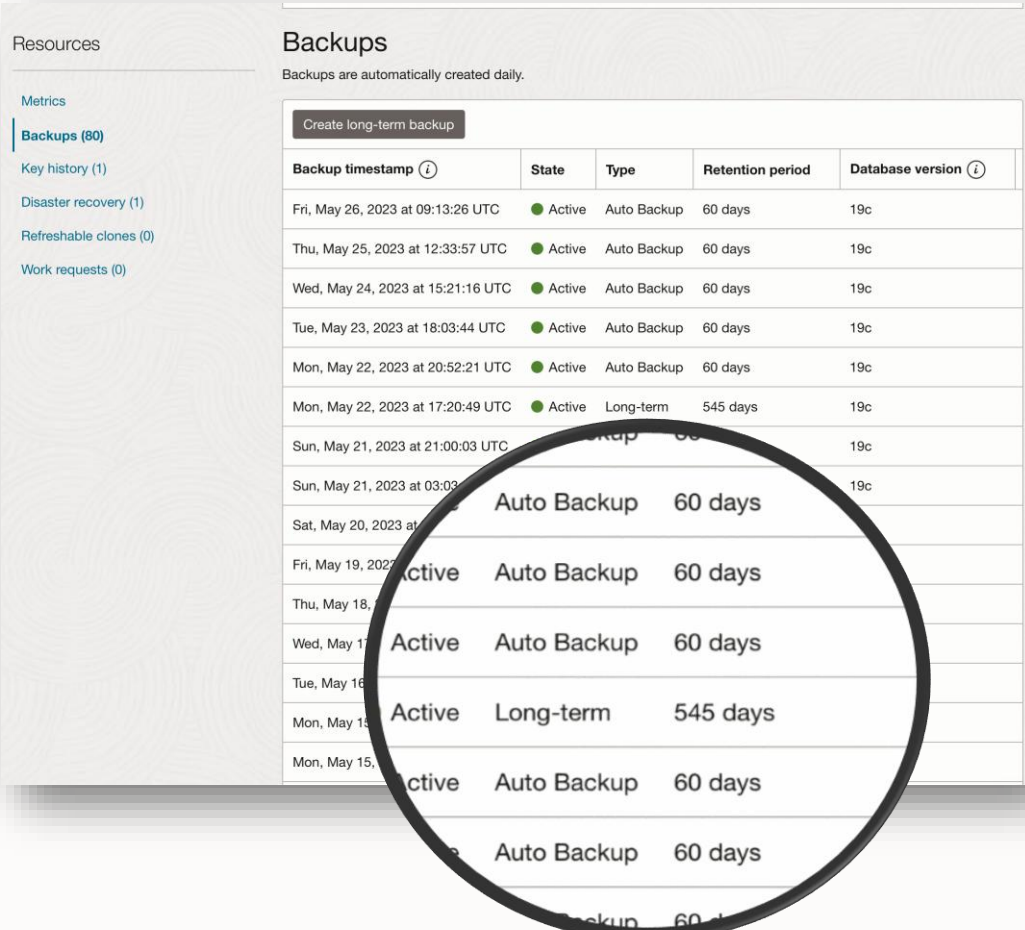




# Minimize risk with **automated backups**

Top reasons why automated backups beat customer-managed backups

- 1 No customer action required**  
Fully automated daily backup  
Configurable retention from 0 to 60 days  
Point in time recovery
- 2 Automated backup testing**  
Validation ensures future data restoration
- 3 Optional long-term backup retention**  
Up to 10 years for regulatory compliance
- 4 Create a new database from backup**  
Deploy to any cloud region  
Restore to any point in time



Resources

Metrics

**Backups (80)**

Key history (1)

Disaster recovery (1)

Refreshable clones (0)

Work requests (0)

### Backups

Backups are automatically created daily.

Create long-term backup

| Backup timestamp ⓘ                | State  | Type        | Retention period | Database version ⓘ |
|-----------------------------------|--------|-------------|------------------|--------------------|
| Fri, May 26, 2023 at 09:13:26 UTC | Active | Auto Backup | 60 days          | 19c                |
| Thu, May 25, 2023 at 12:33:57 UTC | Active | Auto Backup | 60 days          | 19c                |
| Wed, May 24, 2023 at 15:21:16 UTC | Active | Auto Backup | 60 days          | 19c                |
| Tue, May 23, 2023 at 18:03:44 UTC | Active | Auto Backup | 60 days          | 19c                |
| Mon, May 22, 2023 at 20:52:21 UTC | Active | Auto Backup | 60 days          | 19c                |
| Mon, May 22, 2023 at 17:20:49 UTC | Active | Long-term   | 545 days         | 19c                |
| Sun, May 21, 2023 at 21:00:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Sun, May 21, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Sat, May 20, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Fri, May 19, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Thu, May 18, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Wed, May 17, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Tue, May 16, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Mon, May 15, 2023 at 03:03:03 UTC | Active | Long-term   | 545 days         | 19c                |
| Mon, May 15, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |
| Sun, May 14, 2023 at 03:03:03 UTC | Active | Auto Backup | 60 days          | 19c                |

# Oracle Database@Azure BaseDB

# Oracle Base Database Service

## Standard and Enterprise Database Services

### Full-featured Oracle Database cloud instances

- Deploy the features required from Standard Edition 2, Enterprise Edition, and Oracle Database options
- Run single-instance or enhance availability with 2-node RAC

### Efficient database lifecycle management

- Eliminate infrastructure management
- Increase administrator productivity with user-controlled cloud automation
  - Web and API-driven lifecycle operations
  - Provisioning, updates, backups, etc.

### Cloud economics

- User initiated rolling OCPU scaling to align costs with allocated compute resources
- Choice of 4-tiers of Oracle Database License-included options or Bring Your Own License (BYOL)

### Base Database Service



#### Traditional

Oracle Database service  
in VMs on shared compute

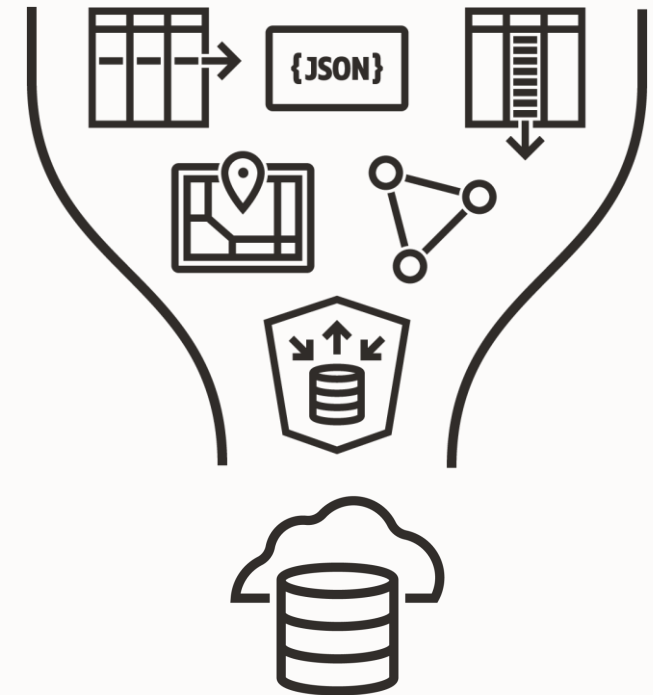
Co-managed

Full administrative control

Supports all DB features incl. RAC

# Oracle Database on Base Database Service

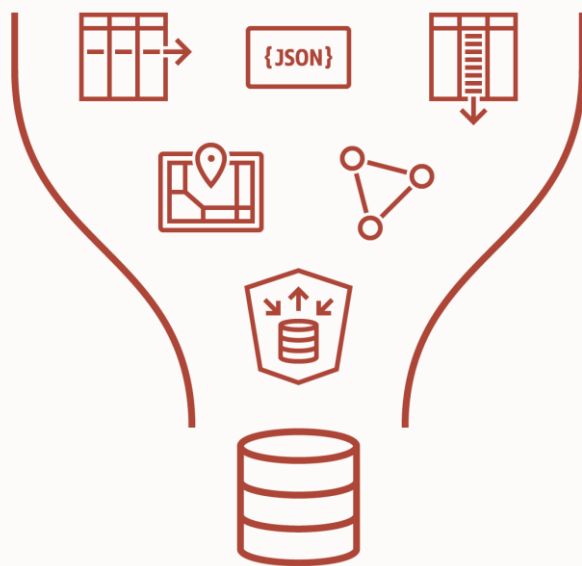
- Any data and workload type
- Application compatibility with on-premises and other cloud deployments
- More productive development
  - Standardized APIs for rapid development
  - Reusability of code, libraries and frameworks
- Maximum availability with RAC and Data Guard
- Defense in-depth security from database to cloud
- Simple database migration with zero downtime
- Consistent pricing model with predictable spend
- Lower TCO



# Oracle Database | World's Best Converged Database Engine

Enterprise-class performance, scalability, reliability & security for all data-driven workloads

## Any Data



Converged, open, SQL database for multiple data types and data models

## Any Workload



Transparent scale-up, scale-out, sharding, in-memory, parallel SQL

## Most Productive



Developers & Architects, Data Analysts/Scientists and DBAs



# License-included Oracle Database Consumption Options

Functionality tiers to meet application-specific requirements

## Standard Database Service

Standard Edition 2 Database features plus

- Tablespace Encryption
- Machine Learning
- Spatial and Graph
- Up to 3 PDBs

## Enterprise Database Service

Enterprise Edition Database features plus

- Data Guard
- Real Application Testing
- Data Masking and Subsetting Pack
- Tuning Pack and Diagnostic Packs
- True Cache

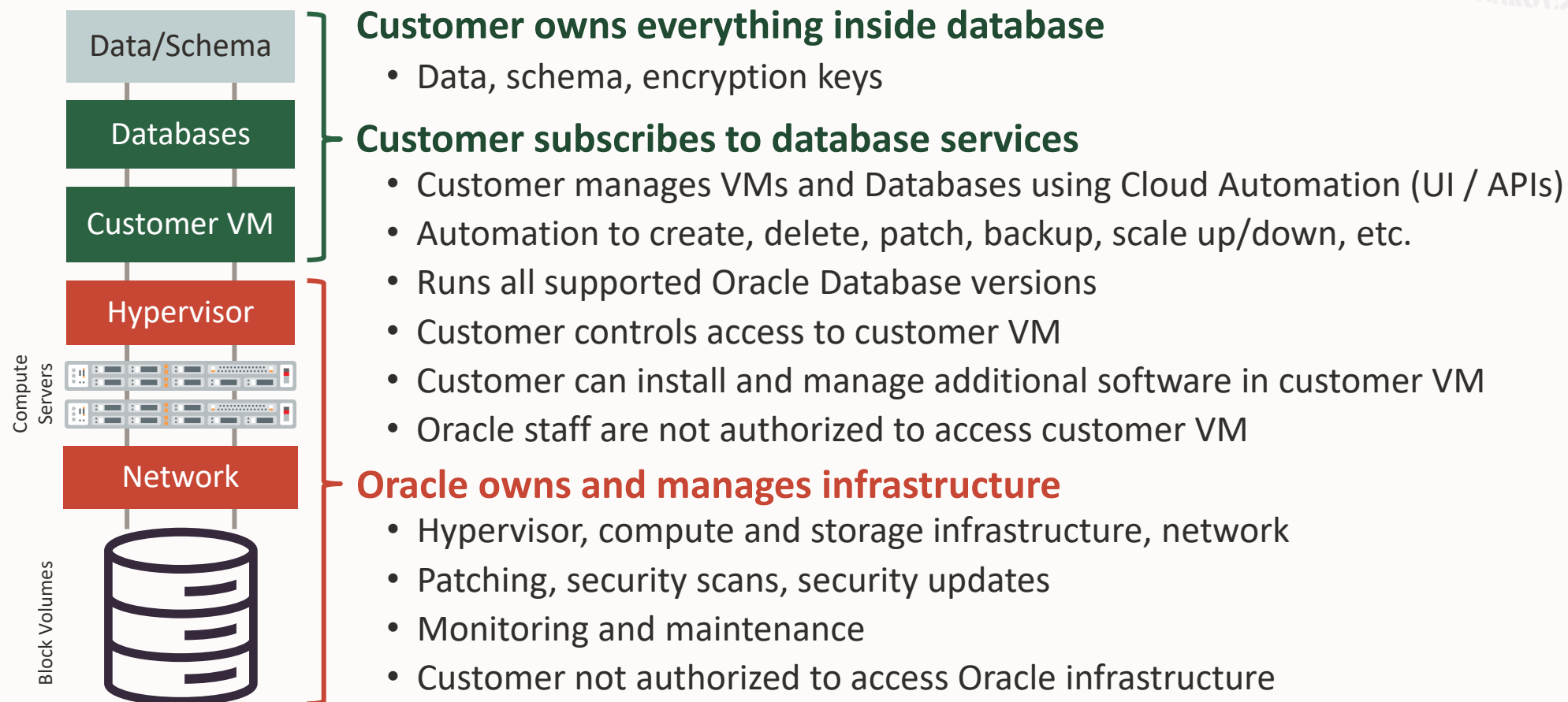
## Enterprise Database Service high performance

Enterprise Database Service features plus

- Partitioning
- Advanced Compression
- Advanced and Label Security, DB Vault
- Multitenant
- OLAP
- Lifecycle and Cloud Management Packs
- SQL Firewall



# Simple Cloud Management Model in Public Cloud





# Deploy, Scale and Manage Databases in Oracle Cloud



[oracle.com/database/base-database-service](https://oracle.com/database/base-database-service)

## 1

### Oracle Database as a Service

- Same database software in Oracle Cloud as on-premises
- Full-stack redundancy and integrated disaster recovery
- Full database control with defense in-depth security
- Simple database migration with zero downtime

## 2

### Simple and Rapid Development

- Single converged database for any data and workload type
- Rapid application development with Oracle APEX and other included tools

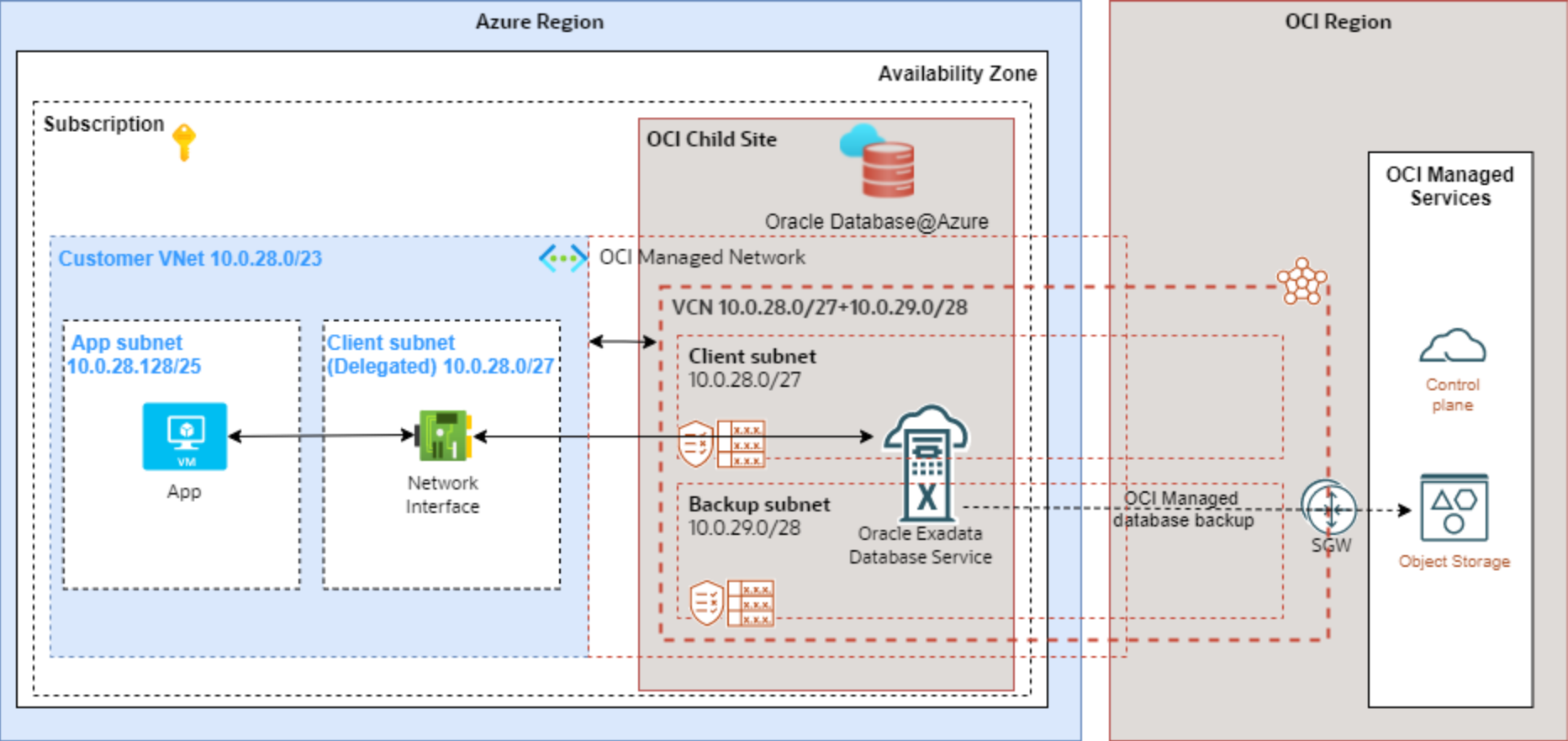
## 3

### Low Cost

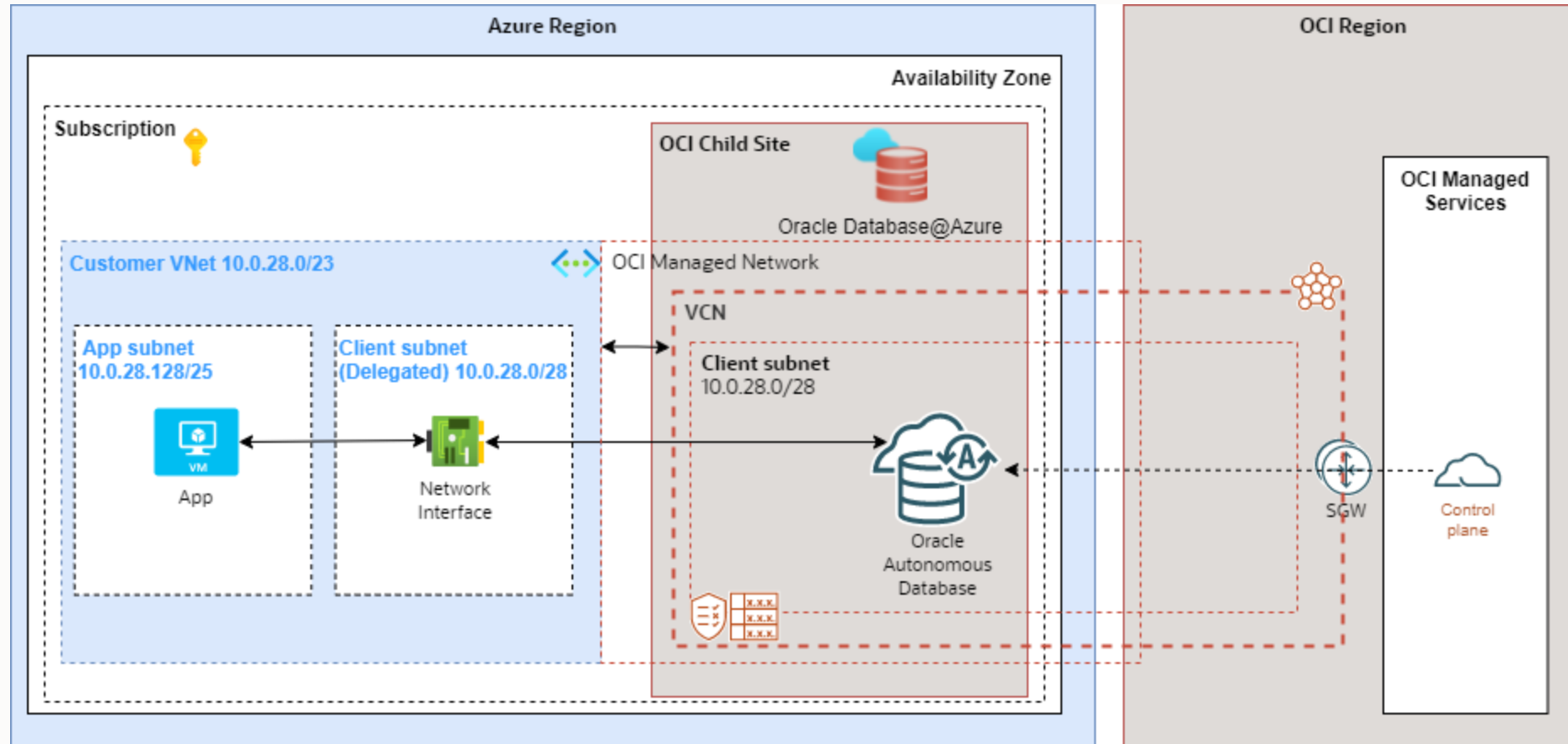
- Flexible shapes with low entry price and pay only for what you use
- License-included or Bring Your Own License (BYOL)
- Cloud automation for database lifecycle management

# Architecture

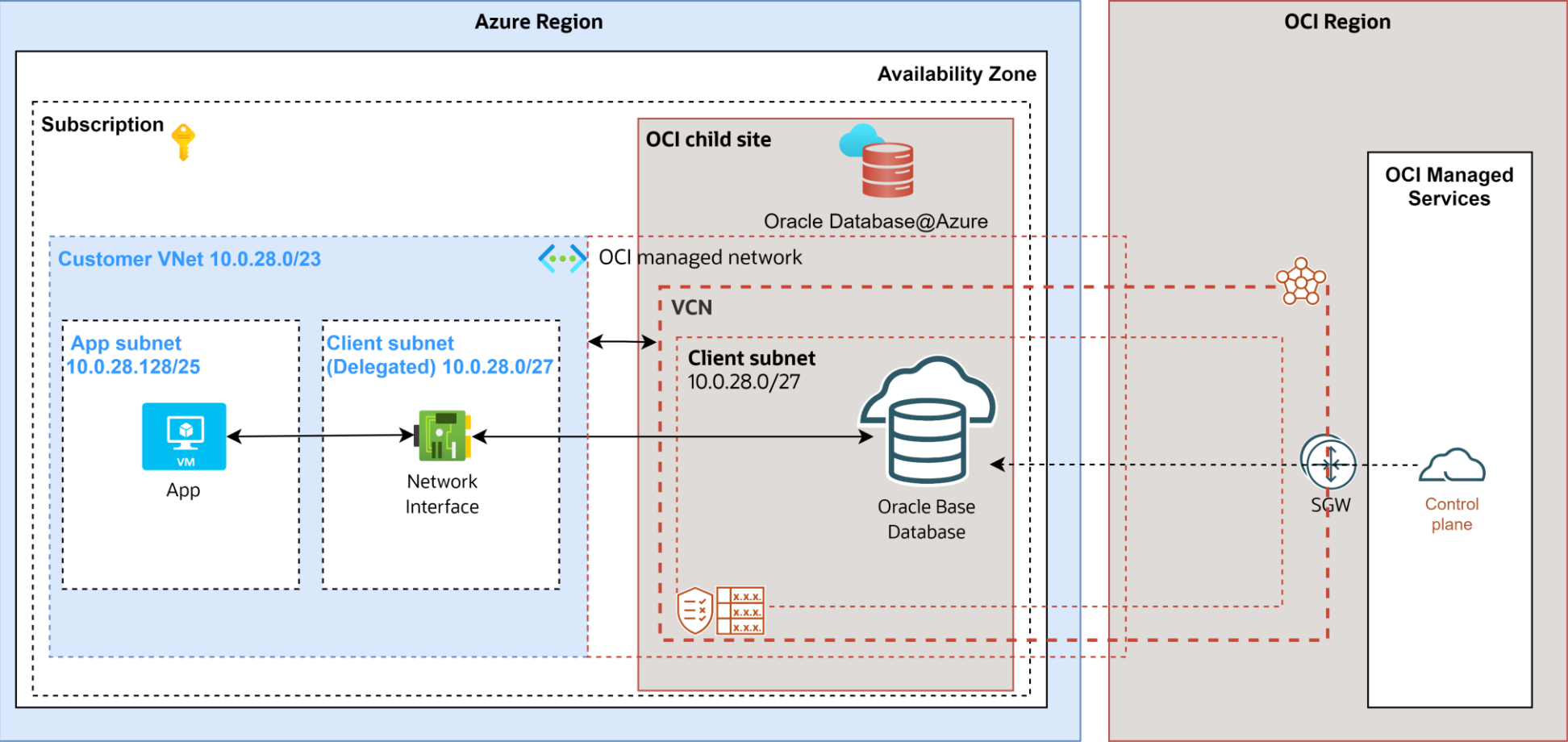
# Exadata Database Service – Single AZ Architecture



# Autonomous Database – Serverless – Single AZ Architecture

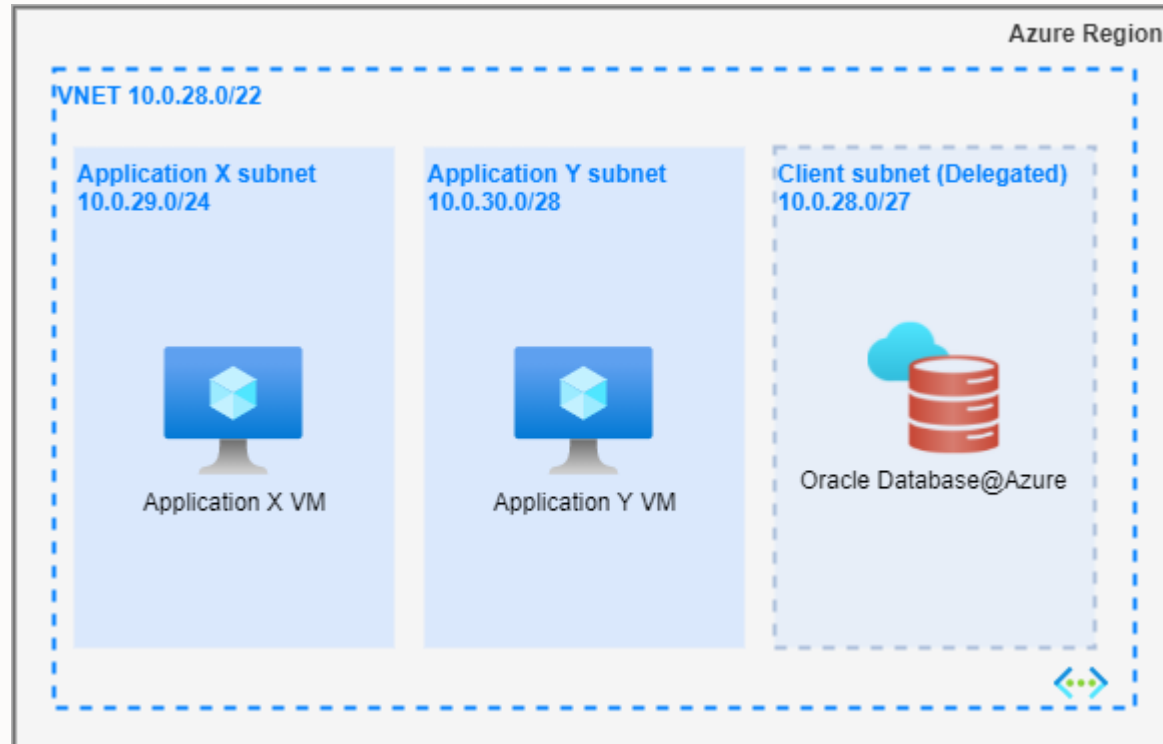


# Base Database – Single AZ Architecture



# Network Topology

# Local VNet Topology

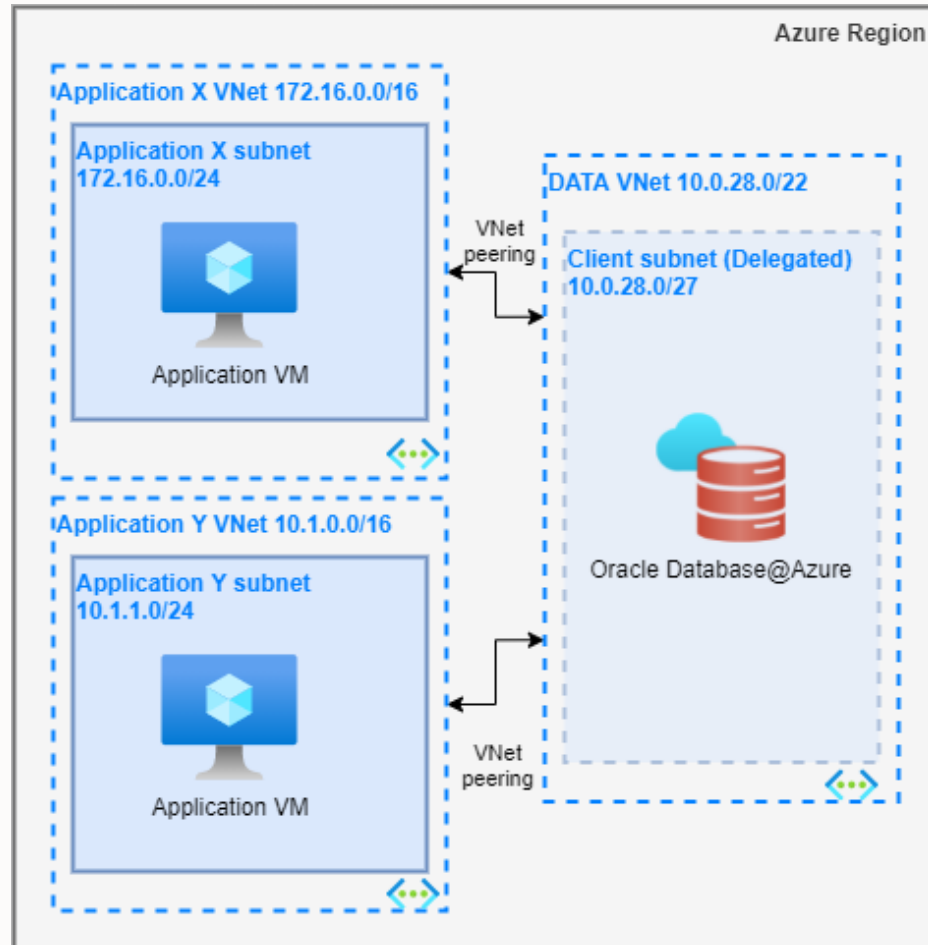


- Simple networking topology connects apps to the database in the same VNet
- App connects to database via client delegated subnet
- Lowest network latency
- No ingress/egress costs





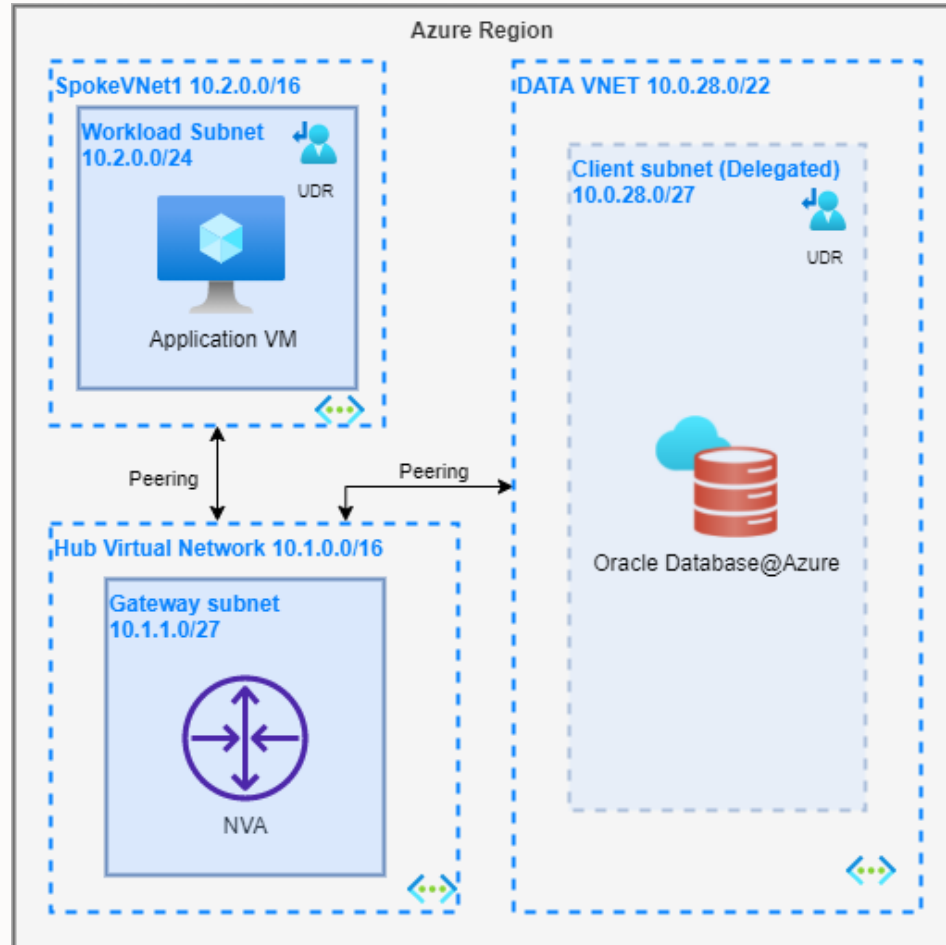
# VNet Peering Topology



- Use local VNet Peering to connect apps in other VNet of the same region to database VNet.
- VNet Peering incurs ingress and egress costs. For more information, see Virtual Network pricing.
- Lowest network latency
- Up to 500 direct peerings possible



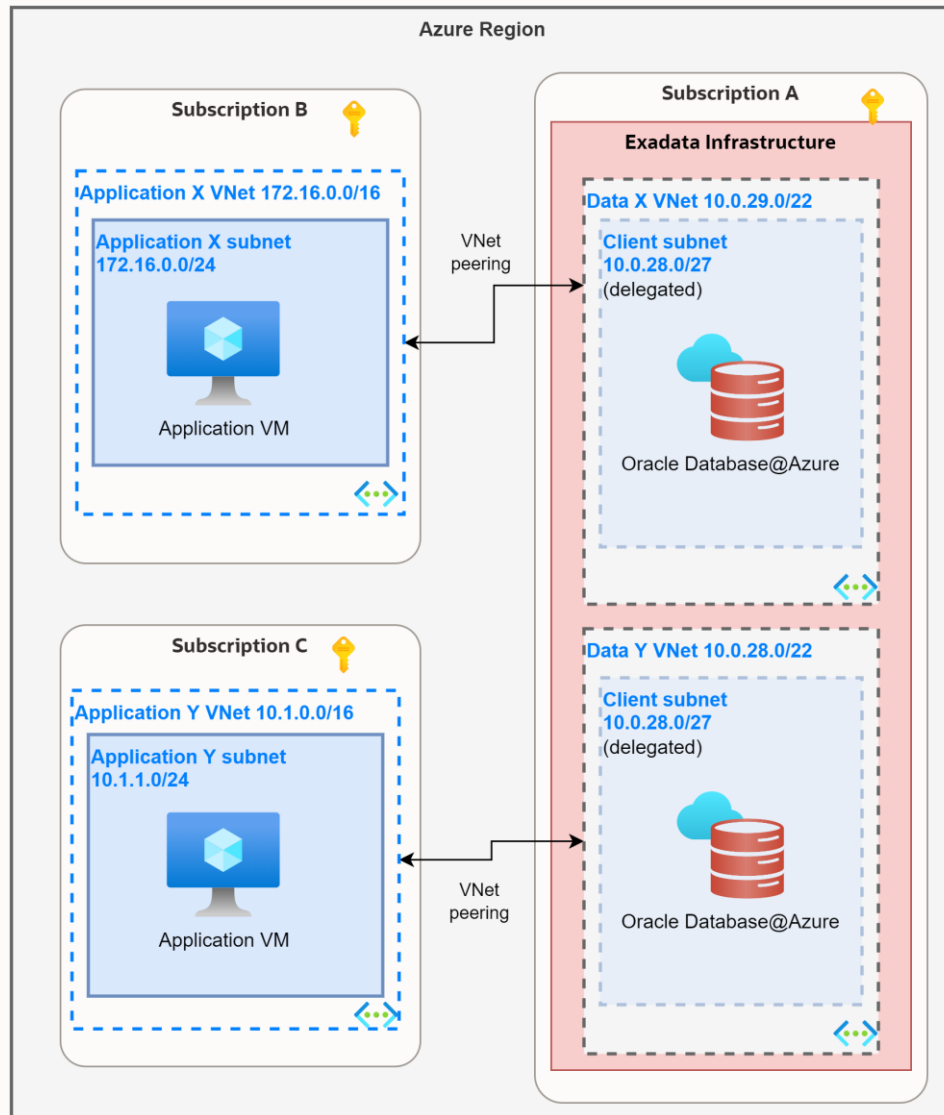
# Hub-spoke VNet Peering Topology



- The hub VNet is a central point of connectivity between apps and database;
- The spoke VNets peer with the hub.
- A local routing device (NVA) is prerequisite for routing traffic from the delegated subnet to hub-and-spoke
- Higher latency

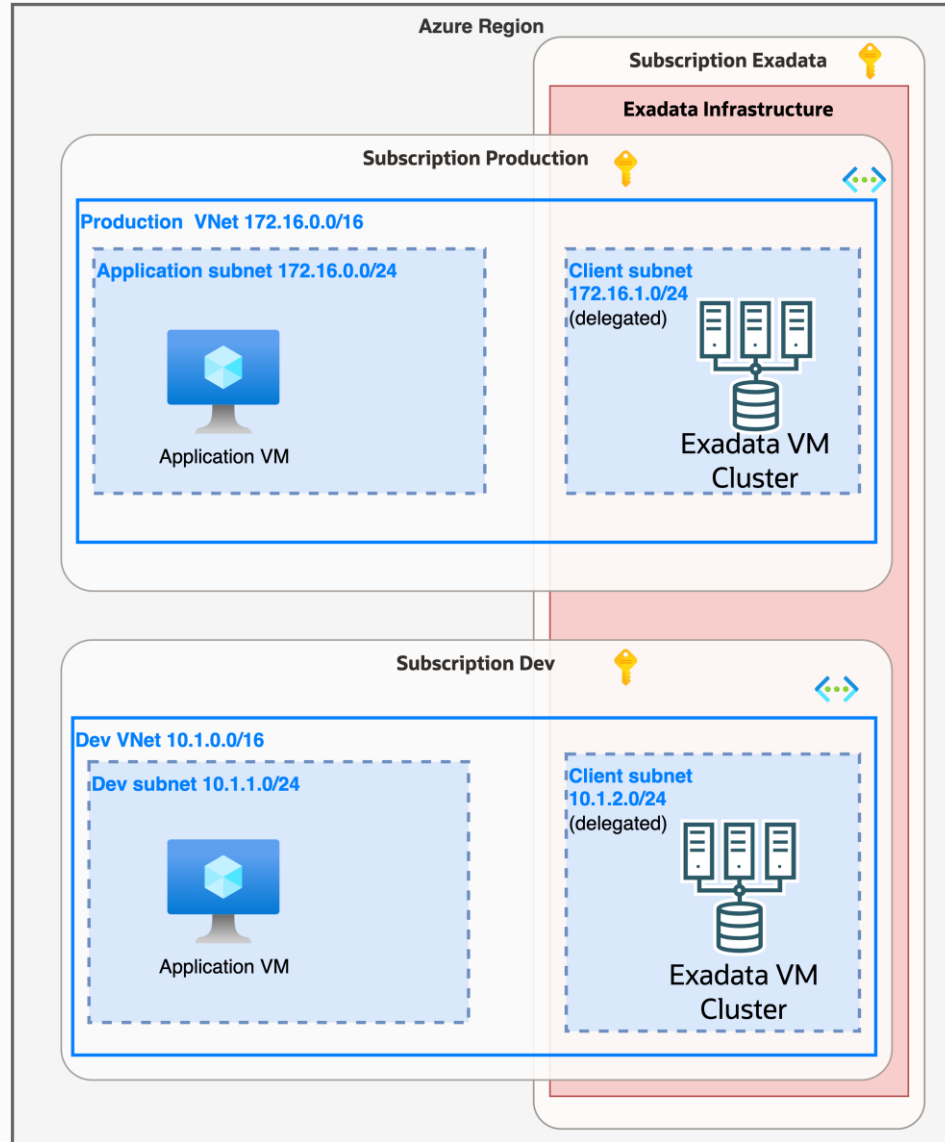


# Centralized DB subscription - VNet Peering Topology



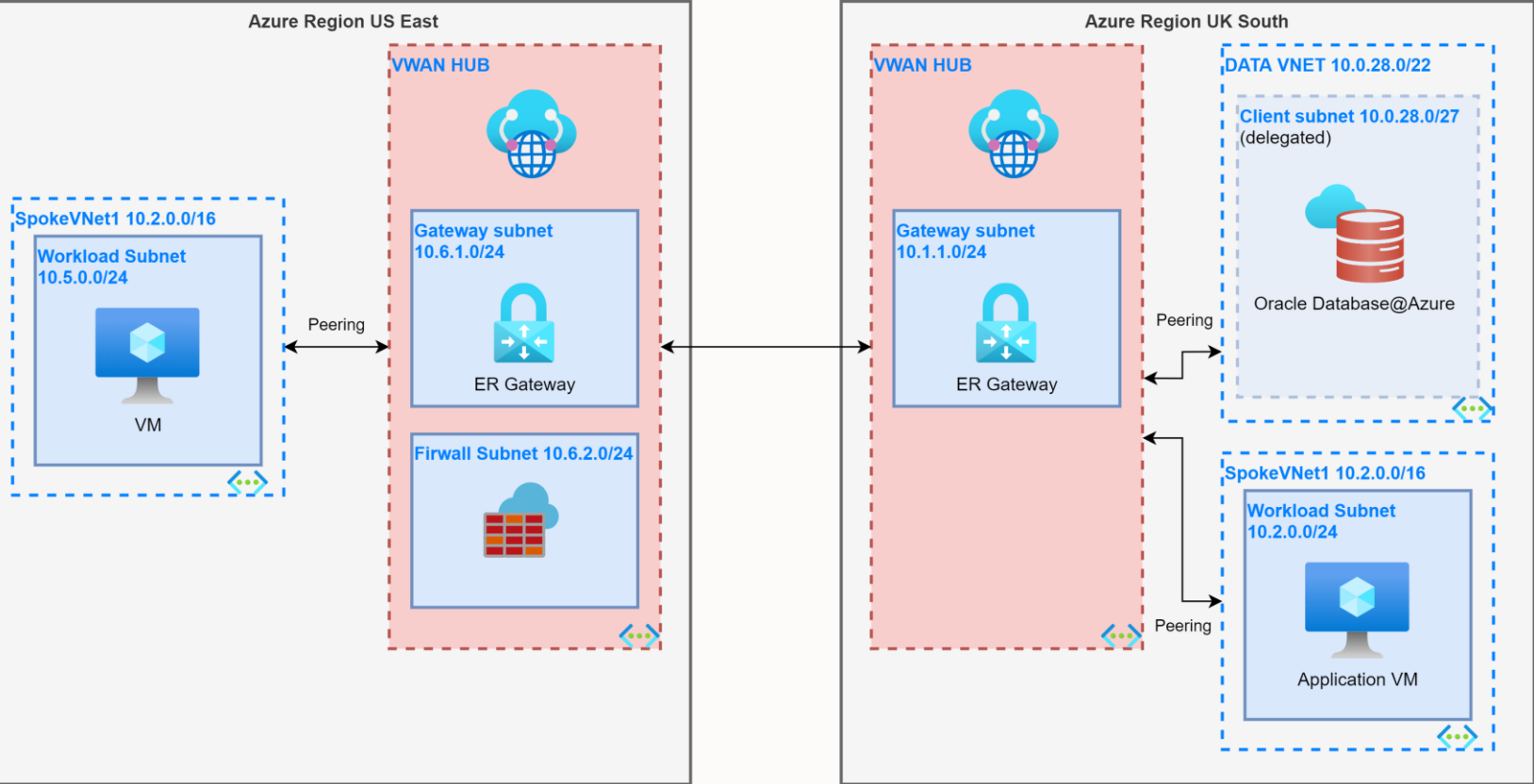
- Databases are consolidated within the same subscription for management.
- Use VNet Peering to connect apps in other VNet of the same region but different subscription to database Vnet
- VNet Peering incurs ingress and egress costs. For more information, see Virtual Network pricing.
- Lowest network latency

# Share Exadata Infrastructure – Multi Subscription topology



- Exadata Infrastructure is consolidated within an Azure subscription for management.
- Each VM Cluster is deployed in a different Azure subscription and Vnet
- VM Cluster are isolated
- Lowest network latency

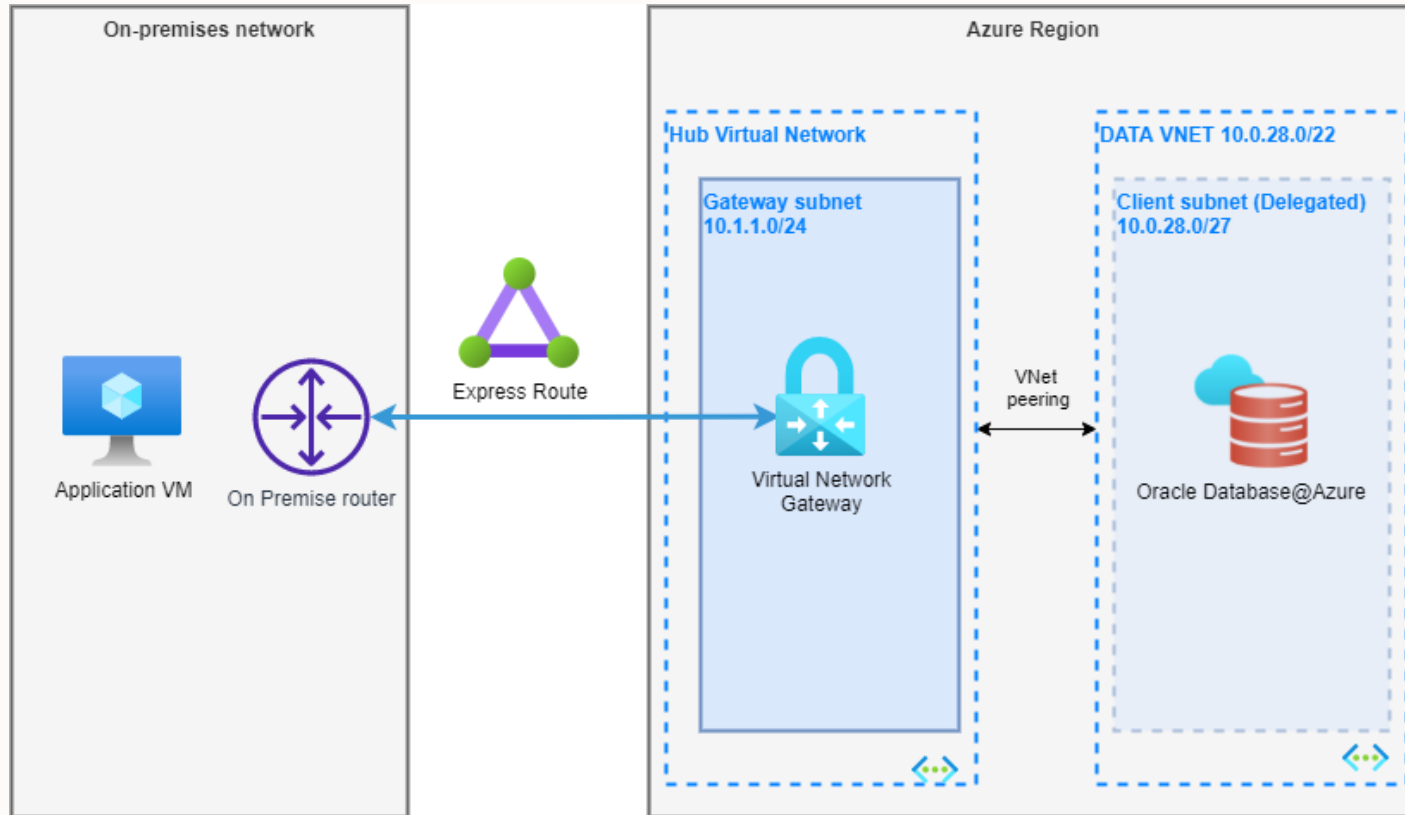
# Global connectivity between regions



- VWAN Hub is configured in both regions to create global connectivity
- Firewall is configured in the Hub to inspect traffic
- Routing is managed by VWAN Hub



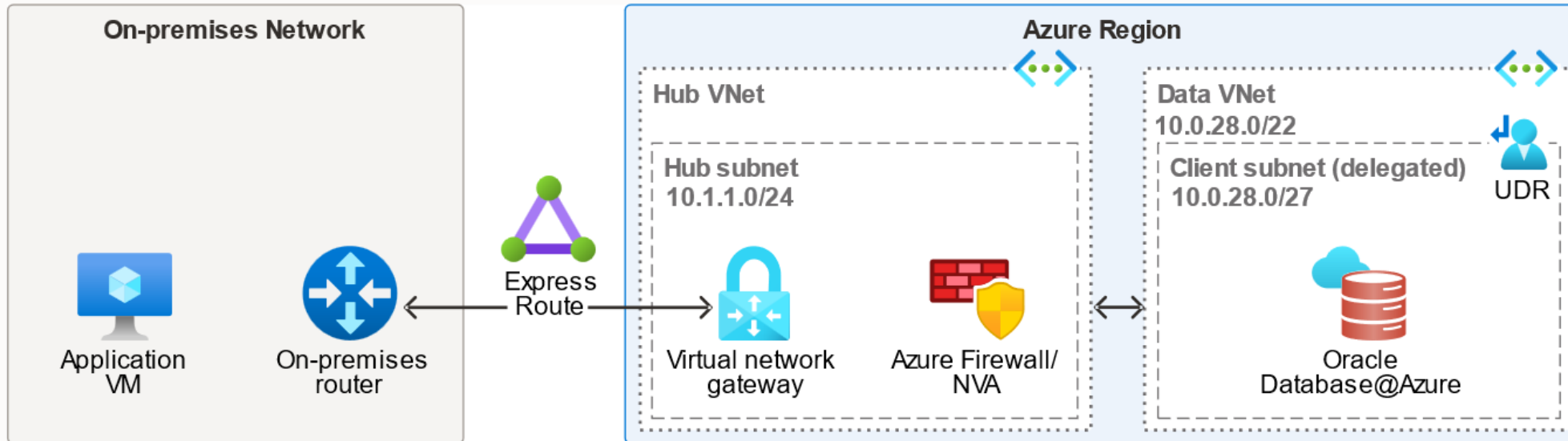
# Hub-spoke On-premises without traffic inspection



- No firewall configured in the Hub
- Routes are learned through transitivity in the Gateway
- Database spoke learns routes through peering.
- VNet Peering incurs ingress and egress costs. For more information, see [Virtual Network pricing](#).



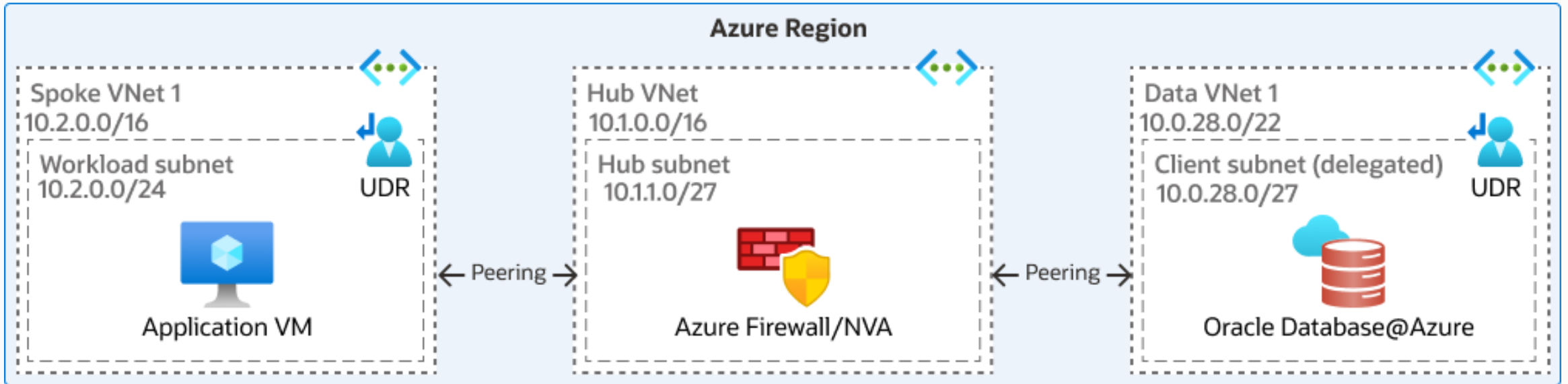
# Hub-spoke On-premises with traffic inspection



- Firewall configured in the Hub
- Routes are learned through transitivity in the Gateway
- Database spoke learns routes through peering.
- Routes are defined on each subnet to route to the Firewall
- VNet Peering incurs ingress and egress costs. For more information, see [Virtual Network pricing](#).

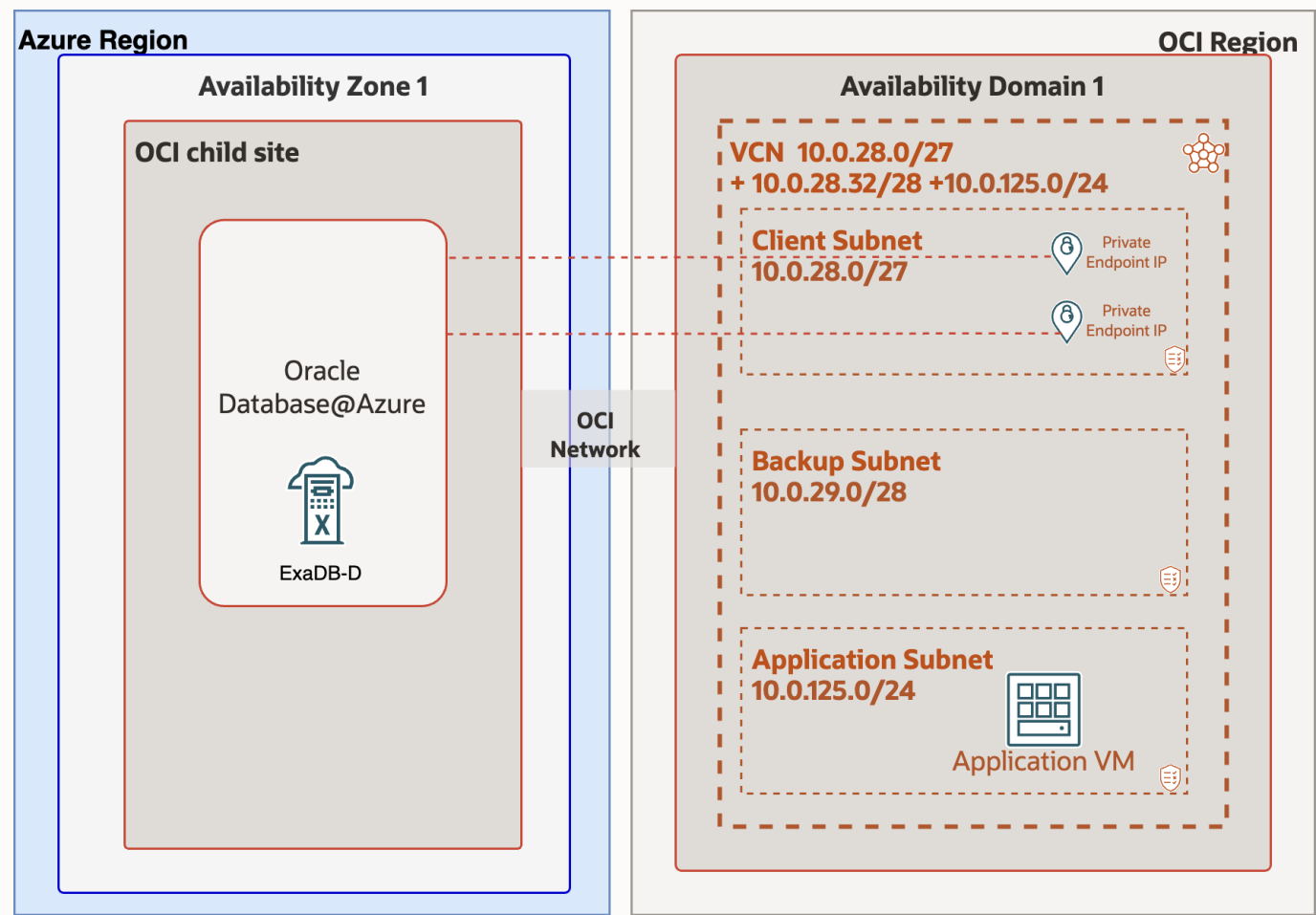


# Hub-spoke with traffic inspection



- Firewall configured in the Hub
- Database spoke learns routes through peering.
- Routes are defined on each subnet to route to the Firewall
- VNet Peering incurs ingress and egress costs. For more information, see [Virtual Network pricing](#).

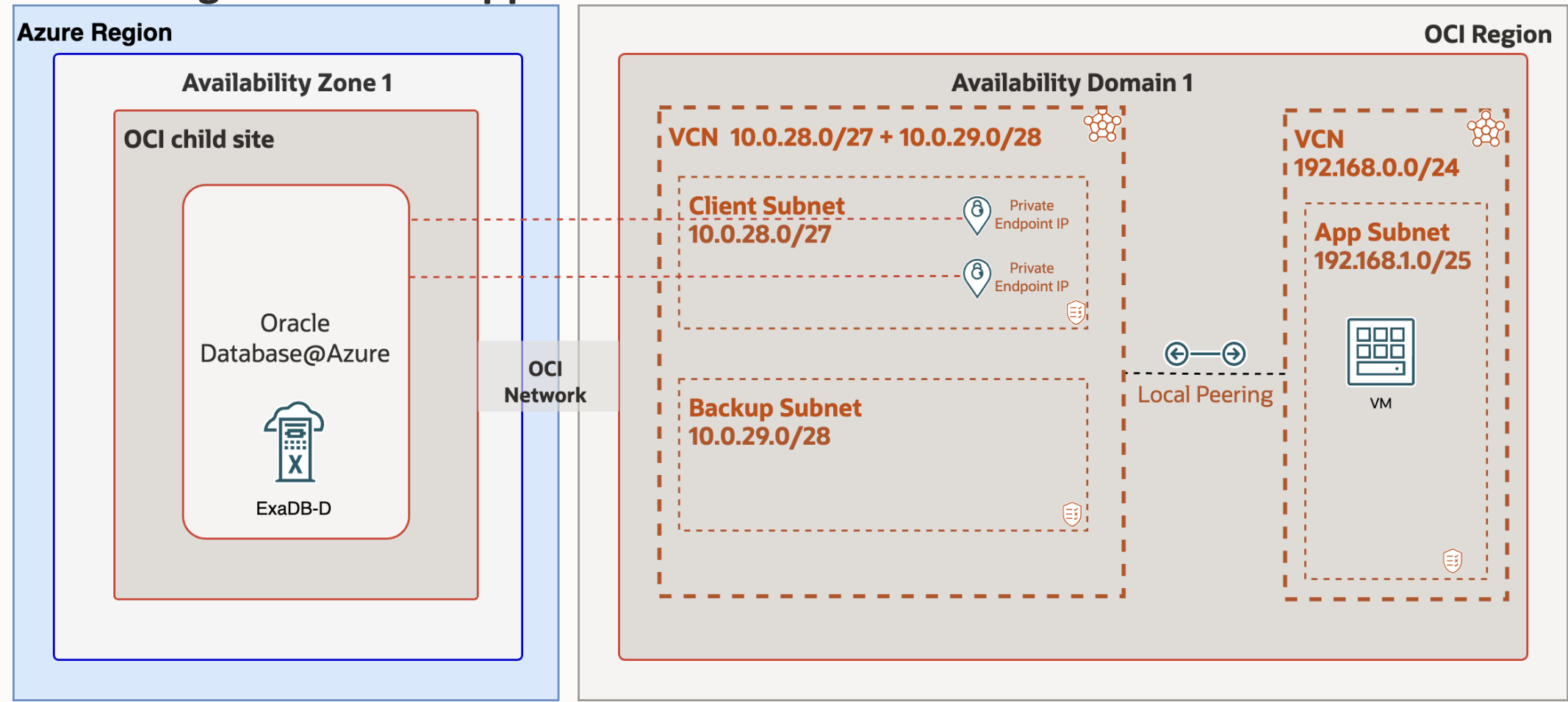
# Connecting Services or application from OCI in Same VCN



OCI customers can natively connect existing resources and services to Oracle Database and Exadata resources by the private endpoint by creating an additional subnet in the shadow VCN and deploy the application in the new Subnet.



# Connecting Services or Application from OCI in Different VCNs

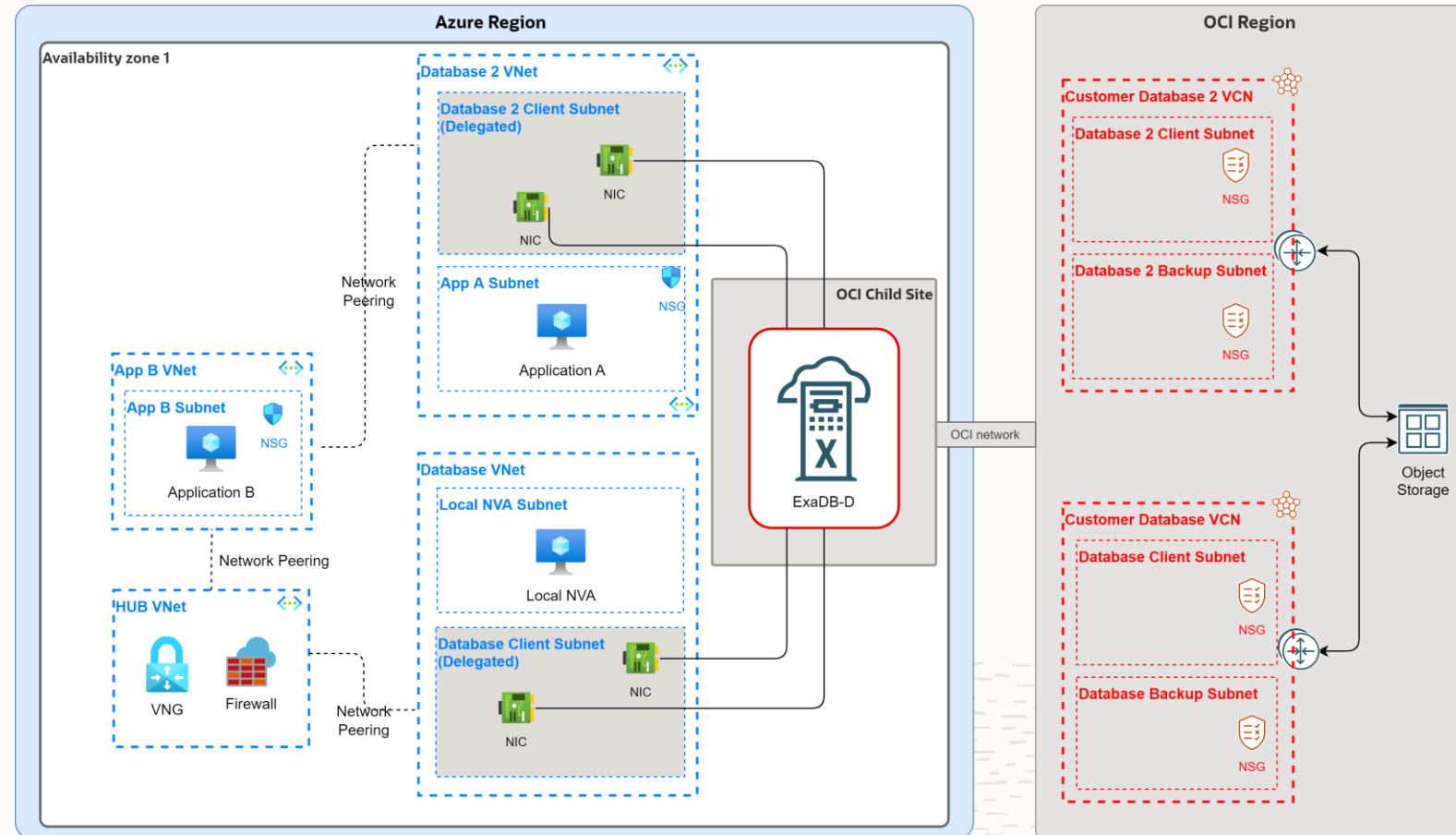


OCI customers can natively connect existing resources and services to Oracle Database and Exadata resources by the private endpoint in the VCN with local peering.



# Azure Networking Examples

1. Same Exadata Cloud Infrastructure
  - 2x VM clusters
  - Can use different VNets/Subnets (this example)
2. Same Exadata Cloud Infrastructure
  - 2x VM clusters
  - Only 1 delegated subnet possible per VNet
    - Can use same VNet and same client subnet when deployed in same AZ
3. Applications can reach DB from DB VNet or other spoke VNet's through direct VNet peering or hub-and-spoke (normal Azure VNet peering charges applies)



Example : 2x VM clusters on same Exadata Cloud Infrastructure



# Network Requirements

# Network

## Oracle Database@Azure requirement

- An Azure VNet available in the region and subscription to deploy the Exadata Cloud VM Cluster
  - The VNet address space should be in RFC 1918.
  - The VNet must at least must contain two subnets, one for database client and another for database backup.
    - Client Subnet cannot overlap 192.168.16.16/28
    - Backup Subnet cannot overlap 192.168.128.0/20
    - For Data Guard client and backup subnets CIDR cannot overlap between Primary and Standby networks
    - DRG is managed by Oracle
  - Need to enable **subnet delegation** for the client subnet to Oracle Database@Azure.
  - DNS resolution for Exadata VM Cluster machines and SCAN IPs are available in Azure DNS.
  - Minimum required subnet sizes, /27

| Rack Size    | Client Subnet: # Required IP Addresses                     | Client Subnet: Minimum Size <a href="#">Note</a> | Backup Subnet: # Required IP Addresses | Backup Subnet: Minimum Size <a href="#">Note</a> |
|--------------|------------------------------------------------------------|--------------------------------------------------|----------------------------------------|--------------------------------------------------|
| Quarter Rack | (4 addresses * 2 nodes) + 3 for <a href="#">SCANs</a> = 11 | /27 (32 IP addresses)                            | 3 address * 2 nodes = 6                | /28 (16 IP addresses)                            |



# Azure Networking – ExaDB-D

Remember to plan for expansion

## ExaDB-D – VM Cluster Networking in Azure

- Needs a VNet with 1x subnet (**Client subnet**) in Azure
  - Client subnet needs to be created and **delegated** to Oracle Database@Azure before ExaDB-D VM cluster deployment
- Need to provide **Backup subnet** CIDR block during VM cluster deployment
  - Backup subnet will be deployed in OCI, not in Azure
- Plan not to overlap network
  - Non overlapping CIDR blocks are needed for each deployment in a DR scenario

The following table shows the minimum required subnet size depending on ExaDB-D configuration you choose. Each database node require 4x IP addresses and in addition 3x addresses are reserved for Single Client Access Names (SCANs.)

| Rack Size                                   | Client Subnet #<br>Required IP Addresses                                                      | Client Subnet :<br>Minimum size *                                                      | Backup Subnet # (Not in Azure)<br>Required IP Addresses | Backup Subnet : (Not in Azure)<br>Minimum size *                                        |
|---------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Flexible infrastructure<br>(X9M and higher) | 4 addresses per database<br>node (minimum 2 nodes)<br>+ 3 for SCANs +<br>13 for ODBAA service | Minimum size determined by<br>total number of IP addresses<br>needed for database node | 3 addresses per database node<br>(minimum 2 nodes)      | Minimum size determined by total<br>number of IP addresses needed for<br>database nodes |

\*The Networking service in Azure reserves five (5) IP addresses in each subnet



# Azure Networking for ExaDB-D – example

ExaDB-D Network examples : 2x or 4x Database nodes

| Size of subnet | Total number of IPs in subnet |
|----------------|-------------------------------|
| /28            | 16                            |
| /27            | 32                            |
| /26            | 64                            |
| /25            | 128                           |
| /24            | 256                           |
| /23            | 512                           |

|                     | Client Subnet #<br>IP Addresses in this example                | Client Subnet :<br>Minimum size in this example                      | Backup Subnet #<br>(not in Azure) |
|---------------------|----------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------|
| Example<br>2x nodes | 4x2 nodes + 3 = 11 IPs<br>ODBAA requires 13 IPs*<br>Total = 24 | /27 minimum                                                          | /28 minimum                       |
| Example<br>4x nodes | 4x4 nodes + 3 = 19 IPs<br>ODBAA requires 13 IPs*<br>Total = 32 | /26 minimum<br>(recommend to to use larger for<br>future expansions) | /27 minimum                       |

\* The Networking service in Azure reserves five (5) IP addresses in each subnet.  
ODBAA service requires eight (12) IP.





# Azure Networking ADB-S

Remember to plan for expansion

- Needs a VNet with 1x subnet (**DB subnet**) in Azure
- DB subnet needs to be created and **delegated** to Oracle Database@Azure before ADB-S deployment
- Can be reused (shared) for additional databases

| DB Subnet #<br>Required IP Addresses                                                                                            | DB Subnet Minimum size                                                 |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1 address Deployed ADB-S instance.<br>+<br>5 addresses reserved for for Azure networking<br>+<br>13 addresses for ODBAA service | Minimum CIDR requirement for 1<br>deployed ADB-S is /27 (16 Addresses) |

\*The Networking service in Azure reserves five (5) IP addresses in each subnet.



# Network delegation

Need to enable **subnet delegation** for client subnet to Oracle Database@Azure.

**Oracle.Database/networkAttachments**

subnet

demo\_vnet\_2

Name

subnet

Subnet address range \* ⓘ

10.12.0.0/24

10.12.0.0 - 10.12.0.255 (251 + 5 Azure reserved addresses)

☐ Add IPv6 address space ⓘ

NAT gateway ⓘ

None

Network security group

None

Route table

None

SERVICE ENDPOINTS

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Services ⓘ

0 selected

SUBNET DELEGATION

Delegate subnet to a service ⓘ

Oracle.Database/networkAttachments

NETWORK POLICY FOR PRIVATE ENDPOINTS

The network policy affects all private endpoints in this subnet. Select the types of network policies that control traffic going to the private endpoints in this subnet. [Learn more](#)

Private endpoint network policy

Save

Cancel

Give feedback



# Oracle Database@Azure - Network considerations

1 delegated subnet by Vnet

Multiples Cluster in the same delegated subnet - shared

Multiples Cluster from **separated** Exadata infra in the **same AZ** in the same delegated subnet

Client subnet - Minimum range /27

- Check documentation to plan accordingly
  - [https://docs.oracle.com/en-us/iaas/Content/multicloud/oaa\\_ip.htm](https://docs.oracle.com/en-us/iaas/Content/multicloud/oaa_ip.htm)

Backup subnet – Minimum range /28

- By default 192.168.252.0/22

Microsoft network planning

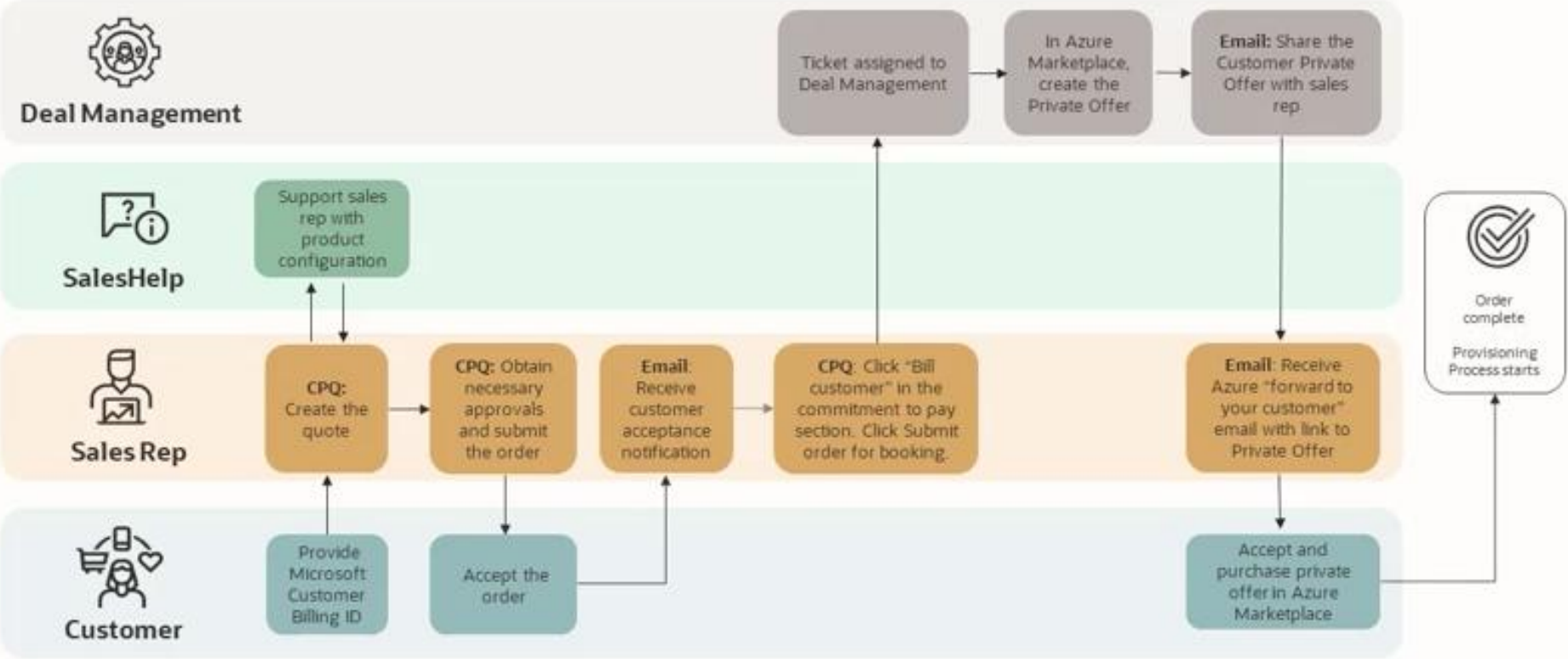
- <https://learn.microsoft.com/en-us/azure/oracle/oracle-db/oracle-database-network-plan>

# Purchase / Enablement

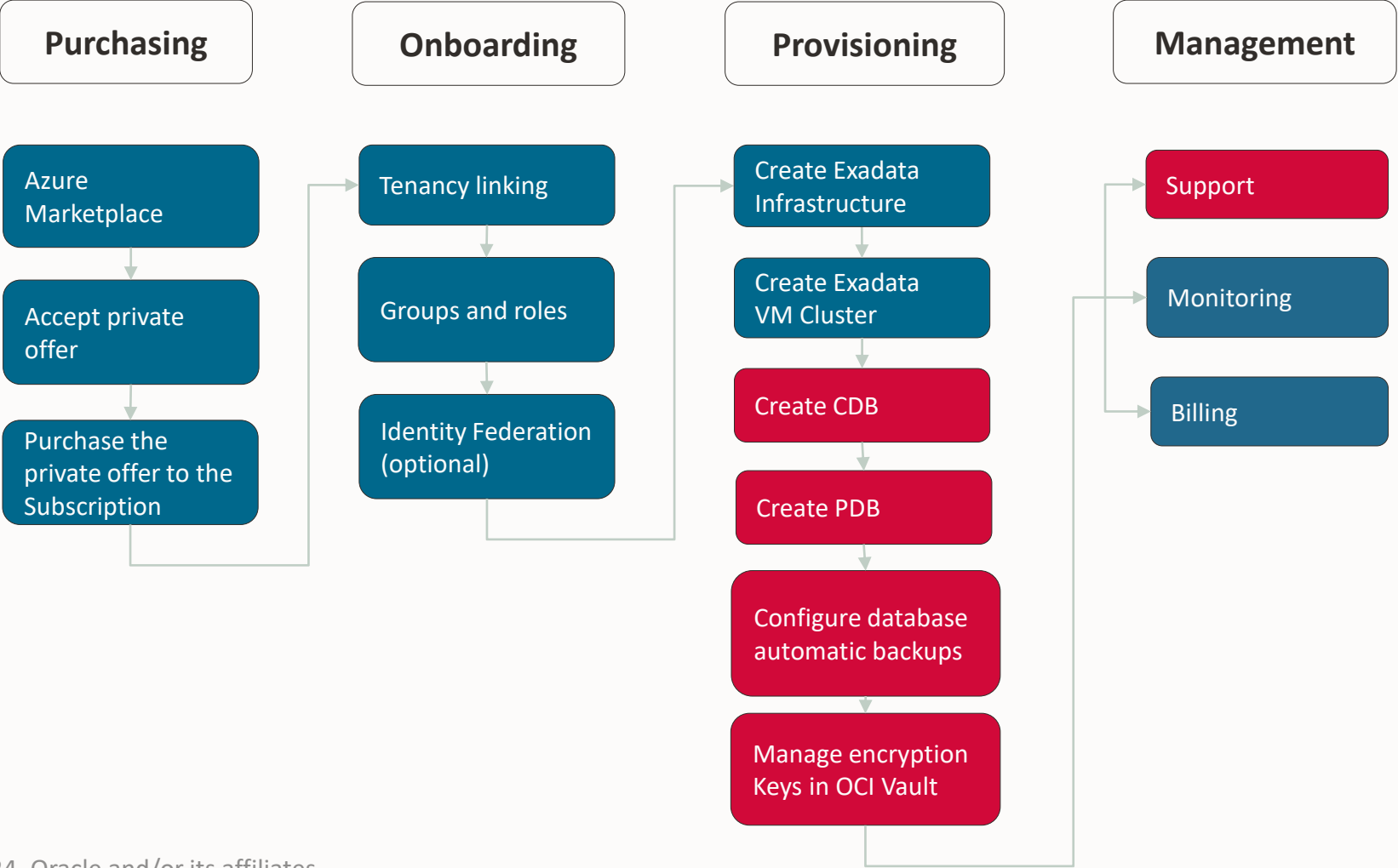
# Sales process

# Oracle Sales Process for Oracle Database@Azure

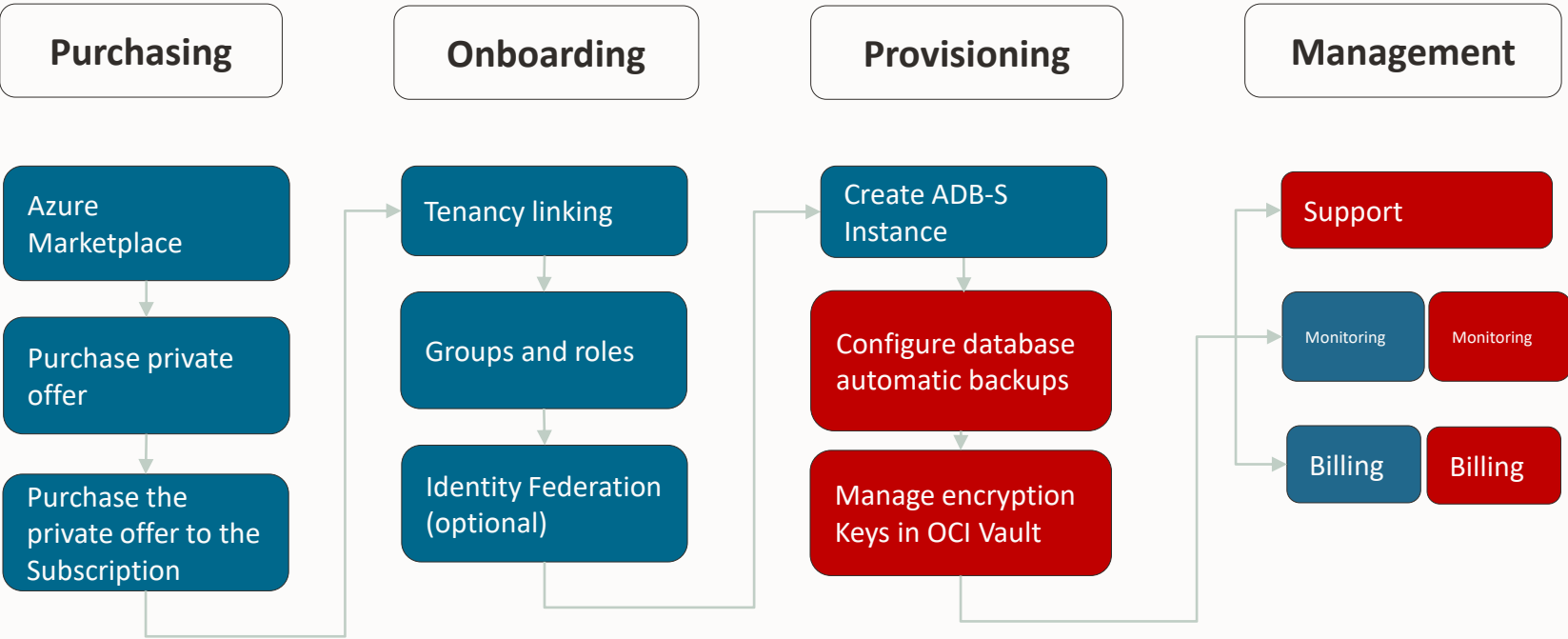
## Quoting Oracle Database@Azure



# User Journey – Private Offer (ExaDB-D)



# User Journey – Private & Pay-As-You-Go Offer (ADB-S)





# Public Offer



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Storage Account

randyw-vision

API Connection

preview-susanappservicebackup

Resource type

randyw-vision-prediction

Cognitive Service

Marketplace

Product name

Product name

Product name

Documentation

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Azure documentation - long title goes here

Azure documentation - long title goes here

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Operating System : All

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Product Type : All

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Oracle America, Inc.



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Subscription ([move](#)) : [orpsandbox8](#) State : Succeeded

Subscription ID : 1e188c58-04a0-47a8-b917-943e13a61e7c

## Purchase details

|                     |                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| Offer name          | Oracle Database@Azure                                                                          |
| Purchase status     |  Subscribed |
| Plan                | oracle_database_at_azure_pay_as_you_go                                                         |
| Purchase start date | 2024-09-09T15:46:35.670Z                                                                       |
| Purchased by        |                                                                                                |
| Billing term        | P1M                                                                                            |

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| <div>Product name</div> Oracle Database@Azure | <div>Status</div> Ready | <div>Plan name</div> PlanName | <div>Publisher</div> Oracle | <div>Product type</div> SaaS | <div>Azure benefit eligible</div> Yes |            |                 |



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|                   |                     |
|-------------------|---------------------|
| Subscription      | MyAzureSubscription |
| Resource group    | myBuyer_group       |
| Plan              | PlanName            |
| Billing term      | 3-year              |
| Recurring billing | Yes                 |
| Subtotal          | \$300,000.00        |

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|      |      |
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 Overview

 Activity log

 Access control (IAM)

 Tags

 **Your deployment is complete**

 Deployment name: Microsoft.Oracle\_1687973914143

Subscription: [MyAzureSubscription](#)

Start time: 6/28/2023, 10:38:39 AM

Correlation ID: 7d5490af-2...6d-be76-514772ce8

> Deployment details

∨ Next steps

[Go to resource](#)

Give feedback

 Tell us about your experience with deployment



# Oracle Database@Azure Deployment

# Pre-requirements for Deployment of Oracle Database@Azure

- [Azure permissions to accept the private offer](#)
- An existing Azure subscription
- An Azure VNet with a subnet delegated to the Oracle Database@Azure service (Oracle.Database/networkAttachments)
- Azure groups and roles
  - odbaa-exa-infra-administrators to provision Exadata Infrastructure
  - odbaa-vm-cluster-administrators to provision VM cluster
- Oracle Cloud account
  - OCI ExaDB-D Service limits
  - OCI IAM Policies for Database Service Resource
  - OCI IAM Policies for Object Storage
  - OCI IAM Policies for OCI Vault
- Identity Federation

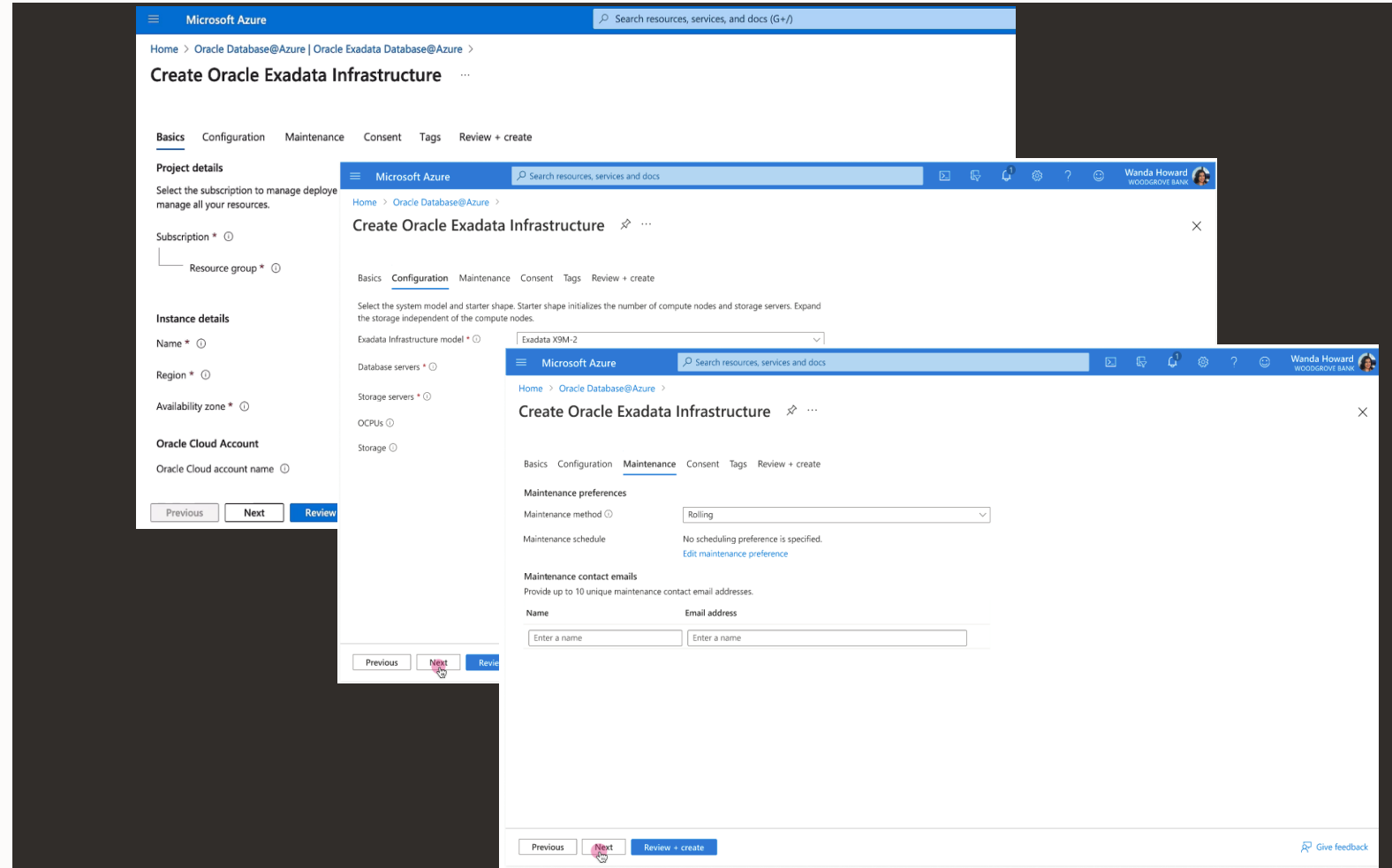


# Exadata Database Service Provisioning



# Steps to provision Exadata Infrastructure

1. Enter basic information like name, AZ, Subscription and resource group you want to deploy in
2. Choose the Exadata Infrastructure with the number of Database Servers and Storage
3. Select the managed patching for the infrastructure as well as the schedule
4. Once these details are entered, you can consent and provision



The screenshot displays the 'Create Oracle Exadata Infrastructure' wizard in the Microsoft Azure portal. The wizard is divided into several tabs: Basics, Configuration, Maintenance, Consent, Tags, and Review + create. The 'Basics' tab is active, showing fields for Subscription, Resource group, Name, Region, Availability zone, and Oracle Cloud Account. The 'Configuration' tab is also visible, showing fields for Exadata Infrastructure model, Database servers, Storage servers, OCPUs, and Storage. The 'Maintenance' tab is active, showing fields for Maintenance method (Rolling), Maintenance schedule (No scheduling preference is specified), and Maintenance contact emails (Name and Email address). The 'Review + create' tab is also visible, showing a summary of the configuration.



# Steps to provision Exadata VM Cluster

1. Enter basic information like name, AZ, subscription and resource group you want to deploy in.
2. Select licensing type (BYOL or License Included)
3. Allocate VM cluster resource (memory, nodes, storage, etc.).
4. Select the VNet it will be deployed and the two **Delegated Subnets**

The screenshot displays the 'Create Oracle Exadata VM Cluster' wizard in the Microsoft Azure portal. The wizard is divided into several tabs: Basics, Configuration, Networking, Diagnostics collection, Consent, Tags, and Review + create. The 'Basics' tab is active, showing fields for Name, Region, Exadata infrastructure, License type, Grid infrastructure Version, Security (SSH public key source), and Key pair name. The 'Configuration' tab is also visible, showing fields for VM cluster allocation (OCPU count per VM, Memory per VM, Total requested OCPU count, Total requested memory, Local storage per VM, Total local storage) and Exadata storage (Usable Exadata storage (TB), Total usable Exadata storage (TB)). The 'Networking' tab is active, showing fields for Virtual network, Client subnet, Backup subnet, DNS (Use private DNS service), Host name prefix, Host domain name, and Host and domain URL. The 'Review + create' tab is also visible, showing a summary of the configuration and a 'Review + create' button.



# Create Database (CDB and PDB) in OCI

- Must be part of the Oracle Database@Azure CDB or PDB admin groups, depending on the operation
- In the Azure VM cluster resource page, click the OCI Database URL **Go To OCI** using the identity federation
- On the OCI console, create CDB and/or PDB

ORACLE Cloud Search resources, services, documentation, and Marketplace Netherlands Northwest (Amsterdam)

Created: Wed, Dec 14, 2022, 15:48:08 UTC  
Cluster name: cl-u6q5dfja  
Time zone: UTC  
License type: Bring Your Own License (BYOL)  
Lifecycle state: Available  
IORM status: Disabled  
Diagnostics collection: Disabled [Edit](#)

Resource allocation  
VM Count: 2  
Enabled OCPUs: 4  
Memory: 2780 GB  
Local Storage: 1800 GB  
Exadata storage: 192 TB  
Storage for local backups: Not enabled  
Storage for Exadata sparse snapshots: Not enabled

Infrastructure  
Exadata cloud infrastructure:

Resources  
Databases (3)  
Database homes (3)  
Virtual IP address (0)  
Virtual Machines (2)  
Metrics  
Work requests (0)

Databases  
Create database

| Name | State     | Database |
|------|-----------|----------|
|      | Available |          |
|      | Available |          |
|      | Available |          |

### Create database

Basic information for the database

Provide the database name  
DB1109

Provide a unique name for the database Optional ⓘ

Select a Database version  
19c

Provide a PDB name Optional ⓘ

Specify a database home

Database Home source  
☒ Select an existing Database Home ☐ Create a new Database Home

Database Home display name  
Select a Database Home for the new database

Create administrator credentials

Username Read-only  
sys

Password ⓘ

Confirm password

Create database Cancel



# DNS

Microsoft Azure

Search resources, services, and docs (G+/)

[Home](#) > [Oracle Database@Azure](#) | [Oracle Exadata Database@Azure](#) >

Create Oracle Exadata VM Cluster

Basics

Configuration

Networking

Diagnostics Collection

Consent

Tags

Review + create

Virtual network \*

Client subnet \*

Backup subnet CIDR

DNS

Custom DNS

Host name prefix \*

vnet-emea-specialists-1

subnet-emea-specialists-delegated

☐

Previous

Next

Review + create

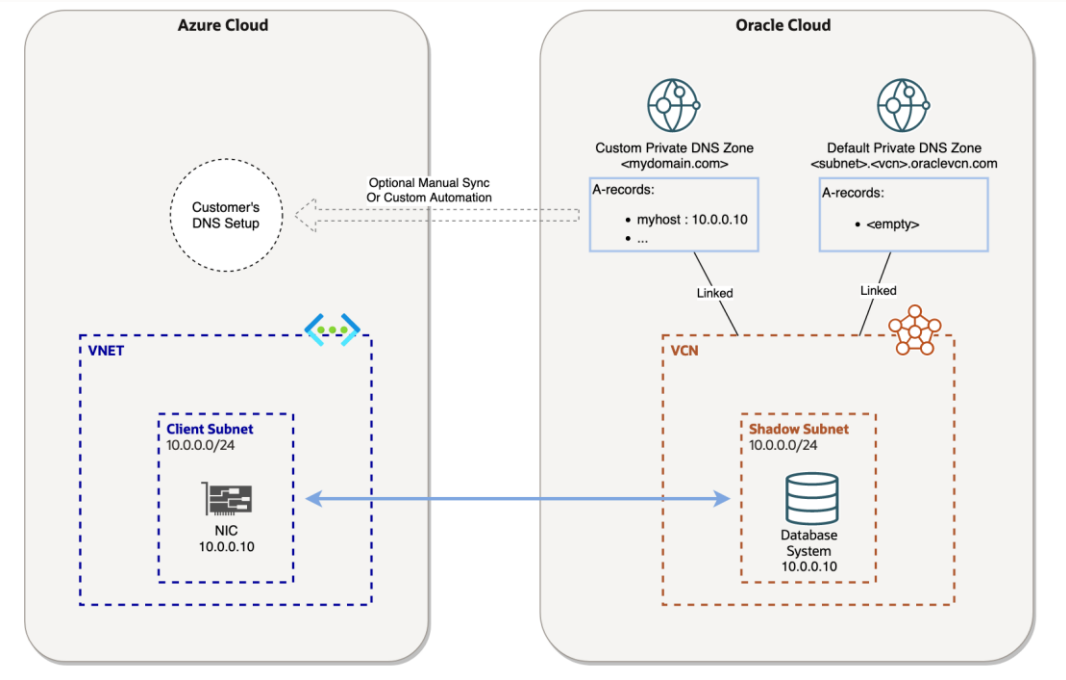
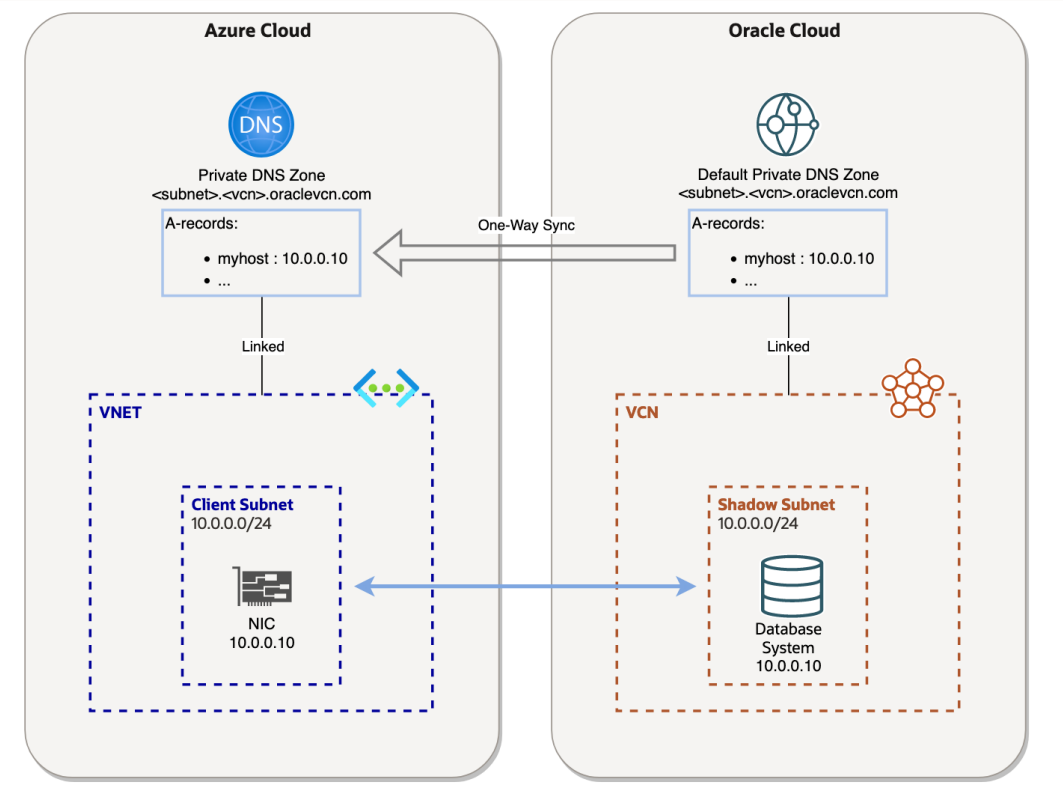


# DNS resolution in OCI and Azure – Use-cases

| Use Cases     | DNS in Azure                         | VM Cluster - Networking     | FQDN            | Azure Subscription        | Steps                                                                                                                                |
|---------------|--------------------------------------|-----------------------------|-----------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Fully managed | DNS servers Default (Azure-provided) | Uncheck private DNS service | *.oraclevcn.com | Same Exadata subscription | <ul style="list-style-type: none"><li>Fully managed in Azure and OCI</li></ul>                                                       |
| Custom FQDN   | DNS servers Default (Azure-provided) | Check private DNS service   | *.acme.com      | Custom subscription       | <ul style="list-style-type: none"><li>Managed in OCI DNS Private Zone</li><li>Create Private DNS zone and records in Azure</li></ul> |
| Custom DNS    | DNS servers Custom (on-prem)         | Check private DNS service   | *.acme.com      | N/A                       | <ul style="list-style-type: none"><li>Managed in OCI DNS Private Zone</li><li>Create Zone and records in custom DNS</li></ul>        |



# DNS setup



# Autonomous Database Provisioning

# Deployment: Basic

Microsoft Azure

Search resources, services, and

[Home](#) > [Oracle Database@Azure](#) | [Oracle Autonomous Database@Azure](#) >

Create Oracle Autonomous Database ...

⚠ Changes on this step may reset later selections you have made. Review all options prior to deployment.

Basics

Configuration

Networking

Maintenance

Consent

Tags

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription \*

omcpmout1

Resource group \*

jsitest1\_group

Create new

Instance details

Name \*

jsiadb1

Region \*

East US

Oracle Cloud Account

Oracle Cloud account name

orpsandbox6

Previous

Next

Review + create

- Resource group
- Name
- Region





# Deployment: Configuration

Microsoft Azure

Search resources, services, and

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database

Basics Configuration Networking Maintenance Consent Tags Review + create

Autonomous Database supports different workload types, including: Data Warehouse and Transaction Processing. Both of these workload types provide performance improvements and additional features that support operations for the specified workload. [Learn more](#)

Workload type \*

Transaction Processing

Database configuration

Database version \*

19c

ECPUs count \*

2

Compute auto scaling

☒

Storage \*

1024

Storage unit size \*

☒ GB

☐ TB

Storage auto scaling

☒

Backup retention

Automatic backups are managed by Oracle. Backup storage is billed separately and in addition to database storage. When a backup ages beyond the backup retention setting, it is deleted. [Learn more](#)

Backup retention period in days \*

60

Microsoft Azure

Search resources, services, and

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database

Storage auto scaling ☒

Backup retention

Automatic backups are managed by Oracle. Backup storage is billed separately and in addition to database storage. When a backup ages beyond the backup retention setting, it is deleted. [Learn more](#)

Backup retention period in days \*

60

Administrator credentials

Set the password for your Autonomous Database ADMIN user here.

Username

ADMIN

Password \*

\*\*\*\*\*

Confirm password \*

\*\*\*\*\*

License

License type \*

License included

Advanced options

☒

Management

Character set

AL32UTF8

National character set

AL16UTF16

Previous

Next

Review + create

- Workload type: ATP or ADW
- Database version: 19c
- ECPUs count
- Auto scaling
- Storage size
- Storage auto scaling
- Backup retention
- Username
- Password
- Licence type
- Advanced options
  - Character set
  - National Character set



# Deployment: Networking

The screenshot shows the 'Create Oracle Autonomous Database' wizard in the Microsoft Azure portal, specifically the 'Networking' tab. The breadcrumb navigation at the top indicates the path: Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >. The wizard has several tabs: Basics, Configuration, Networking (selected), Maintenance, Consent, Tags, and Review + create. Under the 'Networking' tab, the 'Network access' section contains the following fields:

- Access type:** A dropdown menu with the value 'Managed private virtual network IP only'.
- Require mutual TLS (mTLS) authentication:** A checkbox that is currently unchecked.
- Virtual network \*:** A dropdown menu with the value 'jsivnet'.
- Subnet \*:** A dropdown menu with the value 'adb1'.

Below these fields, there is a yellow warning box with a triangle icon and the text: 'The subscription and resource group must be selected in the "Basics" tab in order to list virtual networks.'

At the bottom of the wizard, there are three buttons: 'Previous' (disabled), 'Next' (disabled), and 'Review + create' (active).

- Access type: Managed private virtual network IP only
- Require mutual TLS (mTLS) authentication
- Virtual network
- Subnet

# Deployment: Maintenance

[Home](#) > [Oracle Database@Azure](#) | [Oracle Autonomous Database@Azure](#) >

Create Oracle Autonomous Database ...

Basics

Configuration

Networking

Maintenance

Consent

Tags

Review + create

Maintenance preferences

Maintenance patch level

Regular

Maintenance contact emails

Contacts in this list receive operational notices and announcements as well as unplanned maintenance notifications.

Email address

Enter an email address

Previous

Next

Review + create

- Maintenance patch level
- Maintenance contact emails



# Deployment: Tags

Microsoft Azure

Search resources, ser

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database ...

Basics

Configuration

Networking

Maintenance

Consent

Tags

Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

| Name ⓘ               | Value ⓘ                | Resource            |
|----------------------|------------------------|---------------------|
| <input type="text"/> | : <input type="text"/> | Autonomous Database |

Previous

Next

Review + create

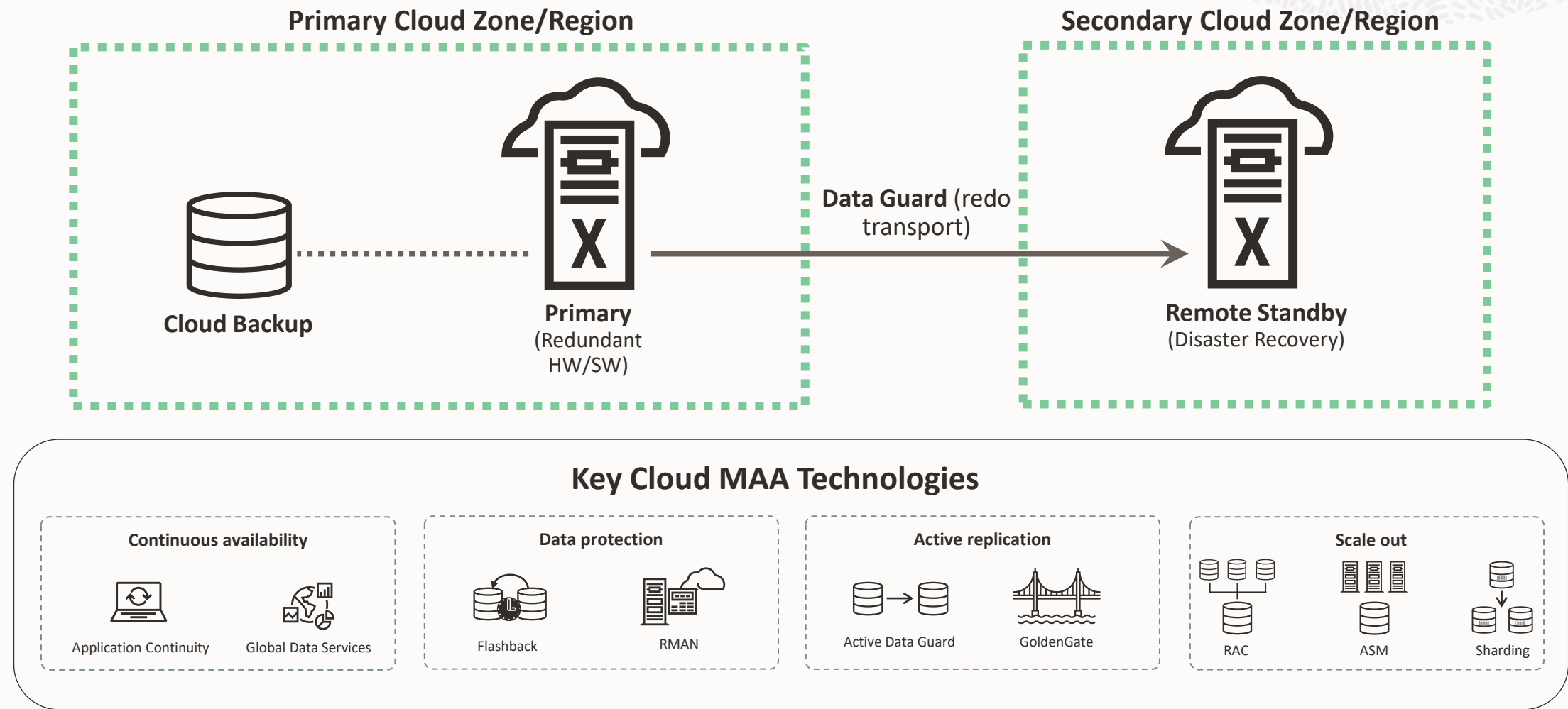
- Tag name and value



# High Availability and Disaster Recovery

# Oracle Maximum Availability Architecture (MAA)

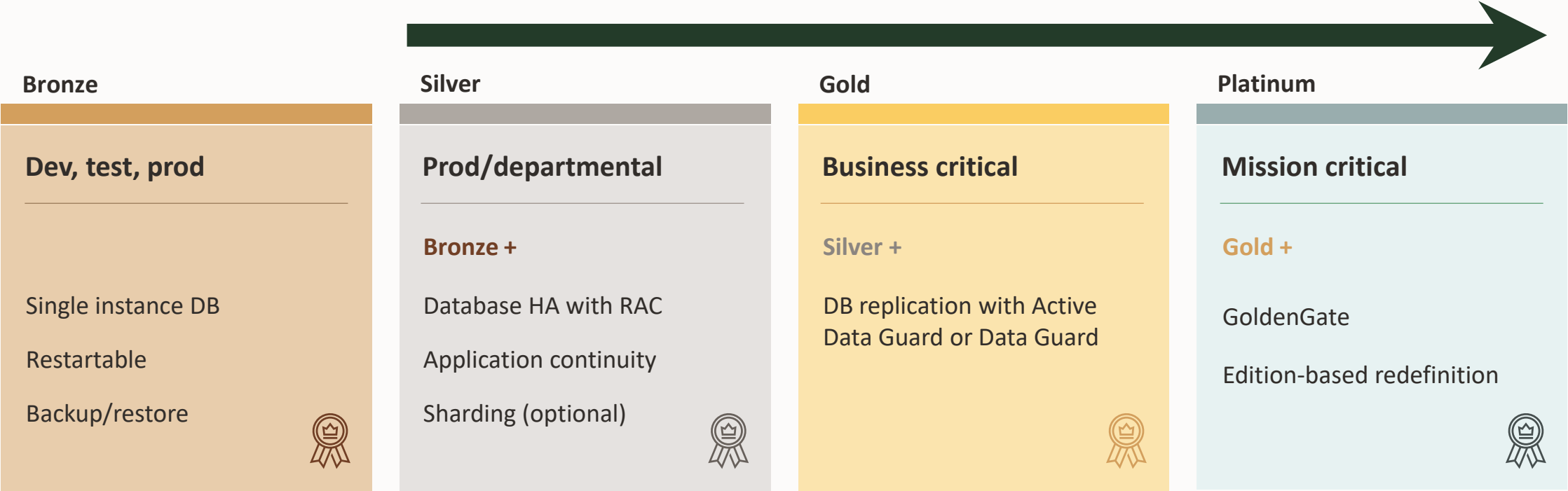
## High-Availability Blueprint in the Cloud



# MAA reference architectures

Availability service levels

Oracle Database@Azure



# ExaDB@Azure / DR different AZ / App DR same AZ

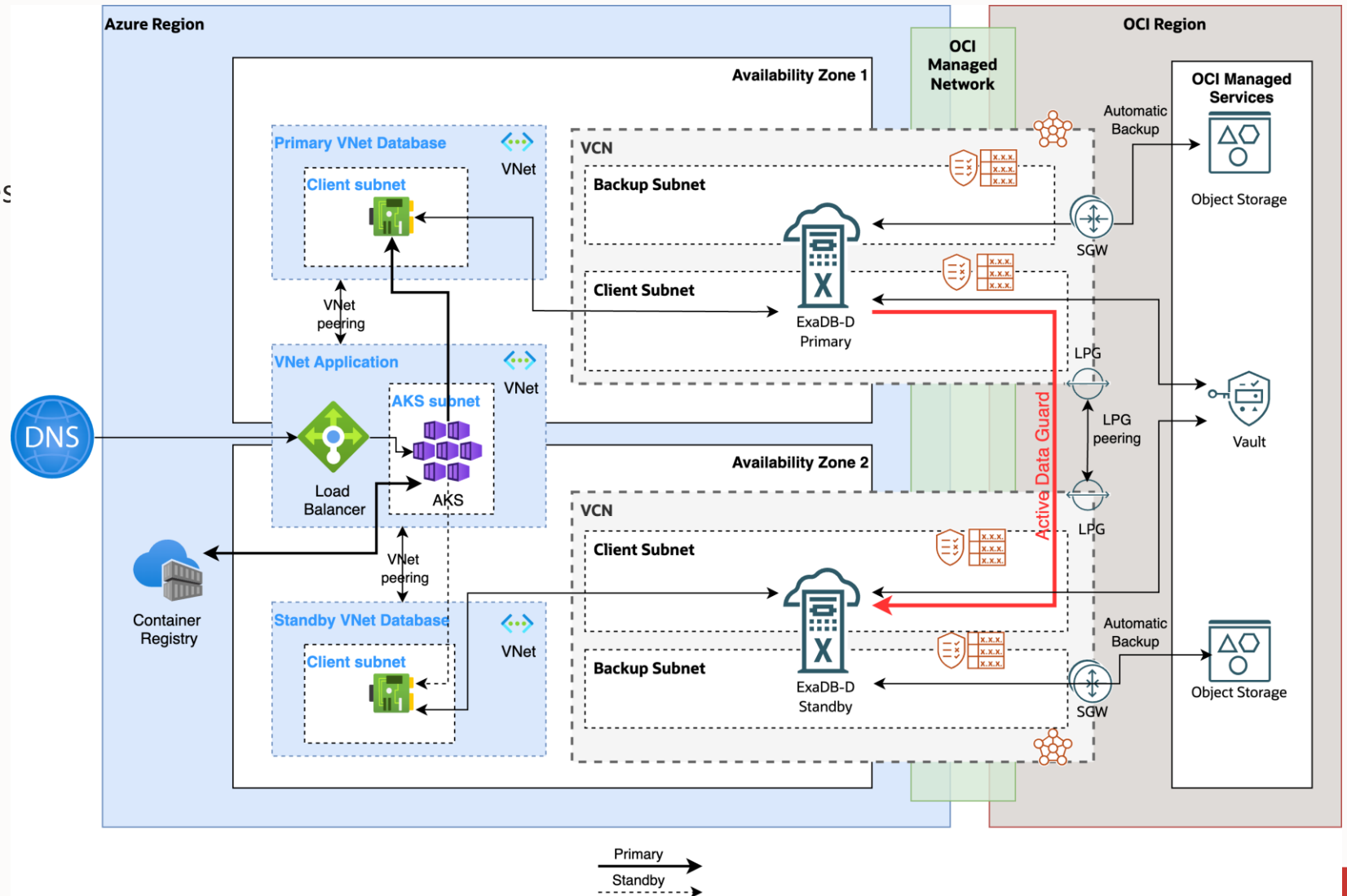
## Gold MAA – OCI network – Low level

### Application layer on Azure

- Application is deployed in Regional Private Azure Kubernetes Services (AKS) multi-AZs
- Vnet Application peered with Primary and Standby Vnet Databases

### Database layer on OCI

- Primary ExaDB in AZ1
- Standby ExaDB in AZ2
- Redo log shipment via the OCI Network (no cross-AZs traffic)
- Routing uses local peering gateway between VCNs





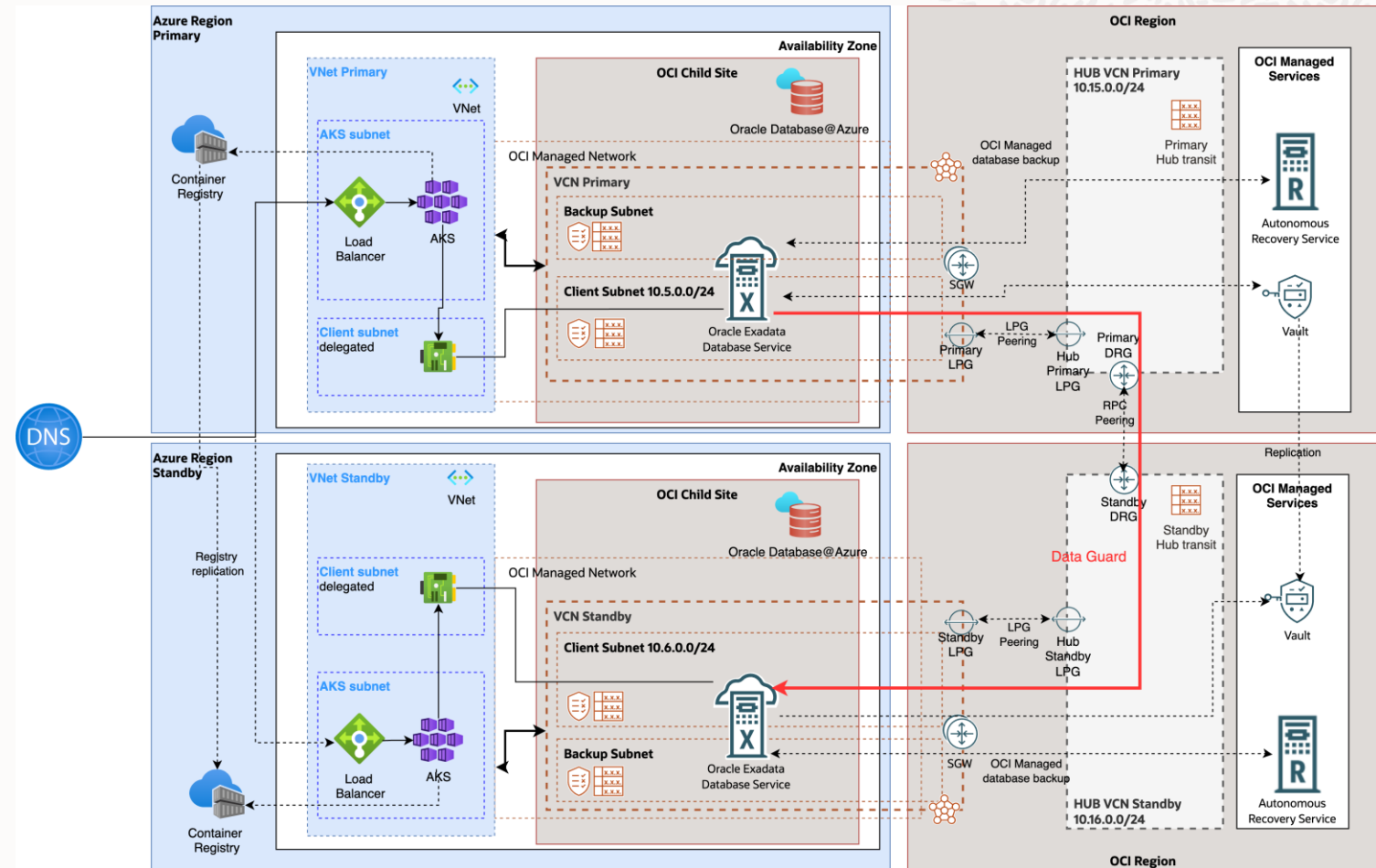
# Oracle Database@Azure | Cross-region Replication Overview

## Application layer on Azure

- Application is deployed in Regional Azure Kubernetes Services (AKS)
- Container images are replicated between regions
- Deploy Dataguard Observer on Azure VM

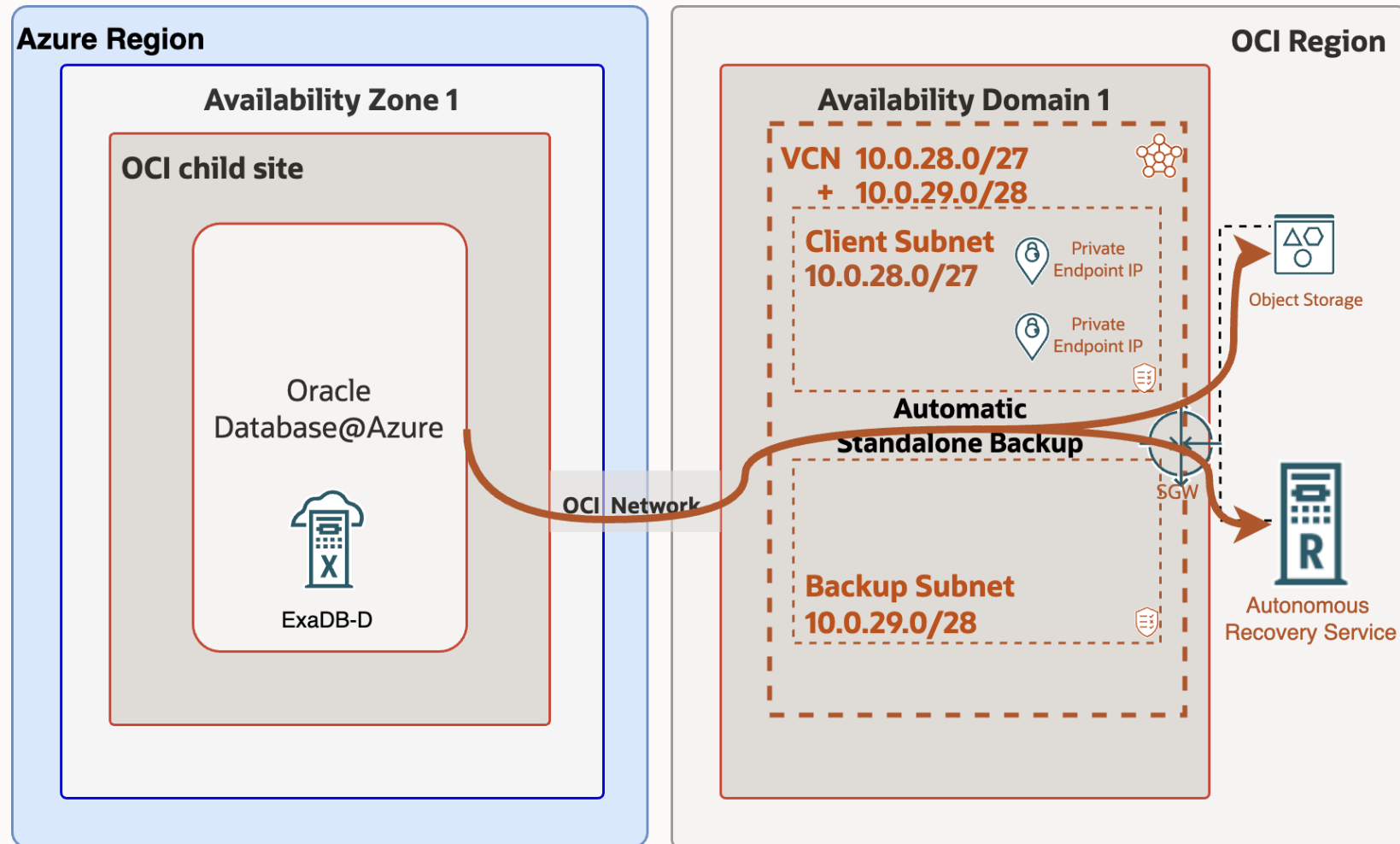
## Database layer on OCI

- Primary DB is hosted in Region 1
- Standby DB is hosted in in Region 2
- Redo log shipment is via the OCI backbone (incurs egress charges)
- Routing uses local peering to hub VCNs and DRGs
- OCI Vault cross region replication (incurs egress charges)
- Observers are place in Azure application subnet.



# Backup

# Automatic and Standalone Backup to OCI Object Storage



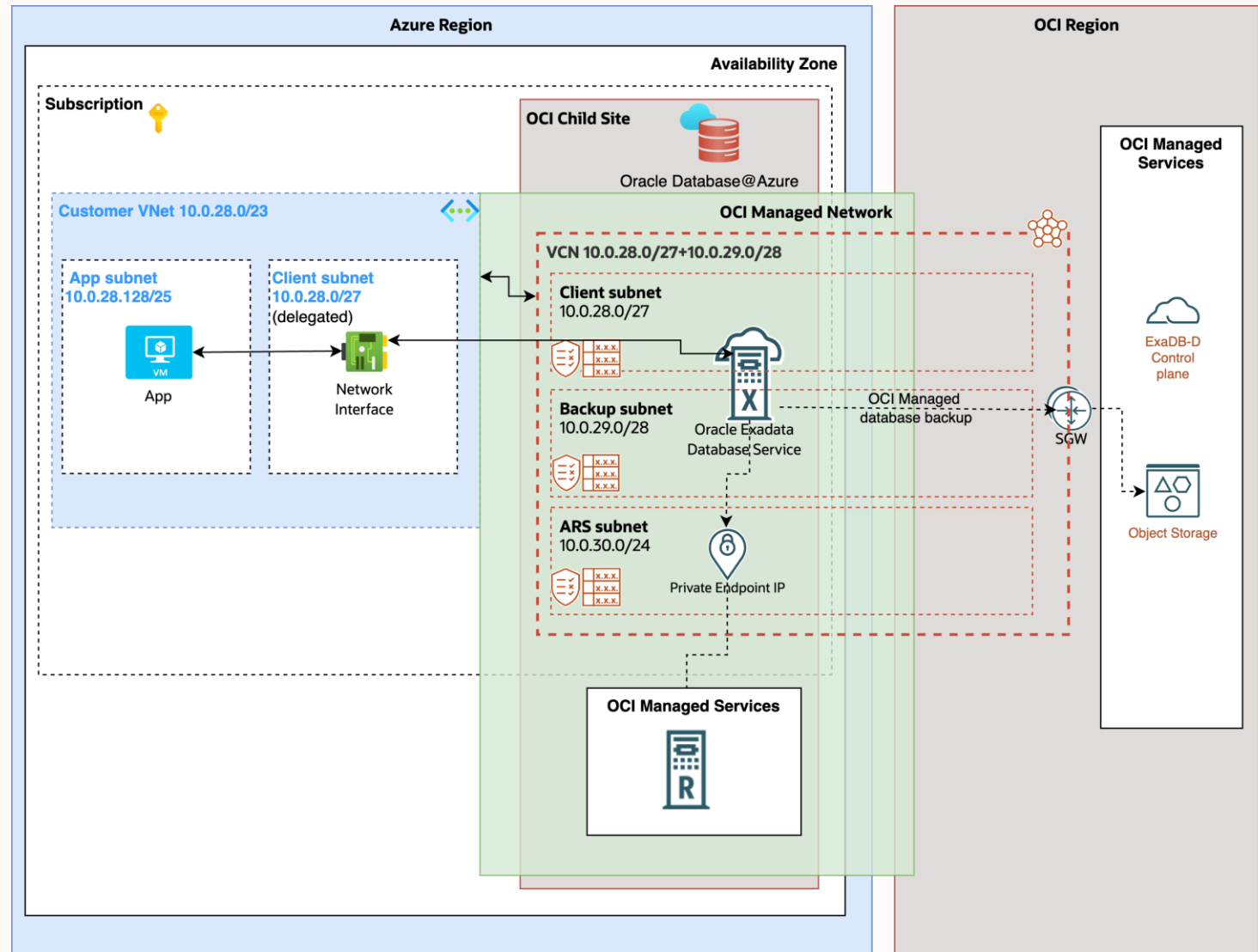
Automatic backup with a Maximum retention period of 60 days to object storage.  
Standalone/Full backup for long term backup retention through OCI API/CLI/Console.



# Autonomous Recovery Service

Oracle Database Autonomous Recovery Service (ARS) is a fully managed, standalone, and centralized cloud backup solution for Oracle Cloud Infrastructure (OCI) databases

- Draw down from Azure commitment
- Dedicated OCI Subnet required, /24 CIDR
- Protection policy
  - Bronze, Silver, Gold, Platinum



# Migration

# Migration Methods

|                                                                  | Migration               | Online                                                                                | Downtime *     |
|------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------|----------------|
| <b>Recommended: ZDM</b><br><b>Oracle Zero Downtime Migration</b> | <b>Physical/Logical</b> |    | <b>Minimal</b> |
| Other methods                                                    |                         |                                                                                       |                |
| Oracle Data Guard                                                | Physical                |    | Minimal        |
| Oracle GoldenGate                                                | Logical                 |    | Minimal        |
| RMAN                                                             | Physical                |    | Required       |
| Oracle Data Pump                                                 | Logical                 |  | Required       |
| Oracle Database Multitenant Movement                             | Physical                |  | Required       |

\* Downtime – The duration depends on the volume and use case.



# Migrating Databases to Oracle Cloud Infrastructure

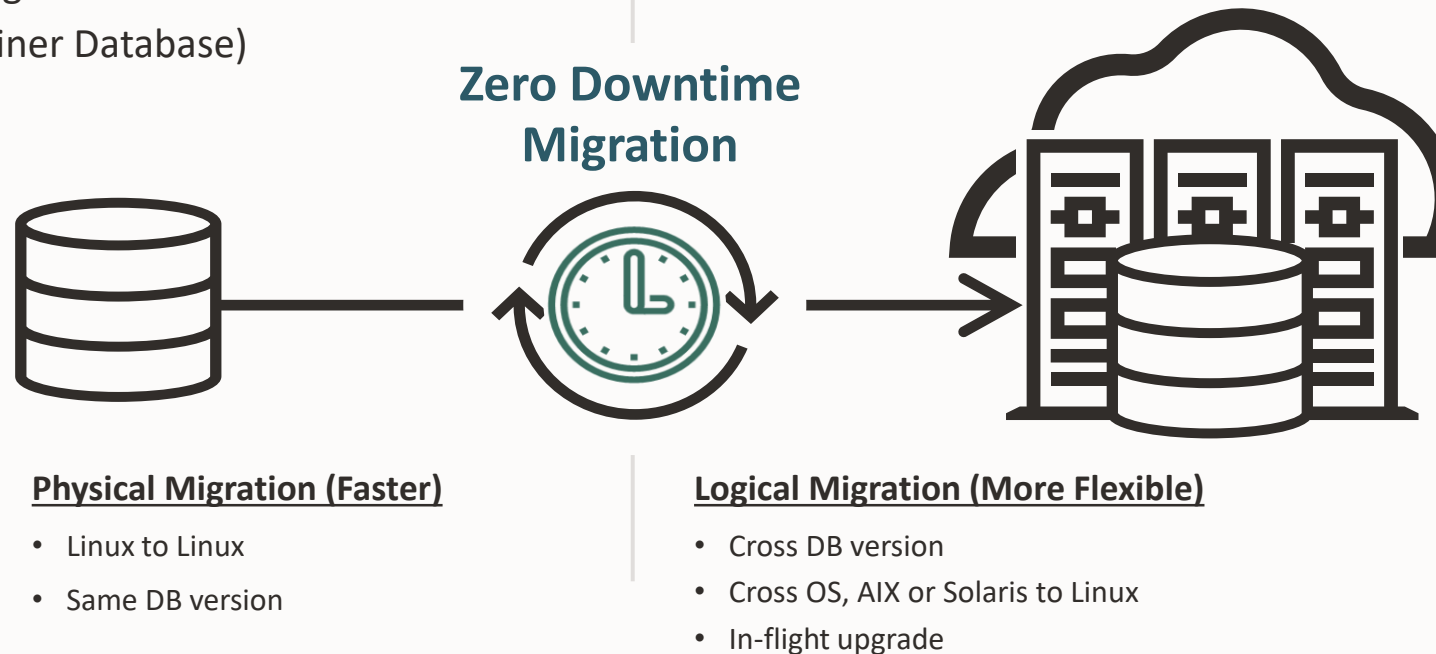
Oracle Zero Downtime Migration – Simple and Free!

## Oracle Database Sources

- On-premises deployments
- Cloud deployments
- Oracle Database Enterprise & Standard Editions
- Linux, AIX, Solaris
- RAC, RAC One Node, Single-Instance
- Non-CDB or CDB (Container Database)

## Oracle Cloud Targets

- Oracle Exadata Database Service
- Autonomous Database Serverless



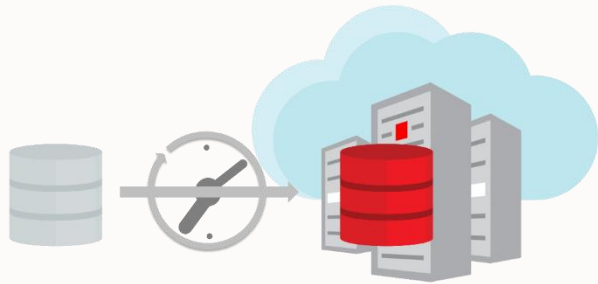


# Oracle Zero Downtime Migration (ZDM)

## Overview

### Simple

- Single-button approach



### Fleet Scale

- Centralized
- Scheduled Operations
- Audit Trail
- Migrations in Parallel
- Jobs Framework

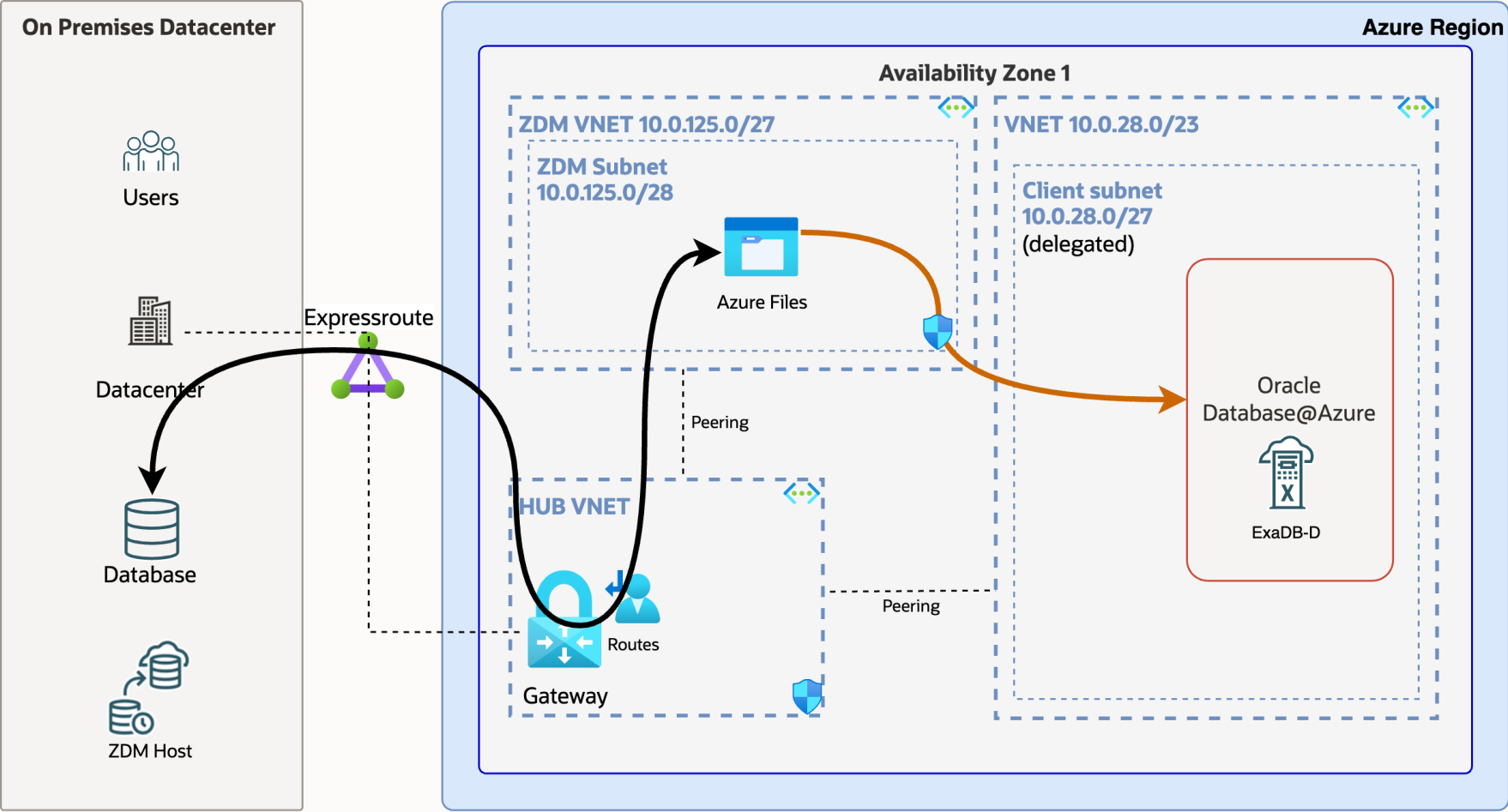
### Comprehensive

- MAA Compliant
- Resumable
- Dry-run option
- Customizable User Actions
- Automated Encryption



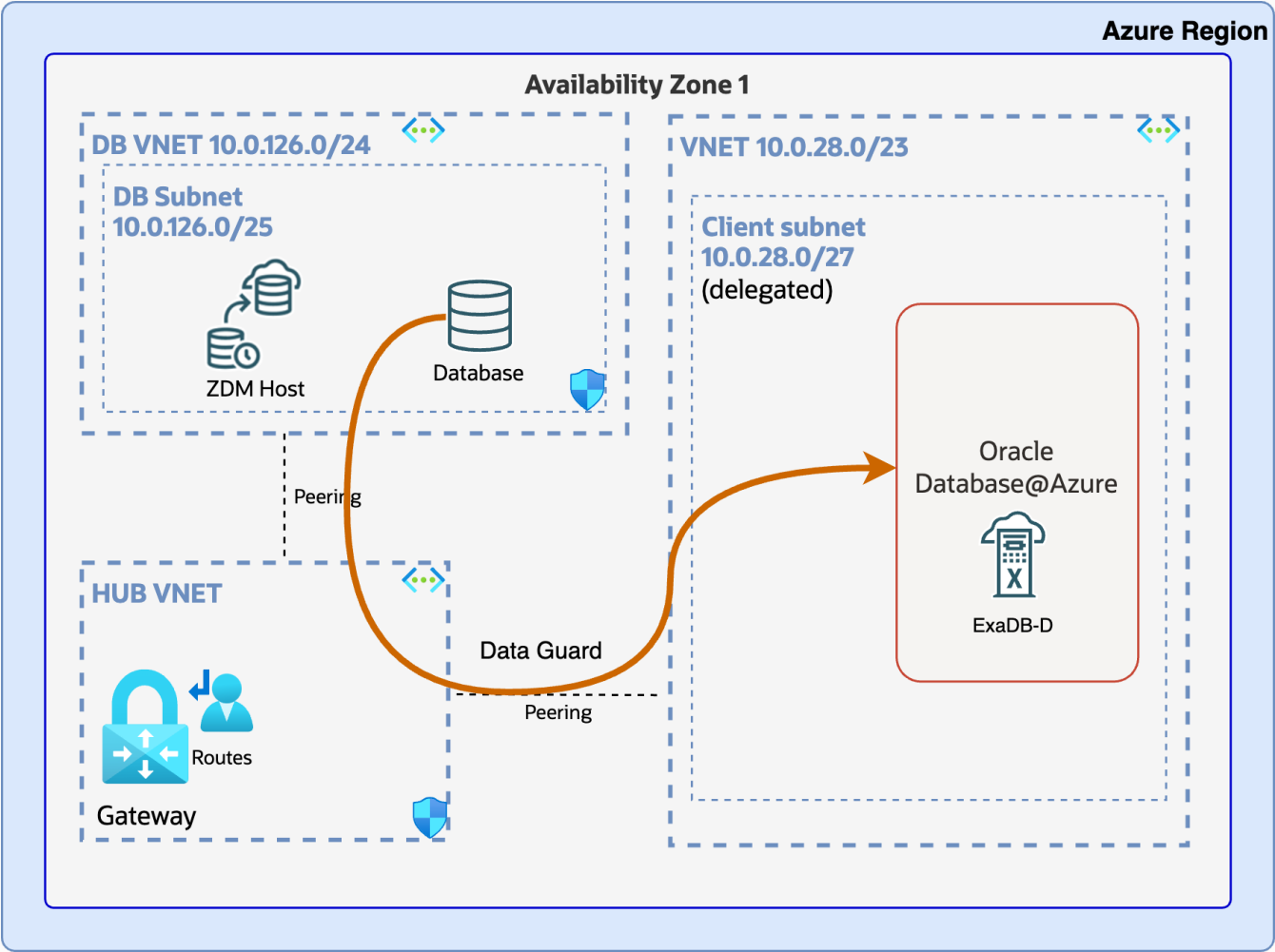
# Migration with ZDM on Azure from On Premises

## Physical Offline Migration

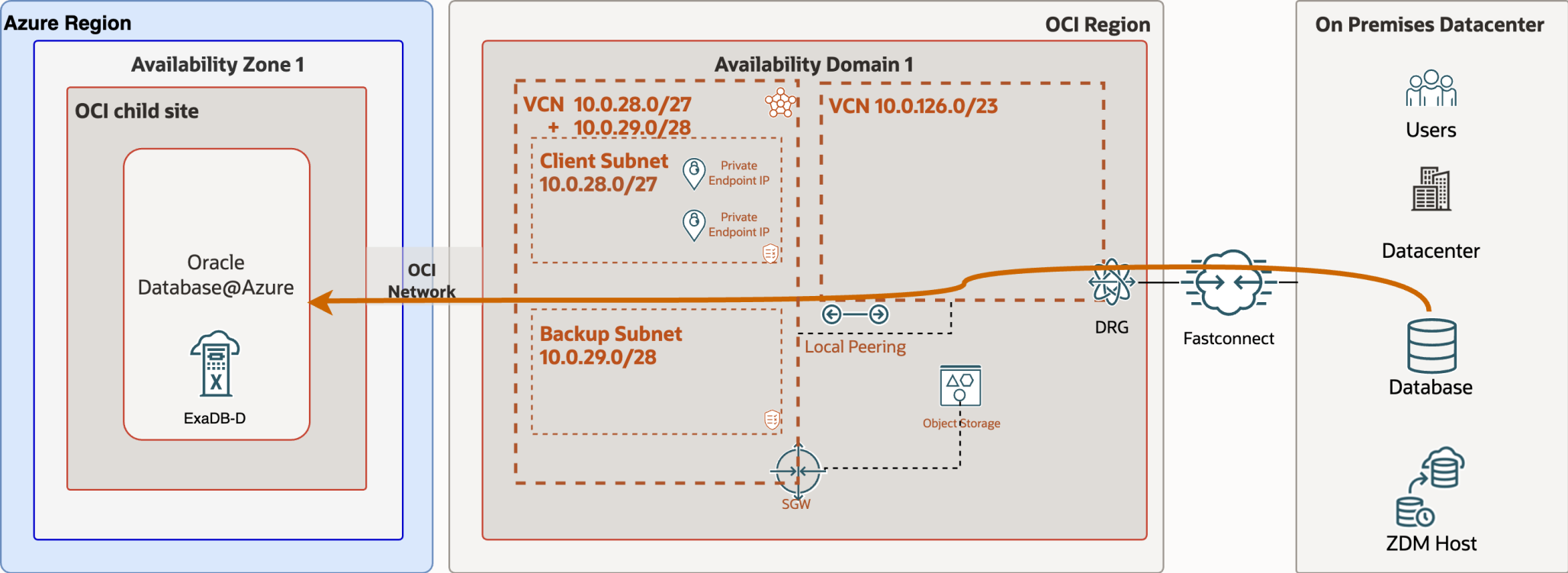


# Migration with ZDM on Azure

Physical Online Migration – direct method (restore from service)



# Migration with ZDM from On Premises through OCI



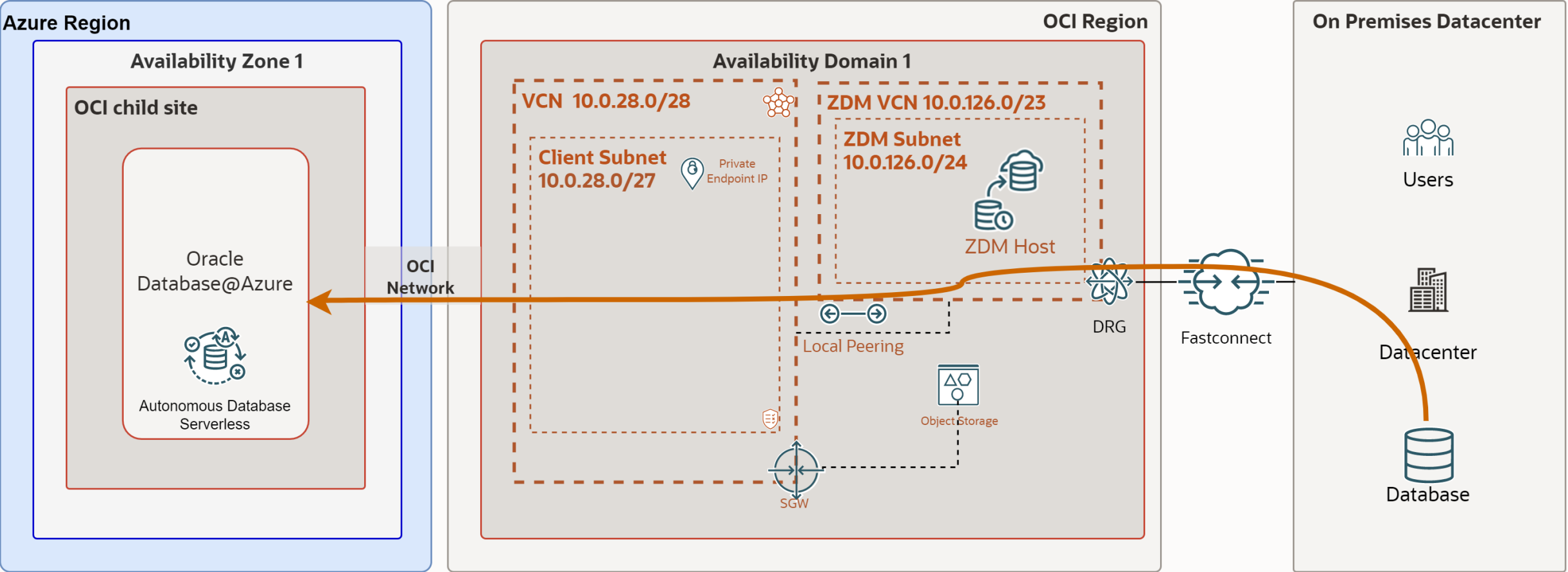
# Migration Methods – ADB-S

|                                                                  | Migration      | Online                                                                              | Downtime *     |
|------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|----------------|
| <b>Recommended: ZDM</b><br><b>Oracle Zero Downtime Migration</b> | <b>Logical</b> |  | <b>Minimal</b> |
| Other methods                                                    |                |                                                                                     |                |
| Oracle GoldenGate                                                | Logical        |  | Minimal        |
| Oracle Data Pump                                                 | Logical        |  | Required       |

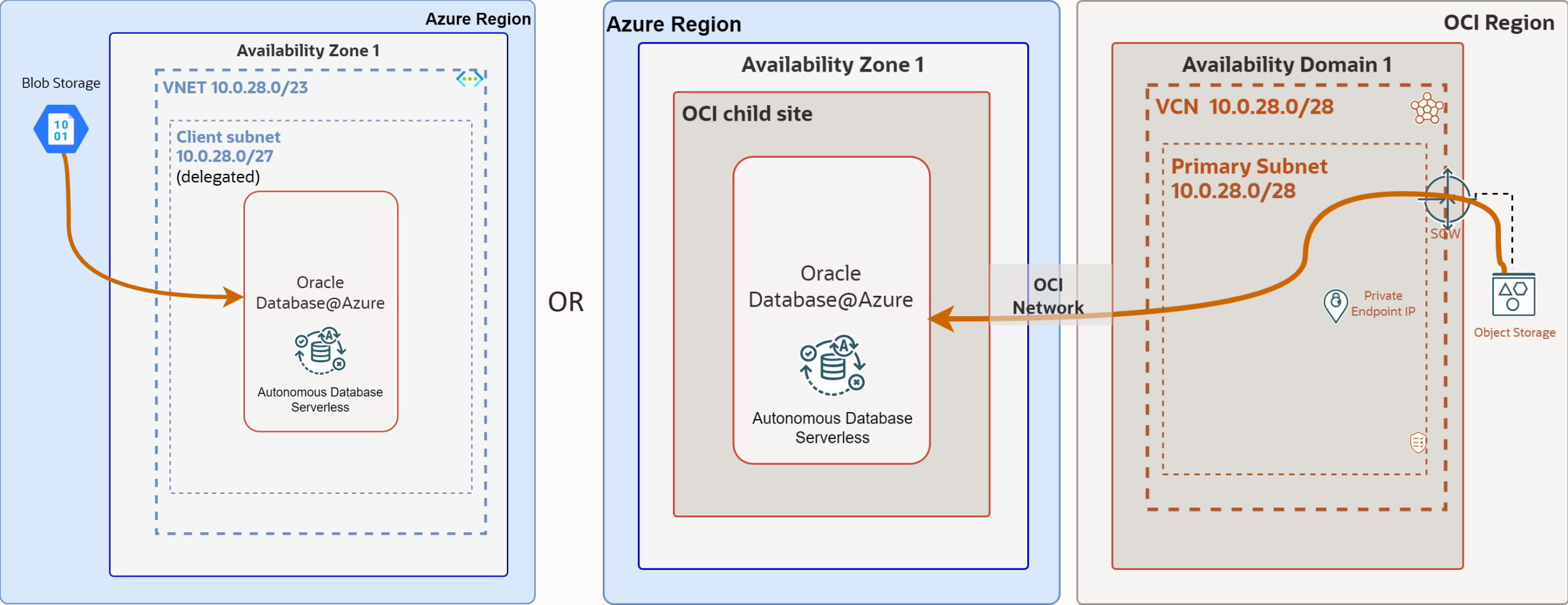
\* Downtime – The duration depends on the volume and use case.



# Migration with ZDM from On Premises through OCI – ADB-S

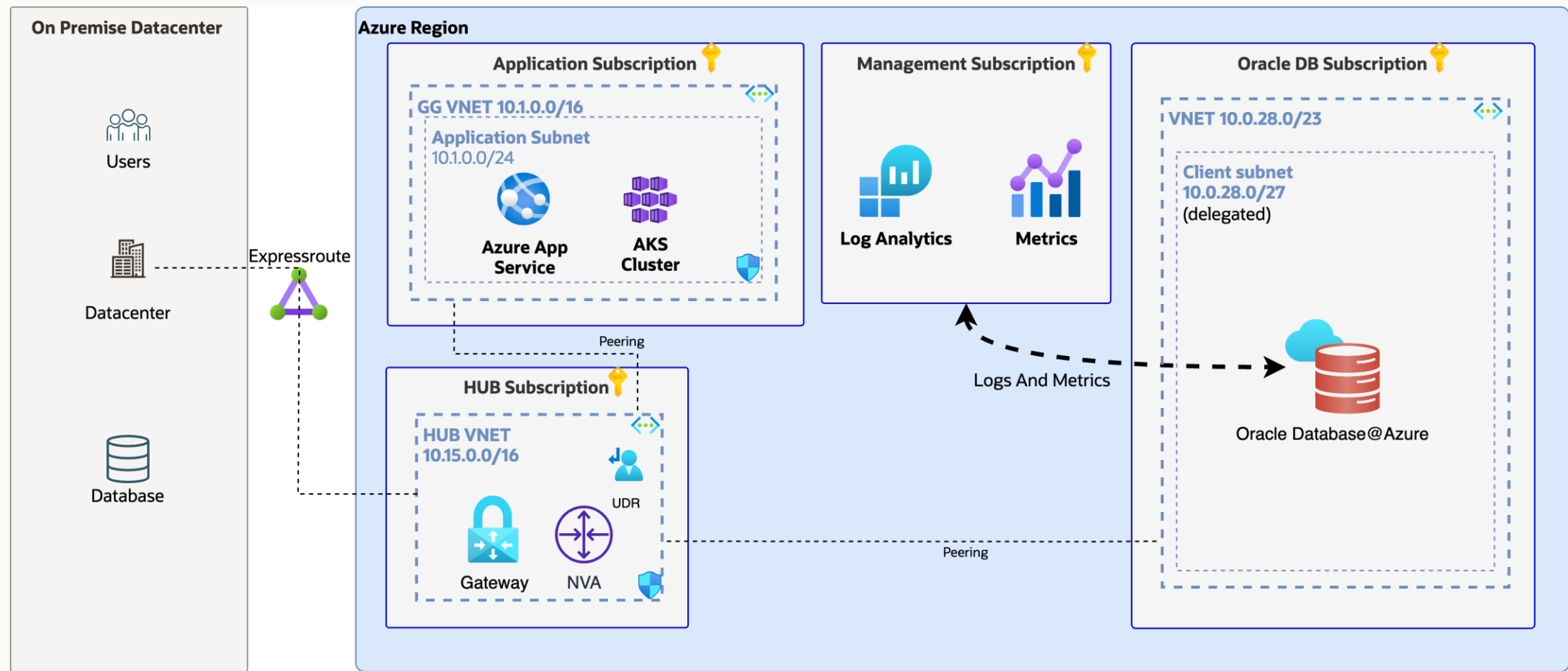


# Migration with Data Pump – ADB-S



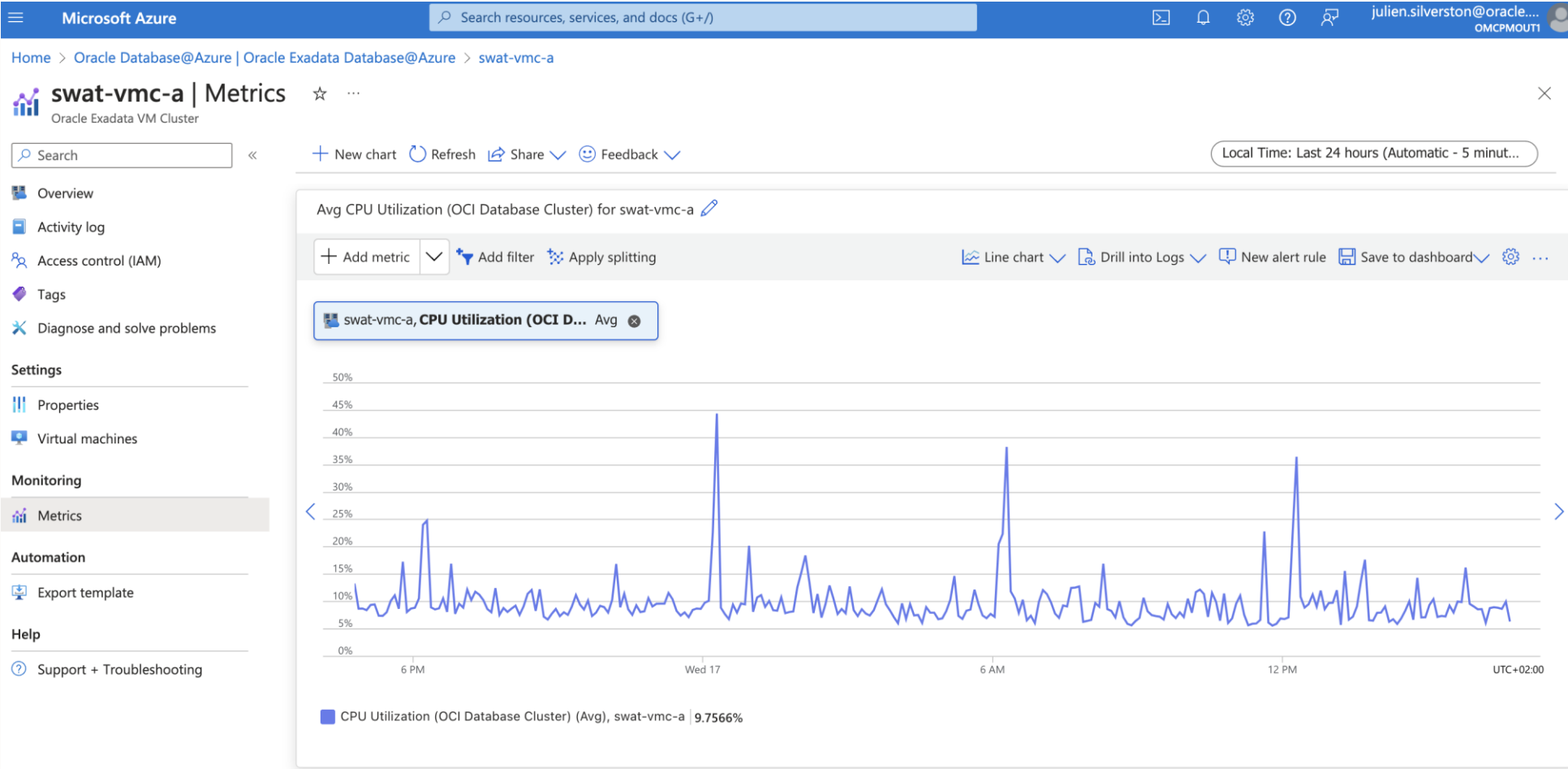
# Monitoring

# Azure Subscription Segmentation: Monitoring subscription





# Azure Monitor Metrics

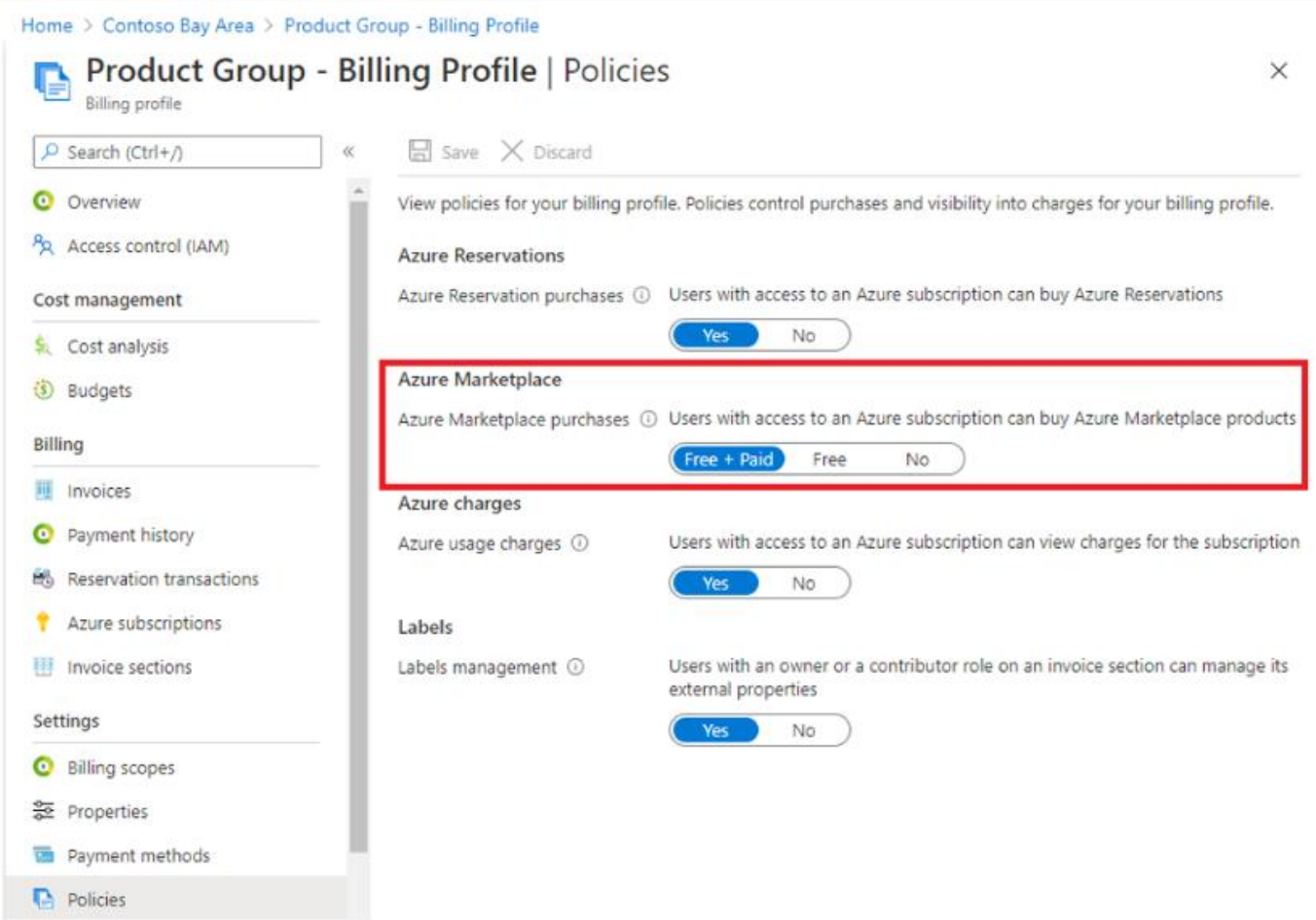


# Identity & Permissions

# Identity - Azure roles

Azure permissions allowing to accept "Free + Paid" offers in Azure Marketplace

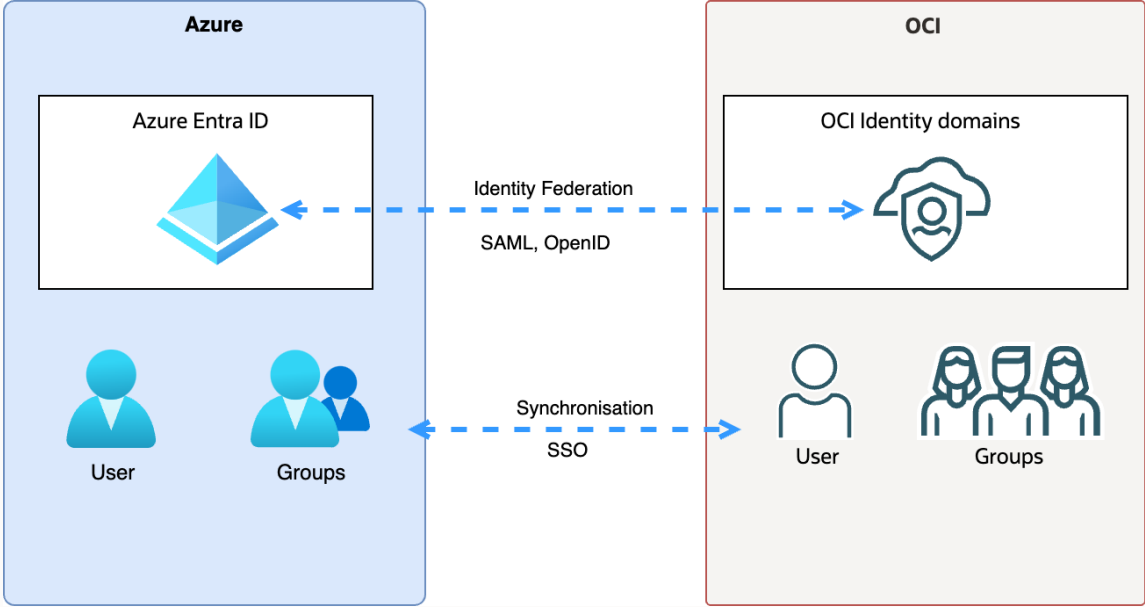
| Agreement type                     | Permissions to accept offer                          | Permissions to purchase or subscribe                           |
|------------------------------------|------------------------------------------------------|----------------------------------------------------------------|
| Microsoft Customer Agreement (MCA) | <a href="#">Billing account owner or contributor</a> | <a href="#">Subscription owner or subscription contributor</a> |
| Enterprise Agreement (EA)          | <a href="#">Enterprise administrator</a>             | <a href="#">Subscription owner or subscription contributor</a> |



# Identity Roles and Groups

| Group name                      | Role Assigned                  | Description                                                                                            |
|---------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------|
| odbaa-exa-infra-administrators  | odbaa-exa-infra-administrator  | User of this group will be able to administrate ExaDB-D resources in Azure                             |
| odbaa-vm-cluster-administrators | odbaa-vm-cluster-administrator | User of this group will be able to administrate VM Cluster resources in Azure                          |
| odbaa-db-family-administrators  | N/A                            | This group is replicate in OCI as part of the federation process, OCI policies will be attached to it. |
| odbaa-db-family-readers         | N/A                            | This group is replicate in OCI as part of the federation process, OCI policies will be attached to it. |
| odbaa-exa-cdb-administrators    | N/A                            | This group is replicate in OCI as part of the federation process, OCI policies will be attached to it. |
| odbaa-exa-pdb-administrators    | N/A                            | This group is replicate in OCI as part of the federation process, OCI policies will be attached to it. |

| Group name                     | Policy                                                   | Description                        |
|--------------------------------|----------------------------------------------------------|------------------------------------|
| odbaa-db-family-administrators | manage database-family                                   | Group to manage DB family actions. |
| odbaa-db-family-readers        | read database-family                                     | Group to read DB family actions.   |
| odbaa-exa-cdb-administrators   | manage db-homes<br>manage databases<br>manage db-backups | Group to manage CDB actions.       |
| odbaa-exa-pdb-administrators   | manage pluggable-databases                               | Group to manage PDB actions.       |



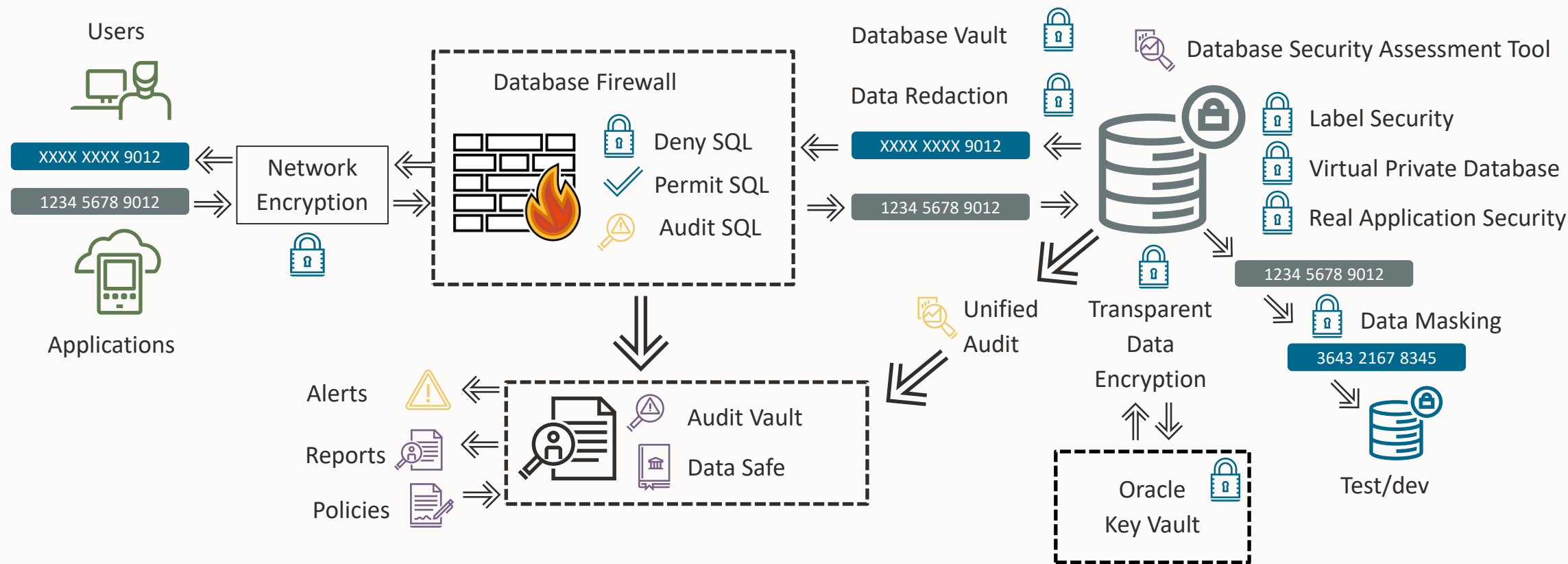
Identity Federation configuration



# Security

# Oracle Maximum Security Architecture

Defense in Depth to protect your data



# Securing your Databases

**Control User and Network Access:** Use passwords, private subnets, and network security groups to control user and network access.

**Restrict Permissions for Deleting Database Resources:** To prevent inadvertent or malicious deletion of databases, grant the delete permissions (DATABASE\_DELETE and DB\_SYSTEM\_DELETE) to a minimum set of users and groups.

**Data Encryption:** All databases created in OCI are encrypted using transparent data encryption (TDE). Ensure that any migrated databases are also encrypted. Rotate the TDE master key every 90 days or less.

**Patching:** Apply Oracle Database security patches (Oracle Critical Patch Updates) to mitigate known security issues, and keep the patch levels up-to-date.

**Use DB Security Tools:** Established Data Masking and Auditing tools to secure your data are available for Oracle Database@Azure.

**Enable Data Safe:** Data Safe is a unified control center for Oracle cloud and on-premises databases. Use Data Safe to assess database and data security configuration, detect associated risk for user accounts, identify existing sensitive data, implement controls to protect data, and audit user activity.





# Securing database traffic – in transit

## Configuring Oracle Database Native Network Encryption and Data Integrity

Configure native Oracle Net Services data encryption and data integrity for both servers and clients.

Oracle Database enables you to encrypt data that is sent over a network

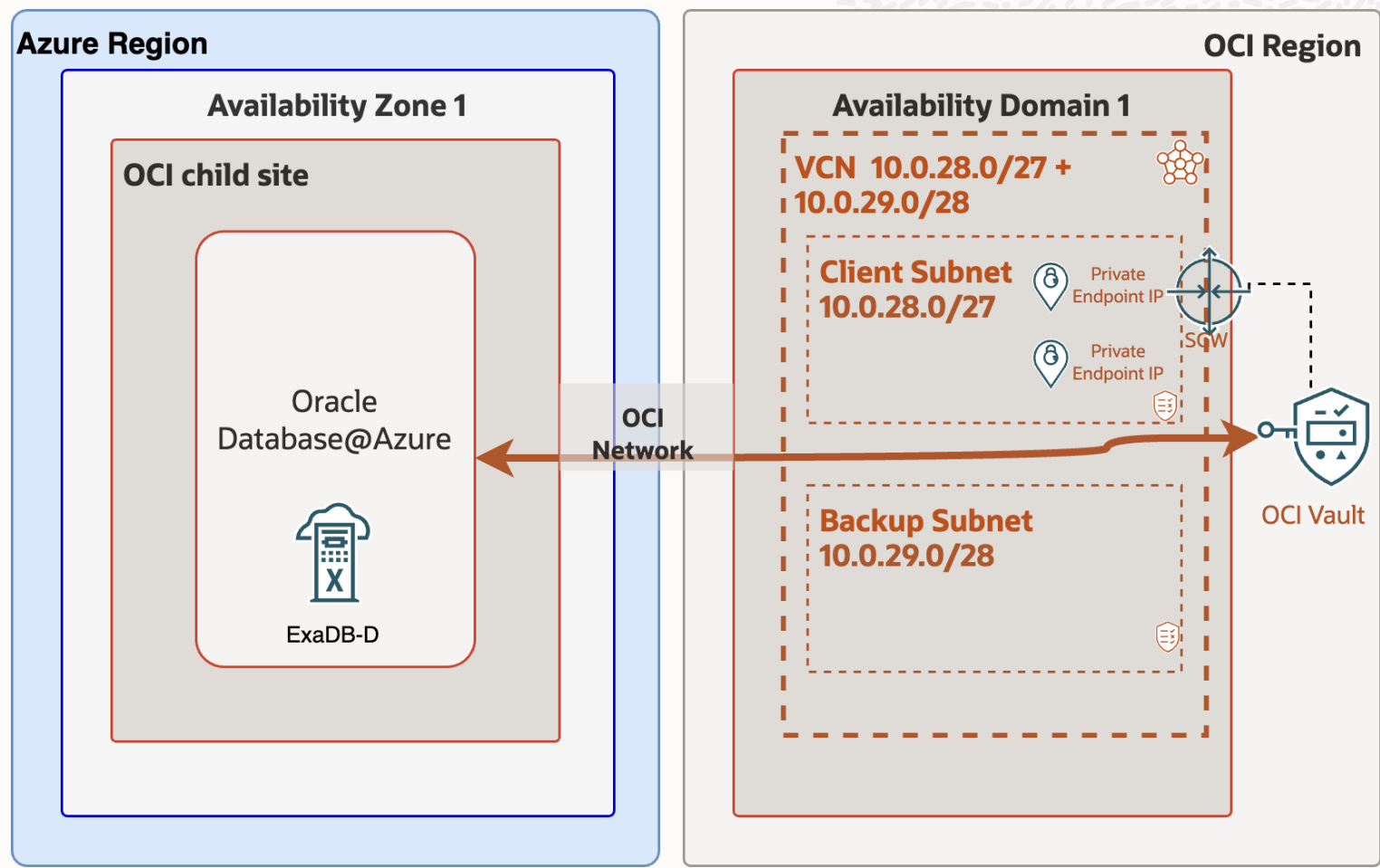
- Oracle Database provides **native data network encryption and integrity** to ensure that data is secure as it travels across the network
- Oracle Database supports the **Federal Information Processing Standard (FIPS)** encryption algorithm, **Advanced Encryption Standard (AES)**
- **Triple-DES encryption (3DES)** encrypts message data with three passes of the DES algorithm
- Oracle offers two ways to encrypt data over the network, native network encryption and **Transport Layer Security (TLS)**
- The Physical layer between the Oracle Database@Azure and OCI are **encrypted**.

<https://docs.oracle.com/en/database/oracle/oracle-database/19/dbseg/configuring-network-data-encryption-and-integrity.html>





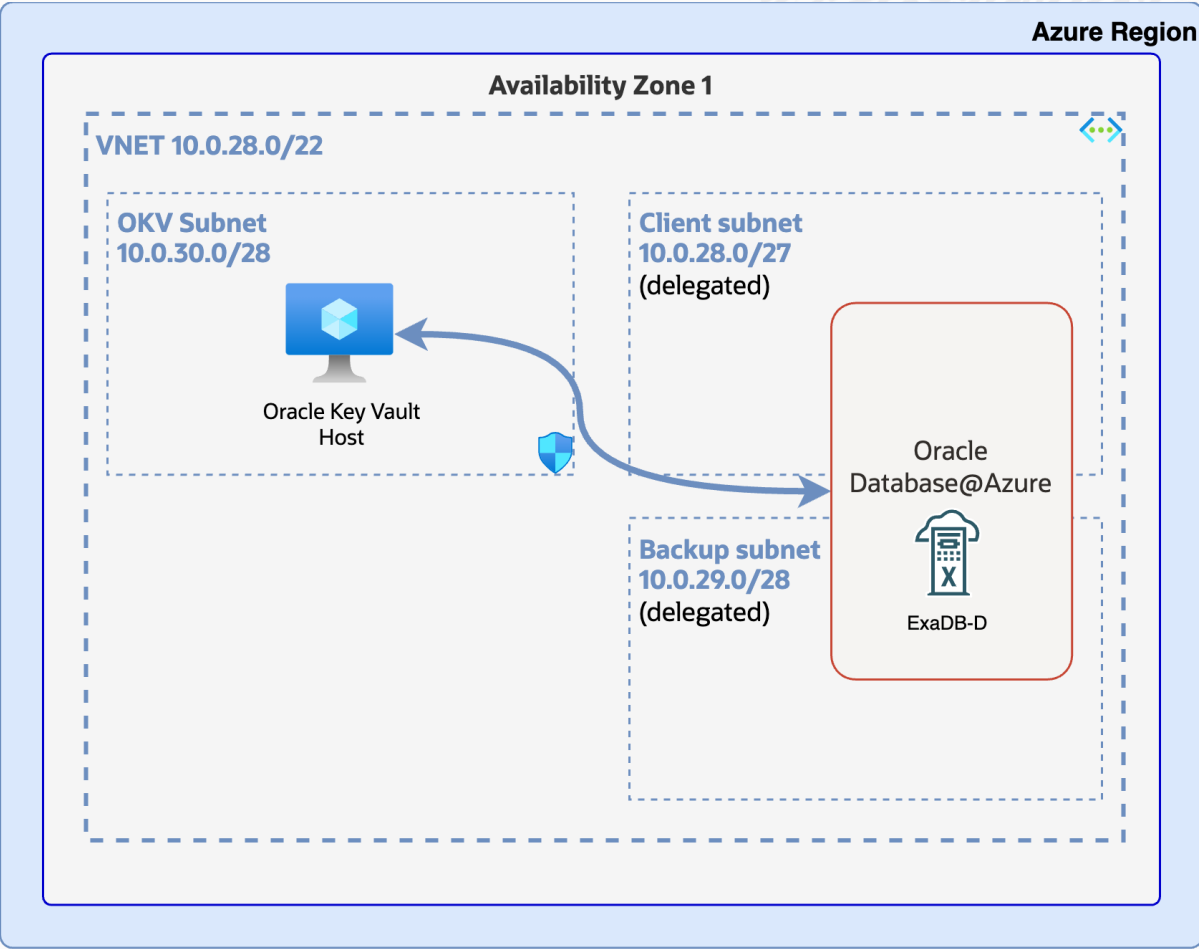
# Encryption using OCI Vault – ExaDB-D



OCI Vault is the default Key Management system for Oracle Database@Azure. For using OCI Vault, the customer needs UCM.



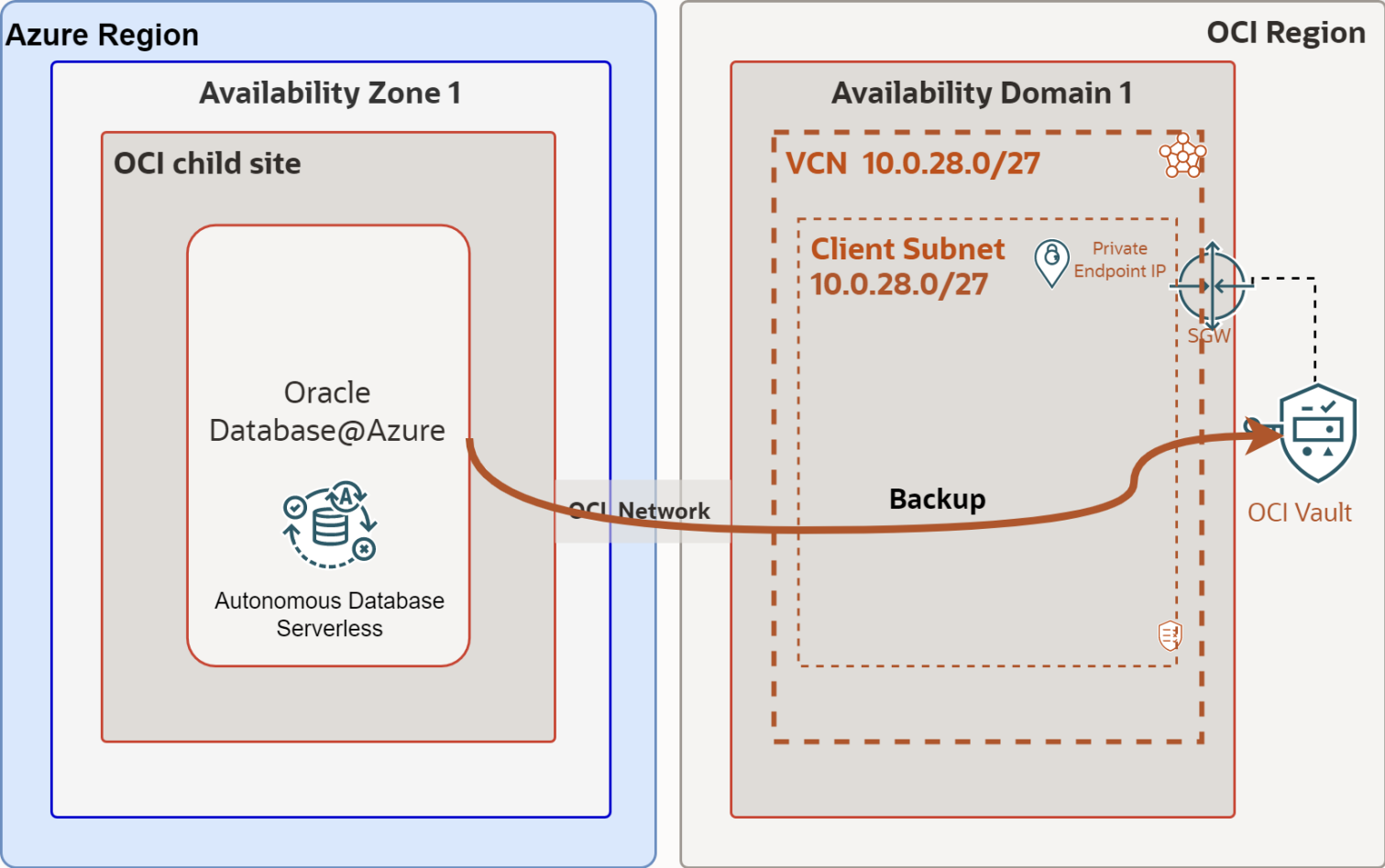
# Encryption using Oracle Key Vault on VM Host – ExaDB-D



Standard images for OKV exist or Install an OKV instance on a Azure VM host.



# Encryption using OCI Vault - ADBS



By default, Oracle Managed Keys and OCI Vault are available for ADB-S.  
To use OCI Vault, the customer needs UCM.



# Compliances

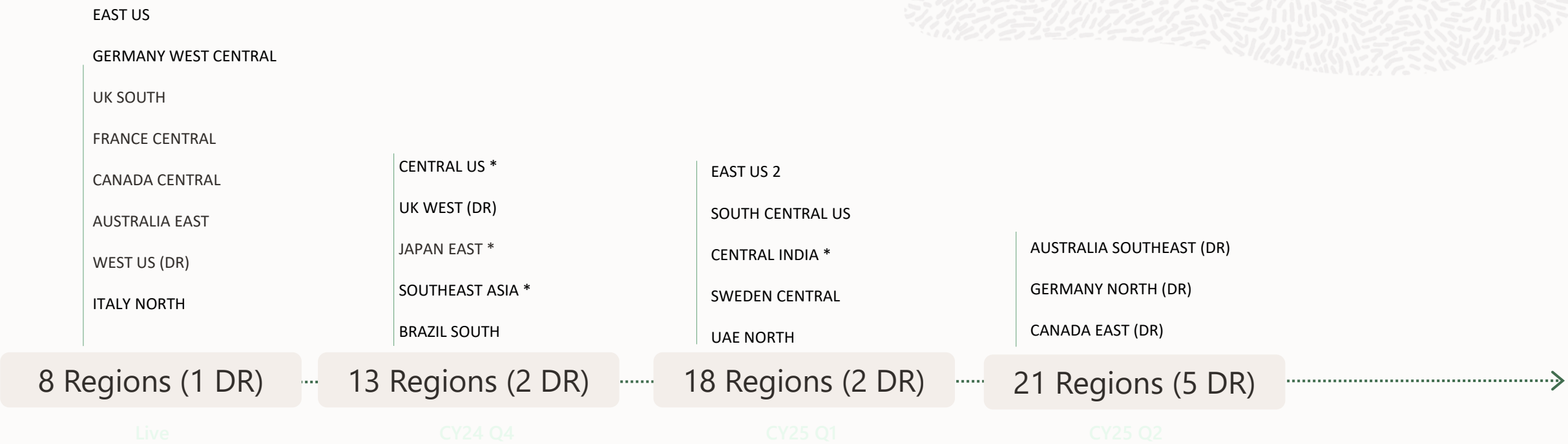


## Certifications

- PCI-DSS, SOC 1,2,3, HIPAA, C5, ISO, CSA STAR, HDS
- Microsoft: <https://servicetrust.microsoft.com/>
- OCI: <https://www.oracle.com/corporate/cloud-compliance/>

# Roadmap

# Oracle Database@Azure Region Roadmap: Onboarding Ready Dates



\* First AZ available to onboard customer

### More Planned Regions

- North Europe
- West Europe
- West US 2
- West US 3
- Spain Central

### More Planned DR Regions

- Brazil Southeast
- France South
- Japan West
- North Central US
- South India



# Automation

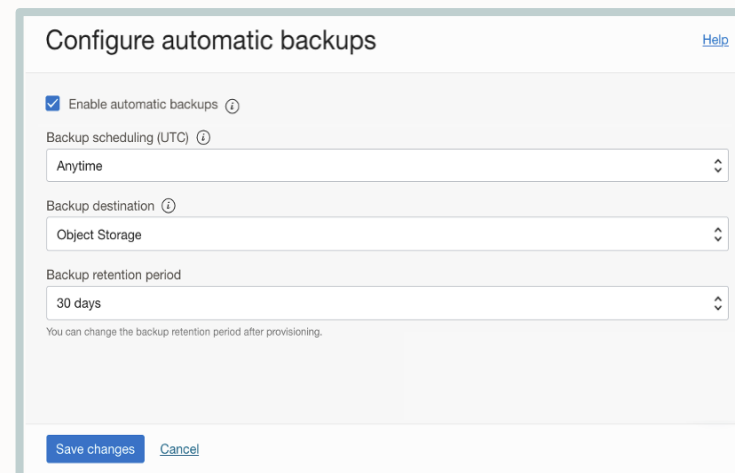
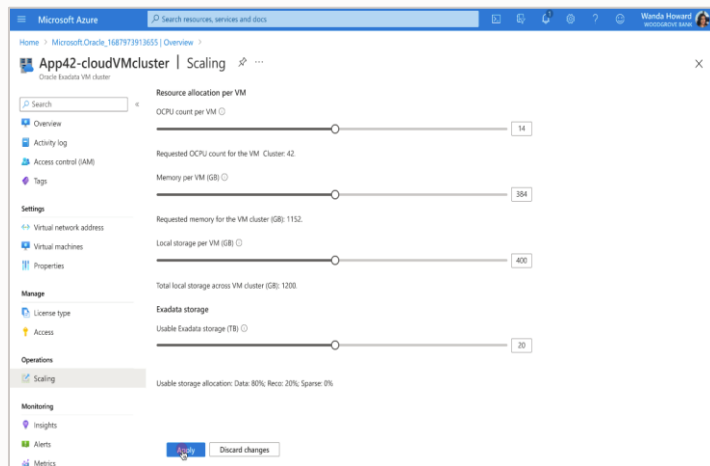
# Cloud Automation for Lifecycle Management

## Microsoft Azure Portal UI REST APIs, SDK, CLI, Terraform

- Create Oracle Exadata Infrastructure and Exadata VM Clusters
- Schedule Infrastructure Maintenance
- Scale Servers and OCPUs

## Oracle Cloud UI REST APIs, SDK, CLI, Terraform

- Create Database Homes and Databases
- Update Operating System, Grid Infrastructure, and Databases
- Backup and Recover Databases
- Enable Data Guard





# Terraform Automation

Combine AZURE and OCI provider to create the Exadata infrastructure, the VM cluster and the database in the same Terraform

```
terraform {
  required_providers {
    azurerm = {
      source = "hashicorp/azurerm"
      version = "= 3.69.0"
    }
    oci = {
      source = "oracle/oci"
      version = "5.10.0"
    }
  }
  required_version = ">= 0.14.9"
}
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:  
+ create

Terraform will perform the following actions:

```
# azurerm_resource_group_template_deployment.exa_cluster will be created
+ resource "azurerm_resource_group_template_deployment" "exa_cluster" {
  + deployment_mode = "Incremental"
  + id              = (known after apply)
  + name            = "Terraform-Deploy-DemoExa-01-VMCluster-01"
  + output_content  = (known after apply)
  + parameters_content = jsonencode(
    {
      + clusterName = {
        + value = "DemoExa-01-VMCluster-01"
      }
    }
  )
}

# azurerm_resource_group_template_deployment.exa_infra will be created
+ resource "azurerm_resource_group_template_deployment" "exa_infra" {
  + deployment_mode = "Incremental"
  + id              = (known after apply)
  + name            = "Terraform-Deploy-DemoExa-01-Infra-02"
  + output_content  = (known after apply)
  + parameters_content = jsonencode(
    {
      + infraName = {
        + value = "DemoExa-01-Infra-02"
      }
    }
  )
}
```

# Terraform Automation

The OCI provider create the database  
in Oracle Database@Azure

```
oci_database_db_home.exa_db_home will be created
resource "oci_database_db_home" "exa_db_home" {
  + compartment_id      = (known after apply)
  + database_software_image_id = (known after apply)
  + db_home_location    = (known after apply)
  + db_system_id        = (known after apply)
  + db_version          = "19.20.0.0"
  + defined_tags        = (known after apply)
  + display_name        = "TFDBHOME"
  + freeform_tags       = (known after apply)
  + id                  = (known after apply)
  + is_desupported_version = (known after apply)
  + kms_key_id          = (known after apply)
  + kms_key_version_id  = (known after apply)
  + last_patch_history_entry_id = (known after apply)
  + lifecycle_details   = (known after apply)
  + source              = "VM_CLUSTER_NEW"
  + state               = (known after apply)
  + time_created        = (known after apply)
  + vm_cluster_id       = (known after apply)

  + database {
    + admin_password      = (sensitive value)
    + backup_id          = (known after apply)
    + backup_tde_password = (sensitive value)
    + character_set       = (known after apply)
    + connection_strings = (known after apply)
    + database_id         = (known after apply)
    + database_software_image_id = (known after apply)
    + db_name             = "TFCDDB"
    + db_unique_name      = (known after apply)
    + db_workload         = "OLTP"
    + defined_tags        = (known after apply)
    + freeform_tags       = (known after apply)
    + id                  = (known after apply)
    + kms_key_id          = (known after apply)
    + kms_key_version_id  = (known after apply)
    + lifecycle_details   = (known after apply)
    + ncharacter_set      = (known after apply)
    + one_off_patches     = (known after apply)
    + pdb_name            = "TFPDB"
    + sid_prefix          = (known after apply)
    + state               = (known after apply)
    + tde_wallet_password = (sensitive value)
    + time_created        = (known after apply)
    + time_stamp_for_point_in_time_recovery = (known after apply)
    + vault_id            = (known after apply)
  }
}
```

**Plan:** 3 to add, 0 to change, 0 to destroy.

**Do you want to perform these actions?**

Terraform will perform the actions described above.  
Only 'yes' will be accepted to approve.

Enter a value:

# Support

# Service Levels



- SLA Oracle Database Exadata 99.95% <https://www.oracle.com/cloud/sla/>
- SLA Oracle Autonomous Database Serverless 99.995% <https://www.oracle.com/cloud/sla/>
- SLO Oracle Database@Azure 99.9% <https://docs.oracle.com/en-us/iaas/Content/General/Reference/servicelevelobjectives.htm>
- Support process Response – support level parity
- [SLA Breach - Service pillars](#)

# Azure support request

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [Oracle Database@Azure | Overview](#) > [OracleSubscription](#) > [default](#) | [Support + Troubleshooting](#) > [Help + support](#) >

## New support request

1. Problem description

2. Recommended solution

3. Additional details

4. Review + create

**Tell us your issue, and we'll help you resolve it.**

Provide information about your billing, subscription, quota management, or technical issue (including requests for technical advice).

Issue type \*

Technical

Subscription \*

omcpmout1 (641531f9-2e12-430b-be35-0d8eba8d957c)

Can't find your subscription? [Show more](#)

Service

☐ My services

☒ All services

Service type \*

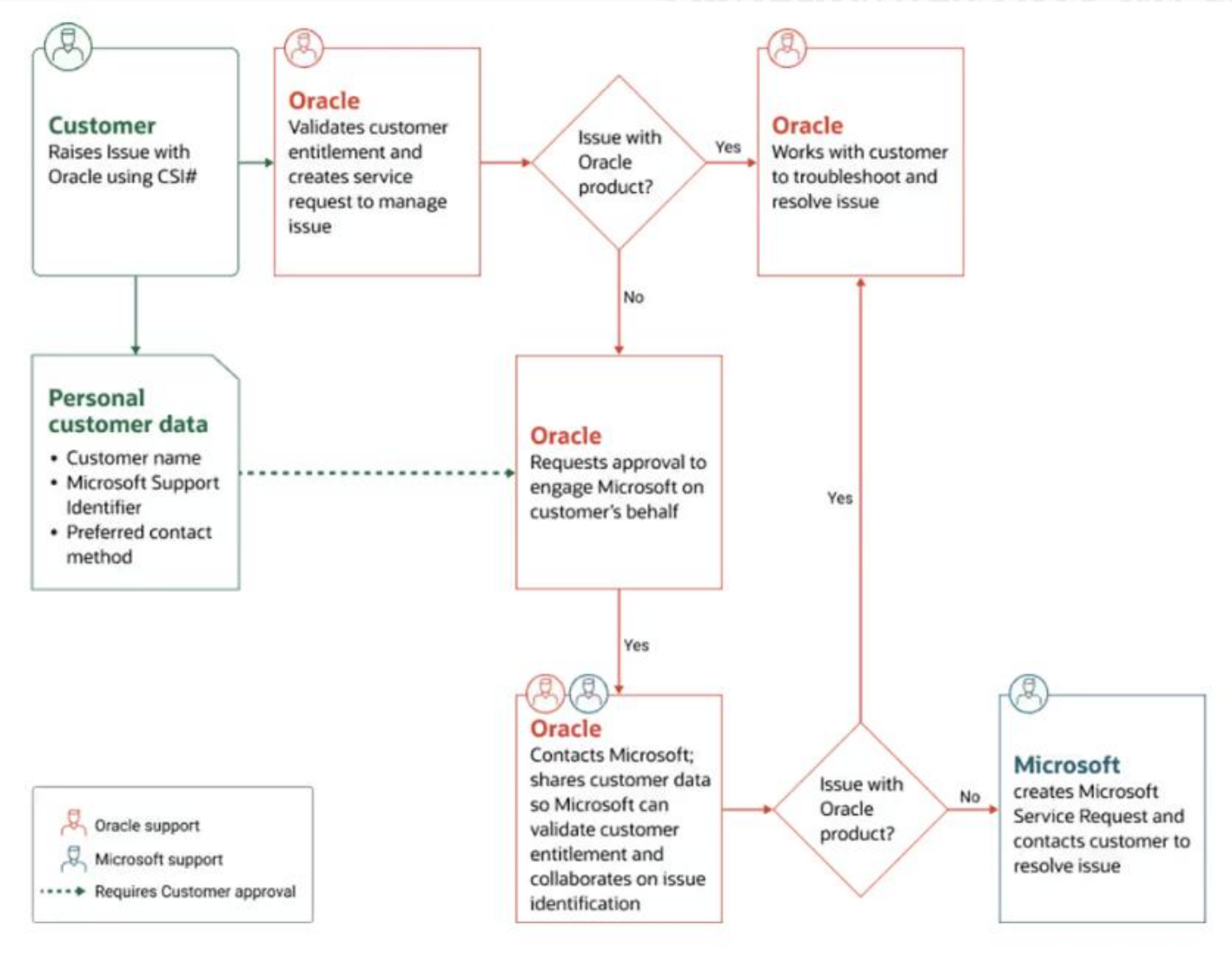
Oracle Database@Azure

Support for Oracle Database@Azure is provided by the Oracle Support team. Please visit the [OCI Support portal](#). If you're a first time user, you will need to register with the Oracle account in the OCI Support portal and have it approved by your Cloud Administrator. For any issues with accessing support for your account, please contact Oracle Support via email or phone, [details here](#)

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# Collaborative Model Support



# Best practices - Troubleshooting



- Failing to deploy Cluster VM
  - Select Azure Subscription – Oracle Database@Azure
  - Select Azure resource group inside Azure Subscription
  - Select VNET in resource group
  - Delegated client subnet inside selected VNET
- If you cannot create resources – make sure there is no Azure lock
- Remove existing resources in client subnet and settings (delegated subnet)
- Can you access the database from the VNET ?
- Can you resolve the FQDN ?
- Make sure OCI Tenancy is IAM Domains enabled
- Cannot go to Go to OCI link
  - Check SSO or OCI credentials

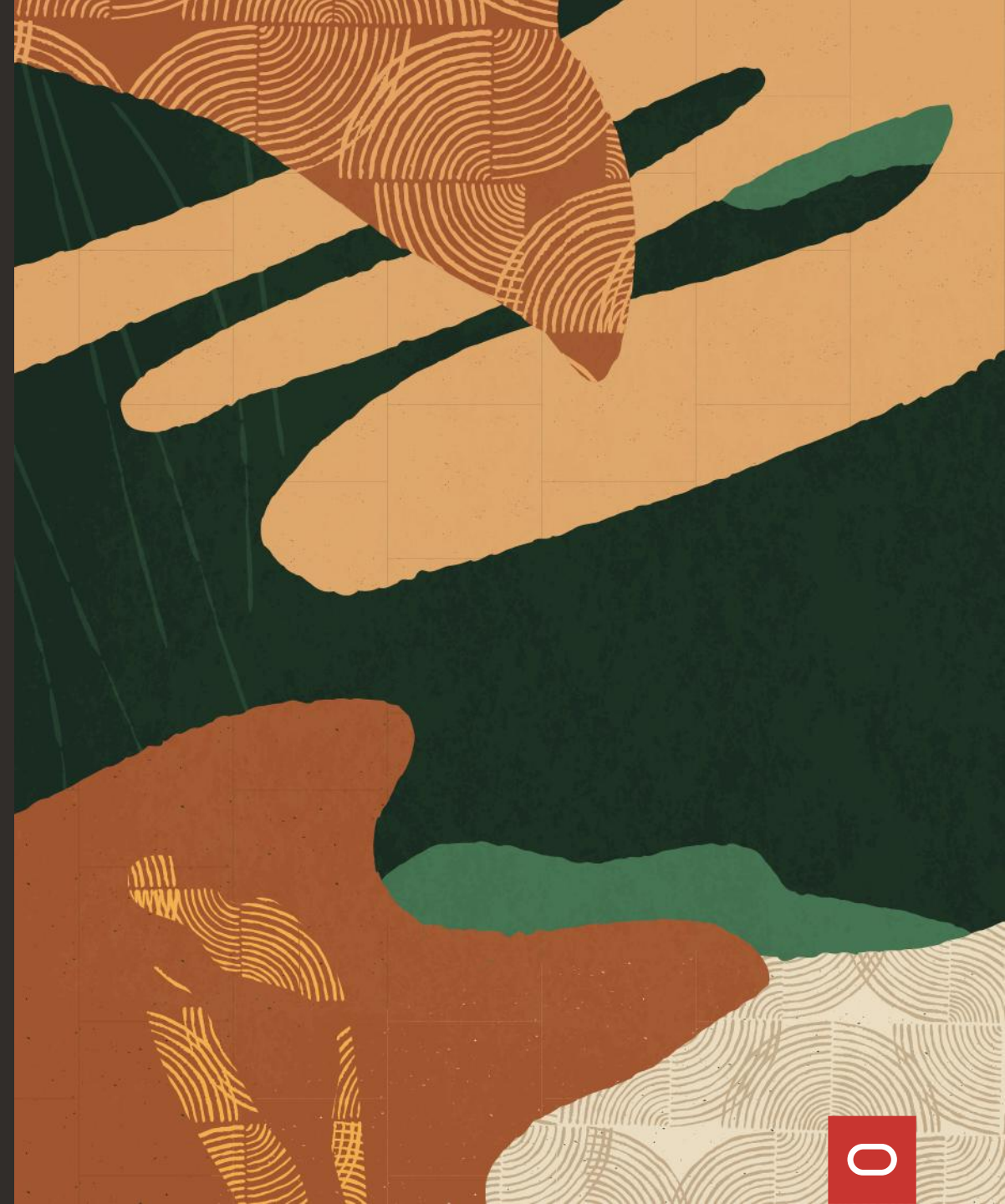


# Thank you & Next Steps

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**OCI Strategic Partnership and Growth**

Ready to assist you in planning your migration





# Oracle Cloud Infrastructure for Microsoft Azure professionals

| Concept                              | Microsoft Azure                  | Oracle Cloud Infrastructure                 |
|--------------------------------------|----------------------------------|---------------------------------------------|
| Cluster of data centers and services | Region                           | <a href="#">Region</a>                      |
| Abstracted data center               | Availability Zone                | <a href="#">Availability Domain</a>         |
| Account                              | Account                          | <a href="#">Tenancy</a>                     |
| Organizing resources                 | Subscriptions<br>Resource Groups | <a href="#">Compartments</a>                |
| Virtual Network                      | Virtual Network (VNET)           | <a href="#">Virtual Cloud Network (VCN)</a> |
| Dedicated Private Connectivity       | ExpressRoute                     | <a href="#">FastConnect</a>                 |
| Identity and Access Management       | Active Directory                 | <a href="#">OCI IAM Domains</a>             |



# Multicloud Services

| Services                      | OCI database services                                   | Oracle Database@Azure                    | Oracle Interconnect for Microsoft Azure                                 | Oracle Database Service for Azure          |
|-------------------------------|---------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------|
| <b>Connectivity</b>           | Native in OCI                                           | Native in Azure                          | Dedicated Tunnel, Customer managed                                      | Dedicated tunnel, Oracle managed           |
| <b>Network cost</b>           | Zero (local in OCI)                                     | Zero (local to Azure)                    | ExpressRoute port and FastConnect port charges                          | Azure Vnet peering egress and ingress cost |
| <b>OCI services available</b> | ExaDB-D, ADB, Base DB, MySQL Heatwave                   | ExaDB-D                                  | All OCI services                                                        | ExaDB-D, ADB, Base DB, MySQL Heatwave      |
| <b>Common use cases</b>       | App DB full stack in OCI                                | App DB full stack in Azure               | App integration, App DB split stack                                     | App DB split stack                         |
| <b>SLA</b>                    | ExaDB-D 99.95%; ADB 99.995%; BaseDB 99.9%; MySQL 99.95% | ExaDB-D 99.95%; ADB 99.995%;             | Customer managed                                                        | SLO for Availability                       |
| <b>Region availability</b>    | 66 OCI regions                                          | 18 Regions                               | 12 regions                                                              | 12 regions                                 |
| <b>Latency</b>                |                                                         | Azure native                             | < 2ms                                                                   | ~ 2ms                                      |
| <b>Support</b>                | OCI support                                             | OCI and Azure collaborative              | OCI and Azure collaborative                                             | OCI and Azure collaborative                |
| <b>Automation</b>             | Yes                                                     | Yes                                      | Yes                                                                     | No                                         |
| <b>Decision points</b>        | High performance + low latency + no or low MACC         | Large unmet MACC + ExaDB-D + low latency | Azure and OCI multicloud integrations / Need ADB and BaseDB immediately | Rapid deployment for quick POC             |



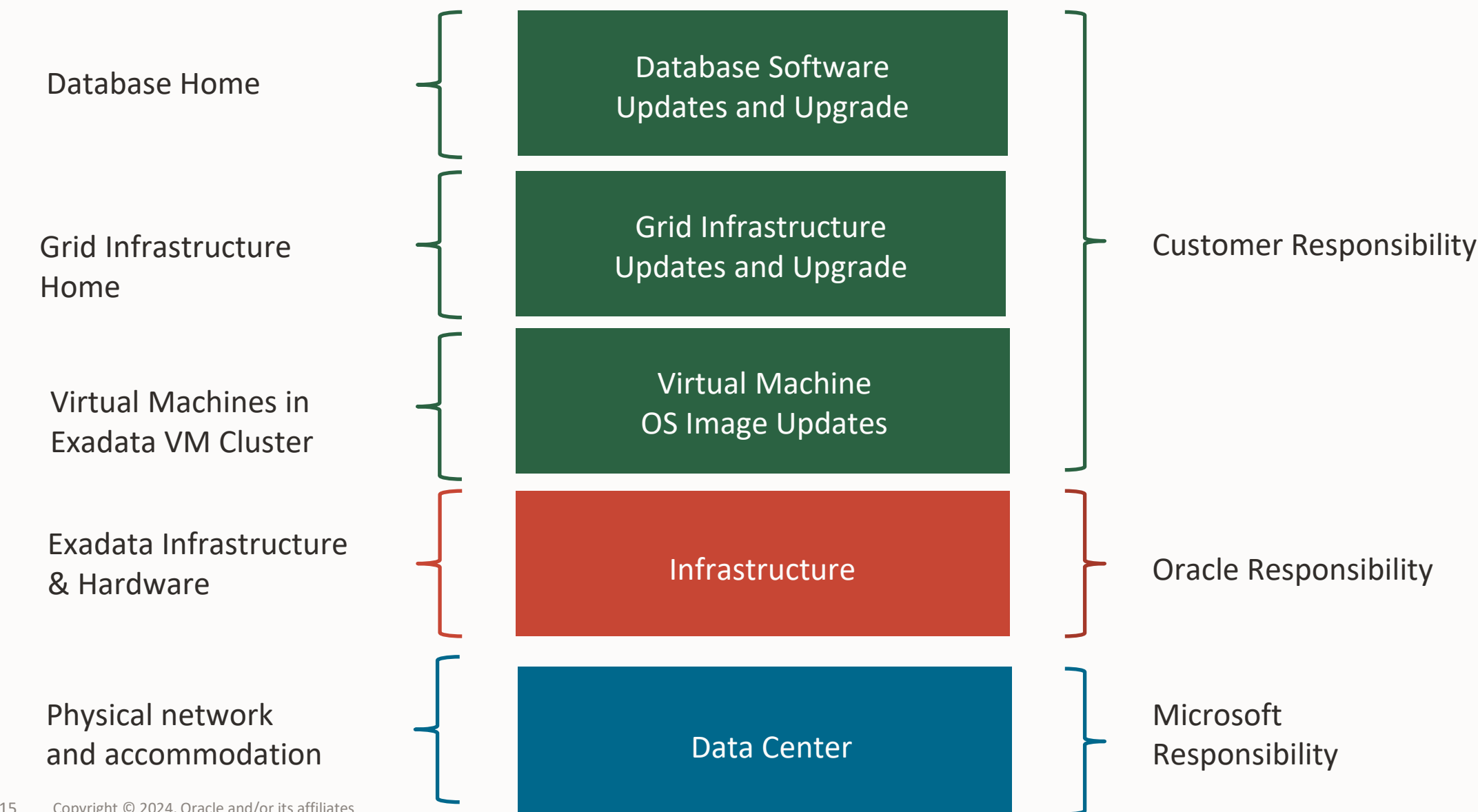
# OCCA Tool for ExaDB-D Sizing



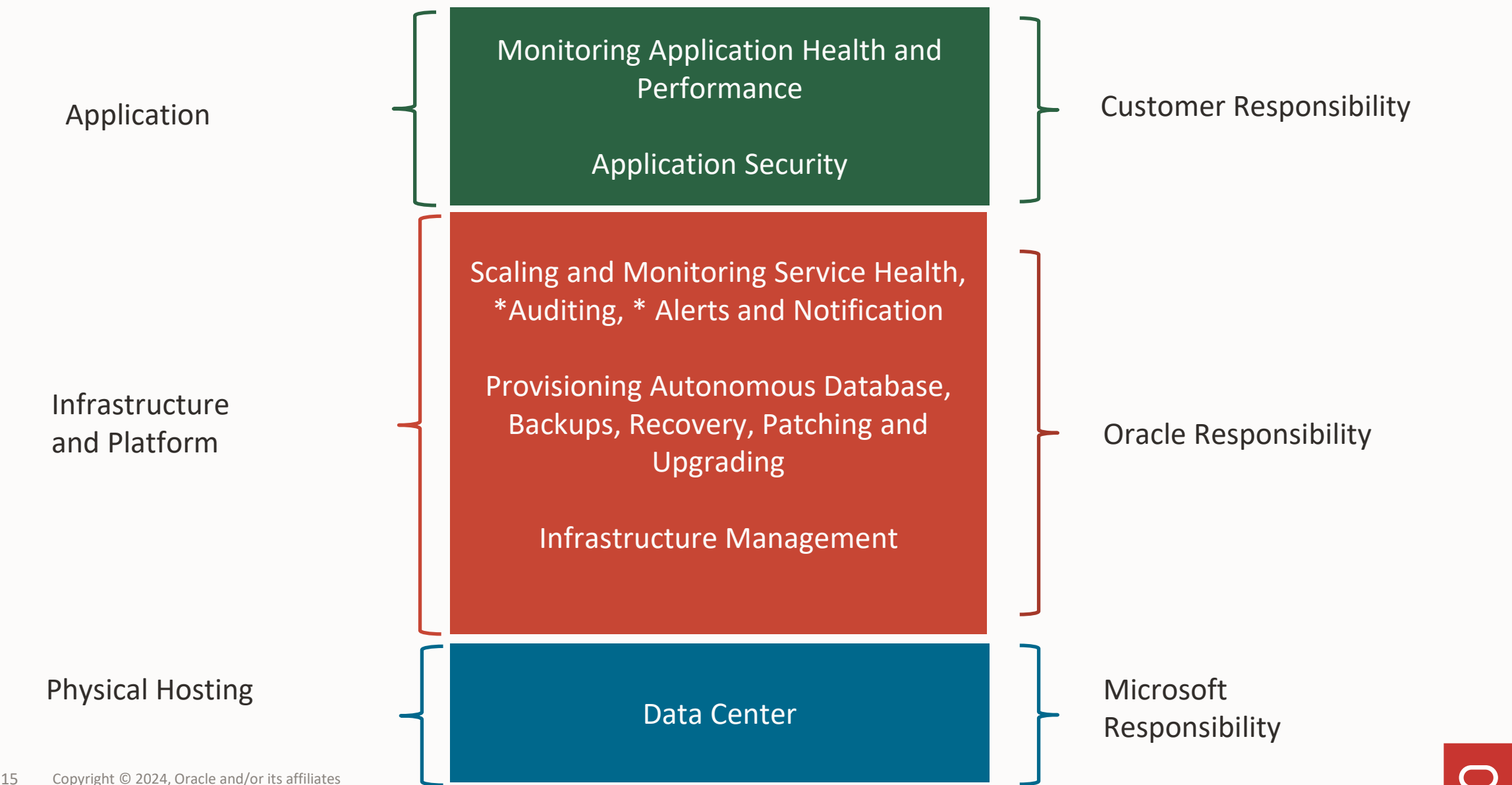
Oracle Cloud Capacity Analytics (OCCA) is available to all Oracle employees internally (VPN required). See the links below to access the OCCA sizing tool and related content including User's Guide, overview presentation, and pre-recorded training.

- [OCCA Web Sizing Tool](#)
- [OCCA User's Guide](#)
- [OCCA Overview Presentation](#)
- [OCCA Slack Channel \(#occa\\_sizing\\_tool\\_users\)](#)
- [OCCA Overview Presentation Video](#)
- [Download AWR Miner](#)
- [Download EMCC Extracts](#)
- [Download OCCA Desktop](#)

# Oracle Database@Azure Responsibility Matrix



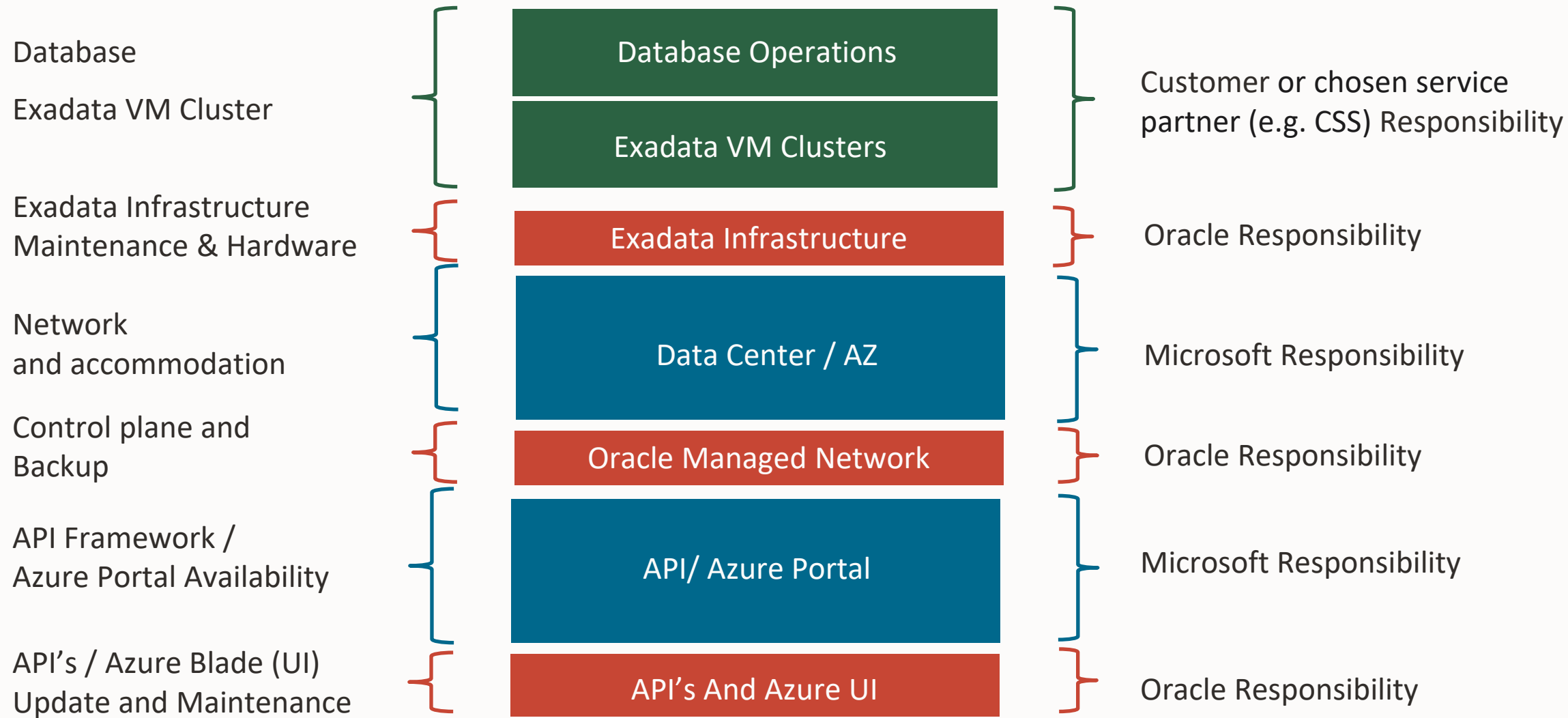
# Oracle Database@Azure ADB-S Responsibility Matrix





Oracle Database@Azure

# Oracle Database@Azure Responsibility Matrix



# Oracle Database@Azure Operational Resources

|                                 | Azure                                                | OCI                                                                                                                    |
|---------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Resource & Lifecycle Management | Exadata instance management<br>(Infra. & VM cluster) | VM Cluster & DB management<br>Scaling, Maintenance<br>DB CRUD, backup, keys,<br>Database connection string information |
| Observability                   | VM and Databases Metrics                             | VM Databases logs and metrics                                                                                          |
| Billing, Costs & Usage          | Invoice<br>Cost management                           | Detailed usage reporting                                                                                               |
| Support                         | Collaboration Service Request first line to OCI      |                                                                                                                        |



# Autonomous Database - Serverless Operational Resources

|                                 | Azure                                           | OCI                                  |
|---------------------------------|-------------------------------------------------|--------------------------------------|
| Resource & Lifecycle Management | ADB-S creation                                  | ADB-S Management<br>Daily operations |
| Observability                   | Database logs and metrics*                      | Databases logs and metrics           |
| Billing, Costs & Usage          | Invoice<br>Cost management                      | Detailed usage reporting             |
| Support                         | Collaboration Service Request first line to OCI |                                      |







Oracle Database@Azure

# Oracle Database@Azure Operational Resource Dependencies

- Oracle Database@Azure requires Client connectivity through Azure network
- Oracle Database@Azure is connected through OCI managed network for Exadata Database Service Control plane and automatic/standalone backup
- Oracle Database@Azure connectivity setup between OCI VCN and Azure edge router (MeetMe).
- Oracle Database@Azure Exadata Database resources are created and managed through API's and Azure Blade (UI) who are created and maintained by Oracle.
- APIs for creating and managing Exadata Database Service resources are published on the Azure API Framework.
- Blade (Azure UI) for Oracle Database@Azure is made available on Azure portal





# Runtime vs Operational Outage Impact and SLA's affected

|                                                                       | Microsoft                                                  | OCI                                                               |
|-----------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Operations – Manageability is affected by following items:</b>     |                                                            |                                                                   |
| <b>Exadata Resource Management- Azure Blade(UI) and API Framework</b> | Affected when Portal/API framework is unavailable on Azure | N/A                                                               |
| <b>Database / Exadata Operations - OCI Console and API's</b>          | N/A                                                        | Affected when OCI control plane is down – OCI Manageability SLA's |
| <b>Runtime – Availability is affected by the following items:</b>     |                                                            |                                                                   |
| <b>Client Connectivity on Azure Vnet to Exadata Database Service</b>  | Affected – Azure SLA's                                     | N/A                                                               |
| <b>Backend connectivity to OCI region</b>                             | N/A                                                        | Backup affected – OCI SLA's                                       |
| <b>Connectivity between Azure edge router and OCI VCN</b>             | N/A                                                        | Affected OCI – SLA's                                              |
| <b>Azure AZ availability</b>                                          | Affected – Azure SLA's                                     | N/A                                                               |
| <b>OCI AD availability</b>                                            | N/A                                                        | Backup affected – OCI SLA's                                       |

<https://www.oracle.com/uk/cloud/sla/>

<https://www.microsoft.com/licensing/docs/view/Service-Level-Agreements-SLA-for-Online-Services?lang=1>

# What is available in Azure and OCI console

|                       | ADB-S @Azure                                            |                          |
|-----------------------|---------------------------------------------------------|--------------------------|
|                       | Azure portal                                            | OCI console              |
| Type                  | ATP and ADW                                             |                          |
| Network Access        | Private access                                          |                          |
| Autonomous Data Guard | Autonomous Data Guard<br>Backup-based disaster recovery |                          |
| Wallet                | Download                                                | Download<br>Rotate       |
| Connection strings    | Yes                                                     |                          |
| Resource Scaling      | Yes                                                     |                          |
| Elastic Pool          | No*                                                     | Yes                      |
| Start/Stop            | No                                                      | Yes                      |
| NSG                   | No*                                                     | Yes                      |
| Performance Hub       | No                                                      | Yes                      |
| Encryption Keys       | No                                                      | Yes                      |
| Backup/Restore        | No                                                      | Yes                      |
| Database Actions      | No*                                                     | Yes                      |
| Admin Password        | Yes (During Provisioning)                               | Yes (After provisioning) |
| Backup Retention      | Yes (During Provisioning)                               | Yes (After provisioning) |
| Licensing             | Yes (During Provisioning)                               | Yes (After provisioning) |
| Enable mTLS           | Yes (During Provisioning)                               | Yes (After provisioning) |

\* Feature on the roadmap

